

Thermal Heat Flow Analysis

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```
#, include=FALSE}  
#knitr::opts_chunk$set(echo = FALSE, warning = FALSE, message = FALSE)  
# echo = FALSE hides the code blocks in the final PDF  
# warning/message = FALSE keeps the document clean of R's startup text
```

0.1 Section 2: Examine and Edit dataset

```
#, include=FALSE}  
library(tidyverse) #installed tidyverse  
  
## Warning: package 'ggplot2' was built under R version 4.5.2  
## Warning: package 'readr' was built under R version 4.5.2  
## Warning: package 'lubridate' was built under R version 4.5.2  
  
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --  
## v dplyr      1.1.4      v readr      2.2.0  
## v forcats    1.0.1      v stringr    1.5.2  
## v ggplot2    4.0.1      v tibble     3.3.0  
## v lubridate  1.9.5      v tidyr      1.3.1  
## v purrr      1.1.0  
## -- Conflicts ----- tidyverse_conflicts() --  
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag()      masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(nlme)
```

```
##
## Attaching package: 'nlme'
##
## The following object is masked from 'package:dplyr':
##
##      collapse
```

```
library(ggplot2)
library(here)
```

```
## here() starts at /Users/sunilmd/Desktop/vscode/MATH-516/project-2-nikita_zhuo
```

```
library(kableExtra)
```

```
##
## Attaching package: 'kableExtra'
##
## The following object is masked from 'package:dplyr':
##
##      group_rows
```

```
library(modelsummary)
```

```
## Warning: package 'modelsummary' was built under R version 4.5.2
```

```
# Load the dataset
```

```
file_path <- here("analysis_and_plots/heat_flow_ICE_EPFL.csv")
df <- read.csv(file_path)
```

```
df_clean_v1 <- df %>% filter(!is.na(HF_value)) #remove rows with NA in HF_value
print(dim(df_clean_v1)) #2638 rows after removing NAs
```

```
## [1] 2638    7
```

```
print(dim(df)) #2736 rows
```

```
## [1] 2736    7
```

```
# factorize dataset
```

```
df_clean <- df_clean_v1 %>%
  mutate(across(
    c(person, day, session, activity, sex, HF_type),
    as.factor
  ))
```

Now do plots to see trends

```
# Summarize by person, day, session, sensor
```

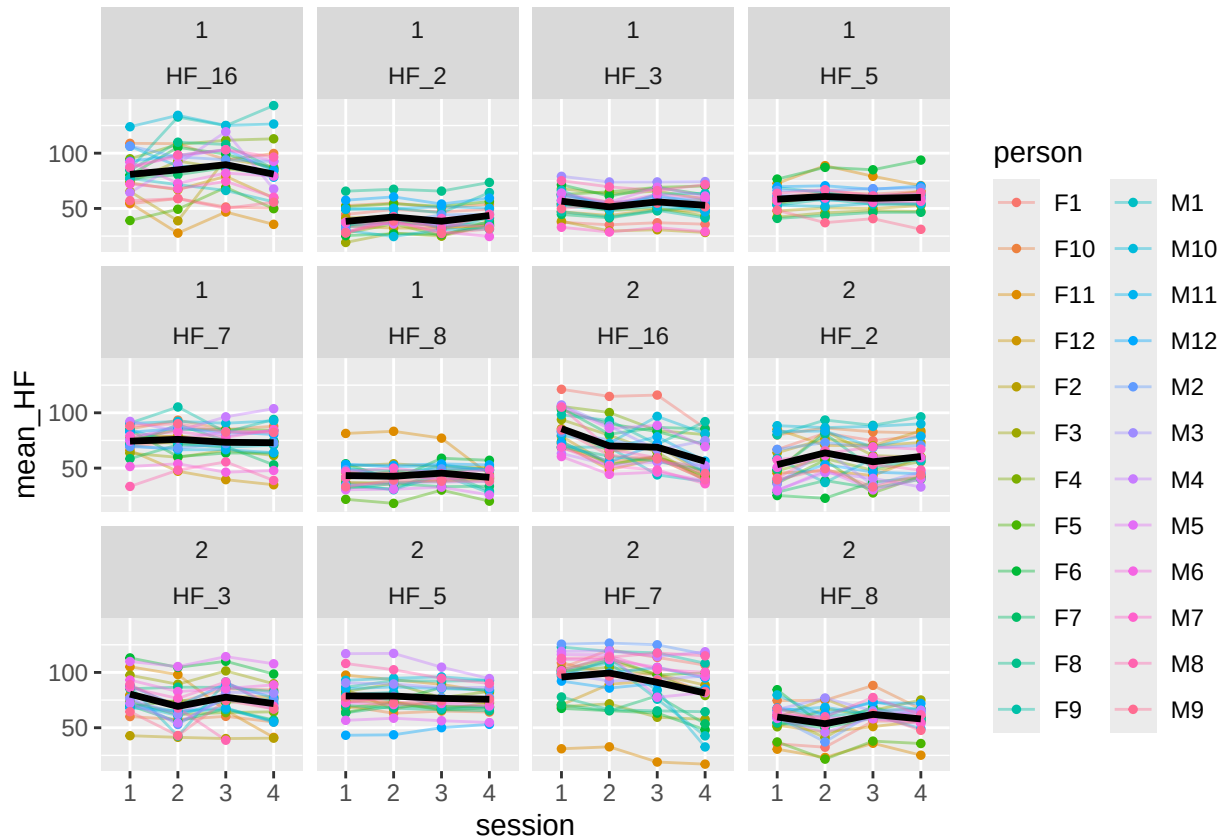
```
df_grid_summary <- df_clean %>%
  group_by(person, day, session, HF_type) %>%
  summarize(mean_HF = mean(HF_value), .groups = 'drop')
```

```
# Create Faceted Spider Plot
```

```
ggplot(df_grid_summary, aes(x = session, y = mean_HF, group = person, color = person)) +
  geom_line(alpha = 0.4) +
```

```
geom_point(size = 1) + stat_summary(aes(group = 1), fun = mean, geom = "line",
  color = "black", size = 1.2) +
facet_wrap (day ~ HF_type) #, scales = "free_y") + # Scales="free_y" lets you see the slopes clearly.
```

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



```
theme_minimal() +
labs(title = "Heat Flow Evolution Across Sessions",
  subtitle = "Rows: Day (1=Cold, 2=Warm) | Columns: Sensor Type",
  x = "Session Number",
  y = "Mean Heat Flow (W/m²)") +
theme(legend.position = "none")
```

```
## <theme> List of 147
## $ line : <ggplot2::element_line>
## ..@ colour : chr "black"
## ..@ linewidth : num 0.5
## ..@ linetype : num 1
## ..@ lineend : chr "butt"
## ..@ linejoin : chr "round"
## ..@ arrow : logi FALSE
## ..@ arrow.fill : chr "black"
## ..@ inherit.blank: logi TRUE
```

```

## $ rect                                     : <ggplot2::element_rect>
##   ..@ fill          : chr "white"
##   ..@ colour        : chr "black"
##   ..@ linewidth     : num 0.5
##   ..@ linetype       : num 1
##   ..@ linejoin       : chr "round"
##   ..@ inherit.blank : logi TRUE
## $ text                                     : <ggplot2::element_text>
##   ..@ family         : chr ""
##   ..@ face           : chr "plain"
##   ..@ italic         : chr NA
##   ..@ fontweight     : num NA
##   ..@ fontwidth      : num NA
##   ..@ colour         : chr "black"
##   ..@ size           : num 11
##   ..@ hjust          : num 0.5
##   ..@ vjust          : num 0.5
##   ..@ angle          : num 0
##   ..@ lineheight     : num 0.9
##   ..@ margin         : <ggplot2::margin> num [1:4] 0 0 0 0
##   ..@ debug          : logi FALSE
##   ..@ inherit.blank : logi TRUE
## $ title                                   : chr "Heat Flow Evolution Across Sessions"
## $ point                                   : <ggplot2::element_point>
##   ..@ colour         : chr "black"
##   ..@ shape          : num 19
##   ..@ size           : num 1.5
##   ..@ fill           : chr "white"
##   ..@ stroke         : num 0.5
##   ..@ inherit.blank : logi TRUE
## $ polygon                                : <ggplot2::element_polygon>
##   ..@ fill          : chr "white"
##   ..@ colour        : chr "black"
##   ..@ linewidth     : num 0.5
##   ..@ linetype       : num 1
##   ..@ linejoin       : chr "round"
##   ..@ inherit.blank : logi TRUE
## $ geom                                    : <ggplot2::element_geom>
##   ..@ ink           : chr "black"
##   ..@ paper         : chr "white"
##   ..@ accent        : chr "#3366FF"
##   ..@ linewidth     : num 0.5
##   ..@ borderwidth   : num 0.5
##   ..@ linetype       : int 1
##   ..@ bordertype    : int 1
##   ..@ family        : chr ""
##   ..@ fontsize      : num 3.87
##   ..@ pointsize     : num 1.5
##   ..@ pointshape    : num 19
##   ..@ colour        : NULL
##   ..@ fill          : NULL
## $ spacing                                : 'simpleUnit' num 5.5points
##   ..- attr(*, "unit")= int 8
## $ margins                                : <ggplot2::margin> num [1:4] 5.5 5.5 5.5 5.5

```

```

## $ aspect.ratio          : NULL
## $ axis.title            : NULL
## $ axis.title.x          : <ggplot2::element_text>
##   ..@ family           : NULL
##   ..@ face             : NULL
##   ..@ italic           : chr NA
##   ..@ fontweight       : num NA
##   ..@ fontwidth       : num NA
##   ..@ colour           : NULL
##   ..@ size             : NULL
##   ..@ hjust            : NULL
##   ..@ vjust            : num 1
##   ..@ angle            : NULL
##   ..@ lineheight       : NULL
##   ..@ margin           : <ggplot2::margin> num [1:4] 2.75 0 0 0
##   ..@ debug            : NULL
##   ..@ inherit.blank    : logi TRUE
## $ axis.title.x.top      : <ggplot2::element_text>
##   ..@ family           : NULL
##   ..@ face             : NULL
##   ..@ italic           : chr NA
##   ..@ fontweight       : num NA
##   ..@ fontwidth       : num NA
##   ..@ colour           : NULL
##   ..@ size             : NULL
##   ..@ hjust            : NULL
##   ..@ vjust            : num 0
##   ..@ angle            : NULL
##   ..@ lineheight       : NULL
##   ..@ margin           : <ggplot2::margin> num [1:4] 0 0 2.75 0
##   ..@ debug            : NULL
##   ..@ inherit.blank    : logi TRUE
## $ axis.title.x.bottom   : NULL
## $ axis.title.y          : <ggplot2::element_text>
##   ..@ family           : NULL
##   ..@ face             : NULL
##   ..@ italic           : chr NA
##   ..@ fontweight       : num NA
##   ..@ fontwidth       : num NA
##   ..@ colour           : NULL
##   ..@ size             : NULL
##   ..@ hjust            : NULL
##   ..@ vjust            : num 1
##   ..@ angle            : num 90
##   ..@ lineheight       : NULL
##   ..@ margin           : <ggplot2::margin> num [1:4] 0 2.75 0 0
##   ..@ debug            : NULL
##   ..@ inherit.blank    : logi TRUE
## $ axis.title.y.left     : NULL
## $ axis.title.y.right    : <ggplot2::element_text>
##   ..@ family           : NULL
##   ..@ face             : NULL
##   ..@ italic           : chr NA
##   ..@ fontweight       : num NA

```

```

## ..@ fontwidth      : num NA
## ..@ colour         : NULL
## ..@ size           : NULL
## ..@ hjust          : NULL
## ..@ vjust          : num 1
## ..@ angle          : num -90
## ..@ lineheight     : NULL
## ..@ margin         : <ggplot2::margin> num [1:4] 0 0 0 2.75
## ..@ debug          : NULL
## ..@ inherit.blank: logi TRUE
## $ axis.text                : <ggplot2::element_text>
## ..@ family          : NULL
## ..@ face            : NULL
## ..@ italic          : chr NA
## ..@ fontweight      : num NA
## ..@ fontwidth       : num NA
## ..@ colour          : chr "#4D4D4DFF"
## ..@ size            : 'rel' num 0.8
## ..@ hjust           : NULL
## ..@ vjust           : NULL
## ..@ angle           : NULL
## ..@ lineheight      : NULL
## ..@ margin          : NULL
## ..@ debug           : NULL
## ..@ inherit.blank: logi TRUE
## $ axis.text.x              : <ggplot2::element_text>
## ..@ family          : NULL
## ..@ face            : NULL
## ..@ italic          : chr NA
## ..@ fontweight      : num NA
## ..@ fontwidth       : num NA
## ..@ colour          : NULL
## ..@ size            : NULL
## ..@ hjust           : NULL
## ..@ vjust           : num 1
## ..@ angle           : NULL
## ..@ lineheight      : NULL
## ..@ margin          : <ggplot2::margin> num [1:4] 2.2 0 0 0
## ..@ debug           : NULL
## ..@ inherit.blank: logi TRUE
## $ axis.text.x.top          : <ggplot2::element_text>
## ..@ family          : NULL
## ..@ face            : NULL
## ..@ italic          : chr NA
## ..@ fontweight      : num NA
## ..@ fontwidth       : num NA
## ..@ colour          : NULL
## ..@ size            : NULL
## ..@ hjust           : NULL
## ..@ vjust           : NULL
## ..@ angle           : NULL
## ..@ lineheight      : NULL
## ..@ margin          : <ggplot2::margin> num [1:4] 0 0 4.95 0
## ..@ debug           : NULL

```

```

## ..@ inherit.blank: logi TRUE
## $ axis.text.x.bottom : <ggplot2::element_text>
## ..@ family : NULL
## ..@ face : NULL
## ..@ italic : chr NA
## ..@ fontweight : num NA
## ..@ fontwidth : num NA
## ..@ colour : NULL
## ..@ size : NULL
## ..@ hjust : NULL
## ..@ vjust : NULL
## ..@ angle : NULL
## ..@ lineheight : NULL
## ..@ margin : <ggplot2::margin> num [1:4] 4.95 0 0 0
## ..@ debug : NULL
## ..@ inherit.blank: logi TRUE
## $ axis.text.y : <ggplot2::element_text>
## ..@ family : NULL
## ..@ face : NULL
## ..@ italic : chr NA
## ..@ fontweight : num NA
## ..@ fontwidth : num NA
## ..@ colour : NULL
## ..@ size : NULL
## ..@ hjust : num 1
## ..@ vjust : NULL
## ..@ angle : NULL
## ..@ lineheight : NULL
## ..@ margin : <ggplot2::margin> num [1:4] 0 2.2 0 0
## ..@ debug : NULL
## ..@ inherit.blank: logi TRUE
## $ axis.text.y.left : <ggplot2::element_text>
## ..@ family : NULL
## ..@ face : NULL
## ..@ italic : chr NA
## ..@ fontweight : num NA
## ..@ fontwidth : num NA
## ..@ colour : NULL
## ..@ size : NULL
## ..@ hjust : NULL
## ..@ vjust : NULL
## ..@ angle : NULL
## ..@ lineheight : NULL
## ..@ margin : <ggplot2::margin> num [1:4] 0 4.95 0 0
## ..@ debug : NULL
## ..@ inherit.blank: logi TRUE
## $ axis.text.y.right : <ggplot2::element_text>
## ..@ family : NULL
## ..@ face : NULL
## ..@ italic : chr NA
## ..@ fontweight : num NA
## ..@ fontwidth : num NA
## ..@ colour : NULL
## ..@ size : NULL

```

```

## ..@ hjust      : NULL
## ..@ vjust      : NULL
## ..@ angle      : NULL
## ..@ lineheight : NULL
## ..@ margin     : <ggplot2::margin> num [1:4] 0 0 0 4.95
## ..@ debug      : NULL
## ..@ inherit.blank: logi TRUE
## $ axis.text.theta : NULL
## $ axis.text.r      : <ggplot2::element_text>
## ..@ family        : NULL
## ..@ face           : NULL
## ..@ italic         : chr NA
## ..@ fontweight     : num NA
## ..@ fontwidth      : num NA
## ..@ colour         : NULL
## ..@ size           : NULL
## ..@ hjust          : num 0.5
## ..@ vjust          : NULL
## ..@ angle          : NULL
## ..@ lineheight     : NULL
## ..@ margin         : <ggplot2::margin> num [1:4] 0 2.2 0 2.2
## ..@ debug          : NULL
## ..@ inherit.blank: logi TRUE
## $ axis.ticks      : <ggplot2::element_blank>
## $ axis.ticks.x     : NULL
## $ axis.ticks.x.top : NULL
## $ axis.ticks.x.bottom : NULL
## $ axis.ticks.y     : NULL
## $ axis.ticks.y.left : NULL
## $ axis.ticks.y.right : NULL
## $ axis.ticks.theta : NULL
## $ axis.ticks.r      : NULL
## $ axis.minor.ticks.x.top : NULL
## $ axis.minor.ticks.x.bottom : NULL
## $ axis.minor.ticks.y.left : NULL
## $ axis.minor.ticks.y.right : NULL
## $ axis.minor.ticks.theta : NULL
## $ axis.minor.ticks.r : NULL
## $ axis.ticks.length : 'rel' num 0.5
## $ axis.ticks.length.x : NULL
## $ axis.ticks.length.x.top : NULL
## $ axis.ticks.length.x.bottom : NULL
## $ axis.ticks.length.y : NULL
## $ axis.ticks.length.y.left : NULL
## $ axis.ticks.length.y.right : NULL
## $ axis.ticks.length.theta : NULL
## $ axis.ticks.length.r : NULL
## $ axis.minor.ticks.length : 'rel' num 0.75
## $ axis.minor.ticks.length.x : NULL
## $ axis.minor.ticks.length.x.top : NULL
## $ axis.minor.ticks.length.x.bottom : NULL
## $ axis.minor.ticks.length.y : NULL
## $ axis.minor.ticks.length.y.left : NULL
## $ axis.minor.ticks.length.y.right : NULL

```



```

## $ axis.minor.ticks.length.theta : NULL
## $ axis.minor.ticks.length.r : NULL
## $ axis.line : <ggplot2::element_blank>
## $ axis.line.x : NULL
## $ axis.line.x.top : NULL
## $ axis.line.x.bottom : NULL
## $ axis.line.y : NULL
## $ axis.line.y.left : NULL
## $ axis.line.y.right : NULL
## $ axis.line.theta : NULL
## $ axis.line.r : NULL
## $ legend.background : <ggplot2::element_blank>
## $ legend.margin : NULL
## $ legend.spacing : 'rel' num 2
## $ legend.spacing.x : NULL
## $ legend.spacing.y : NULL
## $ legend.key : <ggplot2::element_blank>
## $ legend.key.size : 'simpleUnit' num 1.2lines
## ..- attr(*, "unit")= int 3
## $ legend.key.height : NULL
## $ legend.key.width : NULL
## $ legend.key.spacing : NULL
## $ legend.key.spacing.x : NULL
## $ legend.key.spacing.y : NULL
## $ legend.key.justification : NULL
## $ legend.frame : NULL
## $ legend.ticks : NULL
## $ legend.ticks.length : 'rel' num 0.2
## $ legend.axis.line : NULL
## $ legend.text : <ggplot2::element_text>
## ..@ family : NULL
## ..@ face : NULL
## ..@ italic : chr NA
## ..@ fontweight : num NA
## ..@ fontwidth : num NA
## ..@ colour : NULL
## ..@ size : 'rel' num 0.8
## ..@ hjust : NULL
## ..@ vjust : NULL
## ..@ angle : NULL
## ..@ lineheight : NULL
## ..@ margin : NULL
## ..@ debug : NULL
## ..@ inherit.blank: logi TRUE
## $ legend.text.position : NULL
## $ legend.title : <ggplot2::element_text>
## ..@ family : NULL
## ..@ face : NULL
## ..@ italic : chr NA
## ..@ fontweight : num NA
## ..@ fontwidth : num NA
## ..@ colour : NULL
## ..@ size : NULL
## ..@ hjust : num 0

```

```
## ..@ vjust      : NULL
## ..@ angle      : NULL
## ..@ lineheight : NULL
## ..@ margin     : NULL
## ..@ debug      : NULL
## ..@ inherit.blank: logi TRUE
## $ legend.title.position : NULL
## $ legend.position      : chr "none"
## $ legend.position.inside : NULL
## $ legend.direction     : NULL
## $ legend.byrow         : NULL
## $ legend.justification : chr "center"
## $ legend.justification.top : NULL
## $ legend.justification.bottom : NULL
## $ legend.justification.left : NULL
## $ legend.justification.right : NULL
## $ legend.justification.inside : NULL
## [list output truncated]
## @ complete: logi TRUE
## @ validate: logi TRUE
```

0.2 Section 3: Examine Residuals for Linear, Determine transformations (if any)

Look at residuals spread. by session, activity, days. NOT BY sensor or sex!

```
# Base model with all main effects, no interactions, no person random effect
lm_base <- lm(HF_value ~ factor(session) + factor(activity) + factor(HF_type) + factor(day) + factor(sex),
df_clean$resids <- residuals(lm_base)

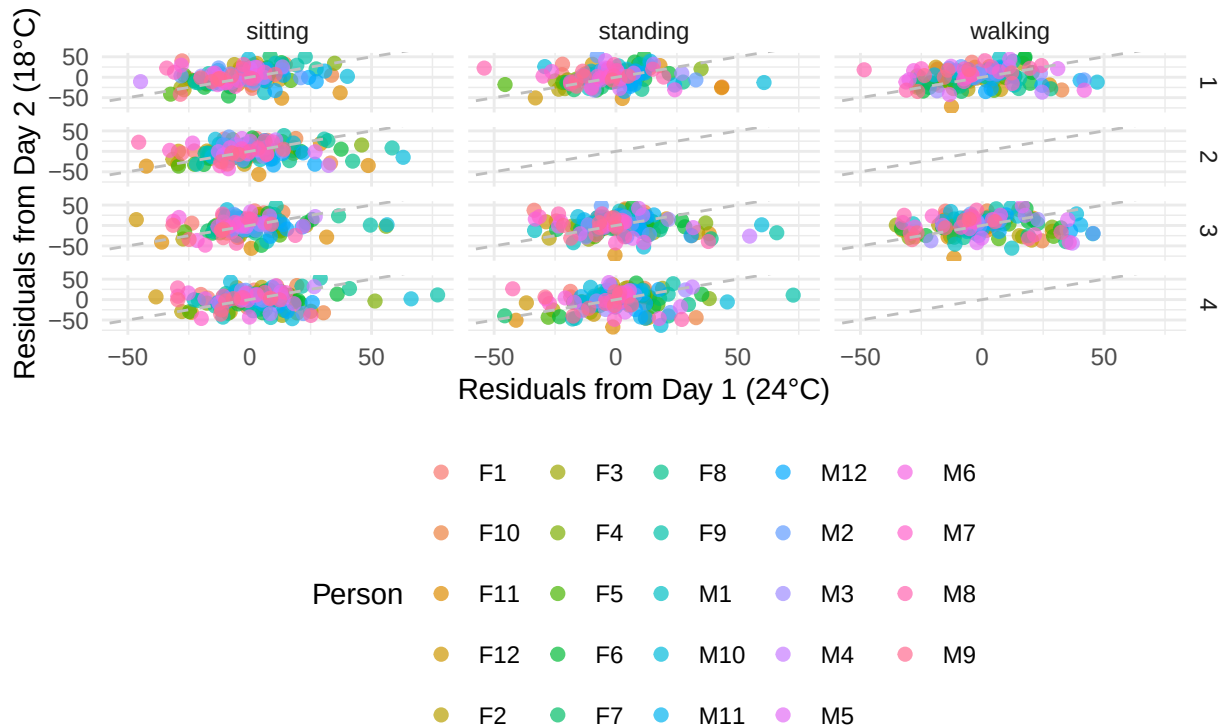
df_pivot_day_narem <- df_clean %>%
  pivot_wider(
    id_cols = c(person, session, activity, HF_type),
    names_from = day,
    values_from = resids,
    names_prefix = "res_day"
  ) %>%
  # Remove cases where data is missing for one of the two days
  filter(!is.na(res_day1) & !is.na(res_day2))

ggplot(df_pivot_day_narem, aes(x = res_day1, y = res_day2, color = factor(person))) +
  geom_point(alpha = 0.7, size = 2) +
  # Dashed line y=x represents perfect correlation
  geom_abline(slope = 1, intercept = 0, linetype = "dashed", color = "gray") +
  facet_grid(session ~ activity) +
  labs(
    title = "Residual Correlation: Day 1 vs Day 2",
    subtitle = "Checking for independence across different temperature environments",
    x = "Residuals from Day 1 (24°C)",
    y = "Residuals from Day 2 (18°C)",
    color = "Person"
  ) +
  theme_minimal() +
```

```
theme(legend.position = "bottom")
```

Residual Correlation: Day 1 vs Day 2

Checking for independence across different temperature environments



Look at residuals spread. by day, activity, and session. Not by sensor or sex!

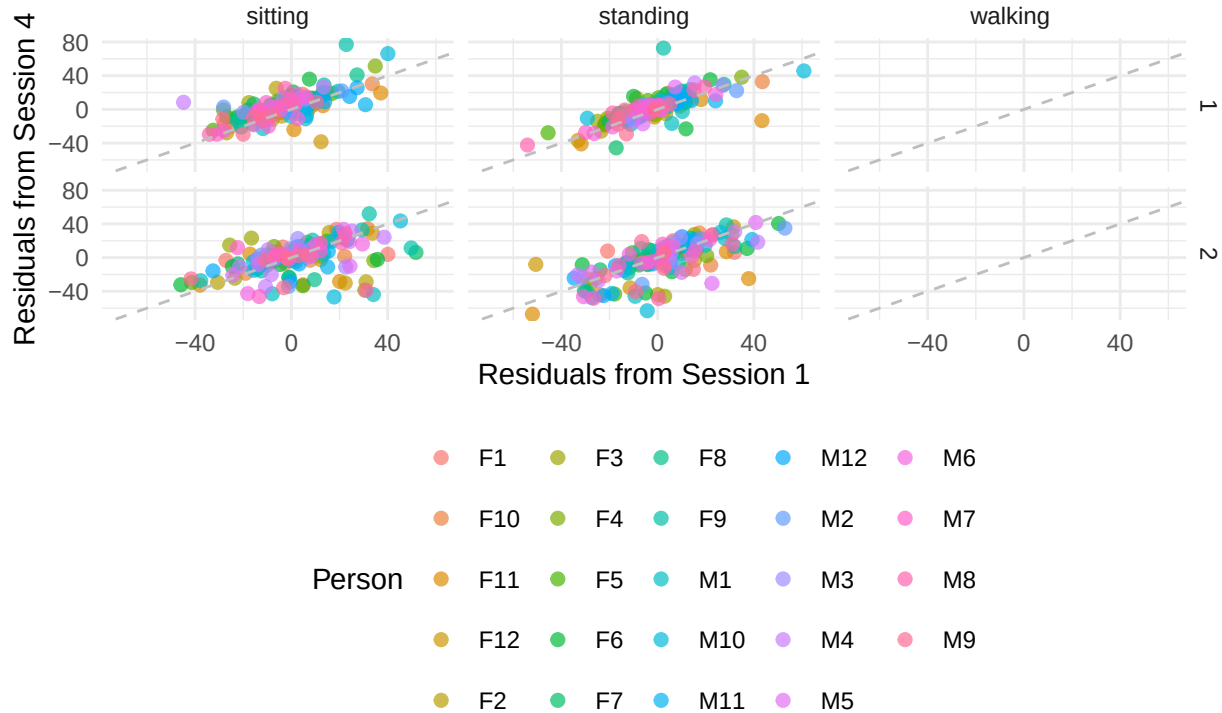
```
df_pivot <- df_clean %>%
  filter(session %in% c(1, 4)) %>%
  pivot_wider(
    id_cols = c(person, day, activity, HF_type),
    names_from = session,
    values_from = resids,
    names_prefix = "res_s"
  )

# Plot residuals of Session 1 vs Session 4 , by sensor and activity, colored by person to check for ind
ggplot(df_pivot, aes(x = res_s1, y = res_s4, color = factor(person))) +
  geom_point(alpha = 0.7, size = 2) +
  geom_abline(slope = 1, intercept = 0, linetype = "dashed", color = "gray") +
  facet_grid(day ~ activity) +
  labs(
    title = "Residual Correlation: Session 1 vs Session 4",
    subtitle = "By Sensor and Activity | Colors represent individual participants",
    x = "Residuals from Session 1",
    y = "Residuals from Session 4",
    color = "Person"
  ) +
  theme_minimal() +
  theme(legend.position = "bottom")
```

```
## Warning: Removed 285 rows containing missing values or values outside the scale range
## (`geom_point()`).
```

Residual Correlation: Session 1 vs Session 4

By Sensor and Activity | Colors represent individual participants



0.3 Section 4: Run model with fixed effects

1. Run the log model with all fixed effects

```
mdl1_log <- gls(
  log(HF_value) ~ day + session + activity + HF_type + sex,
  data = df_clean,
  method = "ML"
)
print(summary(mdl1_log),4)
```

```
## Generalized least squares fit by maximum likelihood
## Model: log(HF_value) ~ day + session + activity + HF_type + sex
## Data: df_clean
## AIC BIC logLik
## 1499.142 1581.431 -735.5711
## Convergence at iteration: 0
##
## Coefficients:
## Value Std.Error t-value p-value
## (Intercept) 4.205999 0.02205313 190.72118 0.0000
## day2 0.207380 0.01262735 16.42311 0.0000
## session2 -0.000208 0.02037319 -0.01022 0.9918
## session3 -0.008779 0.01568979 -0.55955 0.5758
## session4 -0.037588 0.01838417 -2.04459 0.0410
## activitystanding -0.037486 0.01428364 -2.62439 0.0087
```

```

## activitywalking    0.013864 0.01809196    0.76628 0.4436
## HF_typeHF_2       -0.511635 0.02156553 -23.72468 0.0000
## HF_typeHF_3       -0.175129 0.02171339  -8.06548 0.0000
## HF_typeHF_5       -0.106099 0.02181681  -4.86317 0.0000
## HF_typeHF_7        0.062662 0.02148382   2.91671 0.0036
## HF_typeHF_8       -0.417609 0.02162616 -19.31036 0.0000
## sexM              0.036927 0.01248641   2.95740 0.0031
##
## Correlation:
##              (Intr) day2    sessn2 sessn3 sessn4 actvtys actvtyw HF_HF_2
## day2          -0.314
## session2      -0.416  0.114
## session3      -0.352 -0.003  0.382
## session4      -0.382  0.001  0.410  0.424
## activitystanding -0.348  0.081  0.131 -0.001  0.000
## activitywalking -0.371  0.033  0.302 -0.003  0.277  0.396
## HF_typeHF_2    -0.494 -0.001 -0.002 -0.001  0.002  0.001  0.003
## HF_typeHF_3    -0.489 -0.005 -0.002 -0.001  0.004  0.001  0.003  0.501
## HF_typeHF_5    -0.485 -0.013 -0.002 -0.001  0.002  0.001  0.003  0.499
## HF_typeHF_7    -0.496  0.007  0.001 -0.001  0.003  0.001  0.003  0.506
## HF_typeHF_8    -0.492 -0.003 -0.003  0.003  0.001  0.000  0.000  0.503
## sexM           -0.274 -0.015  0.001 -0.001  0.002  0.000  0.002  0.001
##              HF_HF_3 HF_HF_5 HF_HF_7 HF_HF_8
## day2
## session2
## session3
## session4
## activitystanding
## activitywalking
## HF_typeHF_2
## HF_typeHF_3
## HF_typeHF_5      0.495
## HF_typeHF_7      0.503  0.501
## HF_typeHF_8      0.500  0.497  0.505
## sexM             -0.004  0.002 -0.007 -0.001
##
## Standardized residuals:
##              Min          Q1          Med          Q3          Max
## -6.5385520 -0.4525252  0.1159484  0.6284670  2.5561514
##
## Residual standard error: 0.3197864
## Degrees of freedom: 2638 total; 2625 residual

```

2. Run anovas to compare it to models without one of the variables (backwards selection)

```

# print("No day")
mdl1_log_no_day <- gls(
  log(HF_value) ~ session + (activity) + (HF_type) + (sex),
  data = df_clean,
  method = "ML"
)
# print(anova(mdl1_log, mdl1_log_no_day))

# print("No session")
mdl1_log_no_session <- gls(

```

```

log(HF_value) ~ (day) + (activity) + (HF_type) + (sex),
data = df_clean,
method = "ML"
)
# print(anova mdl1_log, mdl1_log_no_session))

# print("No activity")
mdl1_log_no_activity <- gls(
  log(HF_value) ~ (day) + (session) + (HF_type) + (sex),
  data = df_clean,
  method = "ML"
)
# print(anova(mdl1_log, mdl1_log_no_activity))

# print("No sensor")
mdl1_log_no_sensor <- gls(
  log(HF_value) ~ (day) + (session) + (activity) + (sex),
  data = df_clean,
  method = "ML"
)
# print(anova(mdl1_log, mdl1_log_no_sensor))

# print("No sex")
mdl1_log_no_sex <- gls(
  log(HF_value) ~ (day) + (session) + (activity) + (HF_type),
  data = df_clean,
  method = "ML"
)
# print(anova(mdl1_log, mdl1_log_no_sex))

#Option 1: Run the "Drop-One" analysis
anova_summary <- drop1(mdl1_log, test = "Chisq")
print(anova_summary)

```

```

## Single term deletions
##
## Model:
## log(HF_value) ~ day + session + activity + HF_type + sex
##      Df    AIC    LRT Pr(>Chi)
## <none>    1499.1
## day      1 1755.2 258.01 < 2.2e-16 ***
## session  3 1498.2   5.04  0.169116
## activity  2 1505.9  10.78  0.004567 **
## HF_type   5 2438.1 948.97 < 2.2e-16 ***
## sex       1 1505.9   8.77  0.003054 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

#option 2: Combine the anovas

```

#helper function to get anova outputs
get_baseline_metrics <- function(orig, name) {
  data.frame(

```

```

    Model = name,
    Df = attr(logLik(orig), "df"),
    AIC = round(AIC(orig), 4),
    BIC = round(BIC(orig), 4),
    LogLik = round(as.numeric(logLik(orig)), 4),
    stringsAsFactors = FALSE
  )
}

get_anova_outputs <- function(orig, new, name) {
  res <- anova(orig, new)
  data.frame(
    Model = name,
    Df = attr(logLik(new), "df"),
    AIC = round(res$AIC[2], 4),
    BIC = round(res$BIC[2], 4),
    LogLik = round(res$logLik[2], 4),
    LRT_Stat = round(res$L.Ratio[2], 4),
    p_value = round(res$p-value[2], 4)
  )
}

anova_summary_manual_md11 <- bind_rows(
  get_baseline_metrics(md11_log, "Full Model"),
  get_anova_outputs(md11_log, md11_log_no_day, "Drop Day"),
  get_anova_outputs(md11_log, md11_log_no_session, "Drop Session"),
  get_anova_outputs(md11_log, md11_log_no_activity, "Drop Activity"),
  get_anova_outputs(md11_log, md11_log_no_sensor, "Drop Sensor"),
  get_anova_outputs(md11_log, md11_log_no_sex, "Drop Sex")
)

print(anova_summary_manual_md11)

```

##	Model	Df	AIC	BIC	LogLik	LRT_Stat	p_value
## 1	Full Model	14	1499.142	1581.431	-735.5711	NA	NA
## 2	Drop Day	13	1755.157	1831.568	-864.5784	258.0147	0.0000
## 3	Drop Session	11	1498.179	1562.835	-738.0895	5.0369	0.1691
## 4	Drop Activity	12	1505.920	1576.453	-740.9600	10.7779	0.0046
## 5	Drop Sensor	9	2438.115	2491.015	-1210.0575	948.9728	0.0000
## 6	Drop Sex	13	1505.917	1582.328	-739.9585	8.7749	0.0031

The above says to not include session, so we remove it! Then we compare the effects of removing the other variables to that model.

```

# print("No session and no day")
md11_log_no_session_no_day <- gls(
  log(HF_value) ~ (activity) + (HF_type) + (sex),
  data = df_clean,
  method = "ML"
)
# print(anova(md11_log_no_session, md11_log_no_session_no_day))

# print("No session and no activity")
md11_log_no_session_no_activity <- gls(
  log(HF_value) ~ (day) + (HF_type) + (sex),

```

```

data = df_clean,
method = "ML"
)
# print(anova mdl1_log_no_session, mdl1_log_no_session_no_activity))

# print("No session and no sensor")
mdl1_log_no_session_no_sensor <- gls(
  log(HF_value) ~ (day) + (activity) + (sex),
  data = df_clean,
  method = "ML"
)
# print(anova mdl1_log_no_session, mdl1_log_no_session_no_sensor))

# print("No session and no sex")
mdl1_log_no_session_no_sex <- gls(
  log(HF_value) ~ (day) + (activity) + (HF_type),
  data = df_clean,
  method = "ML"
)
# print(anova mdl1_log_no_session, mdl1_log_no_session_no_sex))

anova_summary_manual_mdl1_nosession <- bind_rows(
  get_baseline_metrics(mdl1_log_no_session, "Current Model (No Session)"),
  get_anova_outputs(mdl1_log_no_session, mdl1_log_no_session_no_day, "Drop Day"),
  get_anova_outputs(mdl1_log_no_session, mdl1_log_no_session_no_activity, "Drop Activity"),
  get_anova_outputs(mdl1_log_no_session, mdl1_log_no_session_no_sensor, "Drop Sensor"),
  get_anova_outputs(mdl1_log_no_session, mdl1_log_no_session_no_sex, "Drop Sex")
)

print(anova_summary_manual_mdl1_nosession)

```

##	Model	Df	AIC	BIC	LogLik	LRT_Stat	p_value
## 1	Current Model (No Session)	11	1498.179	1562.835	-738.0895	NA	NA
## 2	Drop Day	10	1754.702	1813.480	-867.3510	258.5231	0e+00
## 3	Drop Activity	9	1508.570	1561.470	-745.2851	14.3913	7e-04
## 4	Drop Sensor	6	2435.527	2470.793	-1211.7634	947.3477	0e+00
## 5	Drop Sex	10	1504.964	1563.742	-742.4821	8.7853	3e-03

0.4 Takeaway: Given the above, we propose a fixed effect model of `mdl1_log_no_session`: $\log(\text{HF_value}) \sim (\text{day}) + (\text{activity}) + (\text{HF_type}) + (\text{sex})$.

- Activity results in a drop in BIC, so we could consider removing it
- Sex barely improves BIC, so we could consider removing it
- For now, including - will re-examine at the end.

0.5 Section 5.1: Run model with interaction terms, v1

Now try the ten possible interaction terms.

```

#HF_type*day
#HF_type*activity
#HF_type*sex

```



```

#HF_type*session
#day*activity
#day*sex
#day*session
#activity*sex
#activity*session
#sex*session
mdl2_log_no_session_all_interactives <- gls(
  log(HF_value) ~ (day) + (activity) + (HF_type) + (sex) + (HF_type * day) + (HF_type * activity) +
  data = df_clean,
  method = "ML"
)
print(summary(mdl2_log_no_session_all_interactives))

## Generalized least squares fit by maximum likelihood
## Model: log(HF_value) ~ (day) + (activity) + (HF_type) + (sex) + (HF_type * day) + (HF_type * activity)
## Data: df_clean
## AIC BIC logLik
## 1110.039 1462.705 -495.0194
##
## Coefficients:
## Value Std.Error t-value p-value
## (Intercept) 4.457507 0.04218401 105.66816 0.0000
## day2 -0.108212 0.03894427 -2.77864 0.0055
## activitystanding -0.091938 0.03684288 -2.49542 0.0126
## activitywalking 0.016619 0.04702707 0.35340 0.7238
## HF_typeHF_2 -0.714008 0.05415672 -13.18411 0.0000
## HF_typeHF_3 -0.634006 0.05476609 -11.57661 0.0000
## HF_typeHF_5 -0.415053 0.05496646 -7.55103 0.0000
## HF_typeHF_7 -0.156907 0.05393795 -2.90903 0.0037
## HF_typeHF_8 -0.895256 0.05433936 -16.47528 0.0000
## sexM -0.016305 0.03908491 -0.41717 0.6766
## session2 -0.032092 0.05198144 -0.61738 0.5370
## session3 -0.023617 0.04090716 -0.57732 0.5638
## session4 -0.134540 0.04775384 -2.81737 0.0049
## day2:HF_typeHF_2 0.489460 0.04019118 12.17829 0.0000
## day2:HF_typeHF_3 0.491191 0.04045301 12.14227 0.0000
## day2:HF_typeHF_5 0.424638 0.04065273 10.44549 0.0000
## day2:HF_typeHF_7 0.339178 0.04004623 8.46965 0.0000
## day2:HF_typeHF_8 0.475592 0.04030759 11.79908 0.0000
## activitystanding:HF_typeHF_2 -0.094054 0.04545878 -2.06900 0.0386
## activitywalking:HF_typeHF_2 -0.376572 0.05756619 -6.54156 0.0000
## activitystanding:HF_typeHF_3 0.187253 0.04580438 4.08811 0.0000
## activitywalking:HF_typeHF_3 0.232791 0.05789764 4.02074 0.0001
## activitystanding:HF_typeHF_5 0.060510 0.04602755 1.31465 0.1887
## activitywalking:HF_typeHF_5 -0.037331 0.05821579 -0.64126 0.5214
## activitystanding:HF_typeHF_7 -0.045989 0.04528678 -1.01551 0.3100
## activitywalking:HF_typeHF_7 -0.119886 0.05728195 -2.09291 0.0365
## activitystanding:HF_typeHF_8 0.181536 0.04562325 3.97903 0.0001
## activitywalking:HF_typeHF_8 0.167564 0.05774291 2.90190 0.0037
## HF_typeHF_2:sexM -0.069921 0.03974431 -1.75927 0.0787
## HF_typeHF_3:sexM 0.080522 0.04000796 2.01265 0.0443
## HF_typeHF_5:sexM 0.019588 0.04020882 0.48715 0.6262
## HF_typeHF_7:sexM 0.117986 0.03958510 2.98056 0.0029

```

## HF_typeHF_8:sexM	0.078800	0.03985482	1.97718	0.0481		
## HF_typeHF_2:session2	0.128495	0.06496050	1.97804	0.0480		
## HF_typeHF_3:session2	0.049616	0.06540500	0.75860	0.4482		
## HF_typeHF_5:session2	0.090241	0.06586480	1.37010	0.1708		
## HF_typeHF_7:session2	0.068287	0.06484533	1.05307	0.2924		
## HF_typeHF_8:session2	0.095574	0.06498626	1.47068	0.1415		
## HF_typeHF_2:session3	0.103204	0.04995292	2.06603	0.0389		
## HF_typeHF_3:session3	0.045357	0.05022253	0.90312	0.3665		
## HF_typeHF_5:session3	0.065132	0.05050486	1.28963	0.1973		
## HF_typeHF_7:session3	0.024555	0.04969513	0.49411	0.6213		
## HF_typeHF_8:session3	0.130943	0.05018505	2.60921	0.0091		
## HF_typeHF_2:session4	0.286563	0.05843853	4.90366	0.0000		
## HF_typeHF_3:session4	0.188869	0.05893512	3.20469	0.0014		
## HF_typeHF_5:session4	0.204479	0.05908829	3.46057	0.0005		
## HF_typeHF_7:session4	0.093884	0.05821942	1.61260	0.1070		
## HF_typeHF_8:session4	0.235584	0.05845098	4.03045	0.0001		
## day2:activitystanding	-0.004552	0.02702696	-0.16842	0.8663		
## day2:activitywalking	-0.016324	0.03348745	-0.48747	0.6260		
## day2:sexM	0.020310	0.02326382	0.87302	0.3827		
## day2:session2	-0.079602	0.03951856	-2.01429	0.0441		
## day2:session3	-0.070403	0.02892327	-2.43412	0.0150		
## day2:session4	-0.115931	0.03388510	-3.42130	0.0006		
## activitystanding:sexM	0.015525	0.02631710	0.58994	0.5553		
## activitywalking:sexM	0.057352	0.03333222	1.72061	0.0854		
## sexM:session2	-0.004924	0.03754240	-0.13116	0.8957		
## sexM:session3	-0.020611	0.02890636	-0.71303	0.4759		
## sexM:session4	-0.021800	0.03387172	-0.64360	0.5199		
##						
## Correlation:						
##	(Intr)	day2	actvtys	actvtyw	HF_tHF_2	HF_tHF_3
## day2	-0.470					
## activitystanding	-0.461	0.167				
## activitywalking	-0.500	0.150	0.395			
## HF_typeHF_2	-0.648	0.206	0.281	0.299		
## HF_typeHF_3	-0.638	0.200	0.277	0.293	0.499	
## HF_typeHF_5	-0.635	0.196	0.276	0.292	0.497	0.492
## HF_typeHF_7	-0.650	0.209	0.282	0.300	0.507	0.501
## HF_typeHF_8	-0.645	0.205	0.279	0.296	0.503	0.497
## sexM	-0.446	0.075	0.120	0.133	0.182	0.176
## session2	-0.555	0.196	0.104	0.298	0.342	0.336
## session3	-0.476	0.121	-0.005	-0.006	0.279	0.274
## session4	-0.519	0.136	0.001	0.280	0.305	0.301
## day2:HF_typeHF_2	0.259	-0.522	-0.055	-0.021	-0.403	-0.200
## day2:HF_typeHF_3	0.257	-0.517	-0.055	-0.021	-0.201	-0.410
## day2:HF_typeHF_5	0.256	-0.516	-0.054	-0.021	-0.200	-0.198
## day2:HF_typeHF_7	0.259	-0.522	-0.056	-0.021	-0.203	-0.201
## day2:HF_typeHF_8	0.258	-0.520	-0.054	-0.020	-0.201	-0.199
## activitystanding:HF_typeHF_2	0.293	-0.046	-0.625	-0.247	-0.450	-0.225
## activitywalking:HF_typeHF_2	0.312	-0.016	-0.248	-0.617	-0.480	-0.240
## activitystanding:HF_typeHF_3	0.289	-0.044	-0.617	-0.244	-0.226	-0.449
## activitywalking:HF_typeHF_3	0.309	-0.014	-0.246	-0.609	-0.242	-0.478
## activitystanding:HF_typeHF_5	0.287	-0.041	-0.614	-0.242	-0.225	-0.222
## activitywalking:HF_typeHF_5	0.307	-0.011	-0.244	-0.605	-0.240	-0.238
## activitystanding:HF_typeHF_7	0.293	-0.048	-0.627	-0.247	-0.228	-0.226

## activitywalking:HF_typeHF_7	0.314	-0.019	-0.249	-0.620	-0.244	-0.242
## activitystanding:HF_typeHF_8	0.291	-0.045	-0.621	-0.245	-0.227	-0.224
## activitywalking:HF_typeHF_8	0.311	-0.016	-0.247	-0.613	-0.242	-0.240
## HF_typeHF_2:sexM	0.228	0.015	0.006	-0.001	-0.352	-0.176
## HF_typeHF_3:sexM	0.226	0.016	0.006	-0.001	-0.177	-0.359
## HF_typeHF_5:sexM	0.224	0.019	0.006	-0.001	-0.176	-0.174
## HF_typeHF_7:sexM	0.230	0.013	0.006	-0.002	-0.178	-0.176
## HF_typeHF_8:sexM	0.227	0.016	0.006	0.000	-0.177	-0.175
## HF_typeHF_2:session2	0.351	-0.055	-0.082	-0.188	-0.541	-0.271
## HF_typeHF_3:session2	0.347	-0.052	-0.081	-0.185	-0.272	-0.539
## HF_typeHF_5:session2	0.345	-0.049	-0.081	-0.184	-0.270	-0.267
## HF_typeHF_7:session2	0.351	-0.057	-0.081	-0.187	-0.274	-0.271
## HF_typeHF_8:session2	0.351	-0.055	-0.082	-0.187	-0.274	-0.270
## HF_typeHF_2:session3	0.294	0.007	0.003	0.003	-0.458	-0.227
## HF_typeHF_3:session3	0.291	0.010	0.003	0.003	-0.228	-0.455
## HF_typeHF_5:session3	0.289	0.012	0.003	0.003	-0.227	-0.224
## HF_typeHF_7:session3	0.296	0.005	0.003	0.003	-0.231	-0.228
## HF_typeHF_8:session3	0.293	0.008	0.002	0.001	-0.228	-0.226
## HF_typeHF_2:session4	0.320	0.005	0.000	-0.173	-0.493	-0.246
## HF_typeHF_3:session4	0.316	0.006	0.000	-0.170	-0.247	-0.492
## HF_typeHF_5:session4	0.314	0.010	0.000	-0.170	-0.246	-0.244
## HF_typeHF_7:session4	0.321	0.003	0.000	-0.174	-0.250	-0.247
## HF_typeHF_8:session4	0.319	0.005	0.000	-0.173	-0.249	-0.246
## day2:activitystanding	0.139	-0.347	-0.317	-0.127	-0.001	0.001
## day2:activitywalking	0.168	-0.383	-0.131	-0.352	0.000	0.002
## day2:sexM	0.145	-0.293	-0.029	-0.010	0.002	0.008
## day2:session2	0.153	-0.404	-0.001	-0.084	0.003	0.006
## day2:session3	0.165	-0.368	0.004	0.005	0.005	0.007
## day2:session4	0.181	-0.398	-0.001	-0.098	0.002	0.003
## activitystanding:sexM	0.163	-0.020	-0.348	-0.136	-0.002	0.001
## activitywalking:sexM	0.174	-0.002	-0.137	-0.343	-0.002	0.000
## sexM:session2	0.196	-0.025	-0.050	-0.106	-0.002	0.002
## sexM:session3	0.166	0.008	0.000	0.000	-0.002	0.001
## sexM:session4	0.179	0.008	0.000	-0.096	-0.003	-0.001
##						
	HF_tHF_5	HF_tHF_7	HF_tHF_8	sexM	sessn2	sessn3
## day2						
## activitystanding						
## activitywalking						
## HF_typeHF_2						
## HF_typeHF_3						
## HF_typeHF_5						
## HF_typeHF_7	0.499					
## HF_typeHF_8	0.496	0.505				
## sexM	0.179	0.179	0.180			
## session2	0.335	0.342	0.340	0.157		
## session3	0.273	0.280	0.280	0.128	0.389	
## session4	0.298	0.306	0.304	0.140	0.420	0.425
## day2:HF_typeHF_2	-0.199	-0.203	-0.201	0.014	-0.067	0.006
## day2:HF_typeHF_3	-0.198	-0.202	-0.200	0.016	-0.066	0.006
## day2:HF_typeHF_5	-0.410	-0.201	-0.199	0.014	-0.066	0.006
## day2:HF_typeHF_7	-0.200	-0.401	-0.202	0.017	-0.068	0.006
## day2:HF_typeHF_8	-0.198	-0.202	-0.404	0.015	-0.067	0.004
## activitystanding:HF_typeHF_2	-0.224	-0.228	-0.227	0.003	-0.083	0.003
## activitywalking:HF_typeHF_2	-0.240	-0.244	-0.242	-0.002	-0.192	0.003

```

## activitystanding:HF_typeHF_3 -0.222 -0.227 -0.225 0.005 -0.082 0.003
## activitywalking:HF_typeHF_3 -0.238 -0.243 -0.241 0.001 -0.189 0.003
## activitystanding:HF_typeHF_5 -0.450 -0.225 -0.224 0.003 -0.083 0.003
## activitywalking:HF_typeHF_5 -0.479 -0.241 -0.240 -0.002 -0.188 0.003
## activitystanding:HF_typeHF_7 -0.225 -0.449 -0.227 0.006 -0.082 0.003
## activitywalking:HF_typeHF_7 -0.241 -0.479 -0.244 0.001 -0.192 0.003
## activitystanding:HF_typeHF_8 -0.223 -0.227 -0.452 0.004 -0.083 0.002
## activitywalking:HF_typeHF_8 -0.239 -0.243 -0.483 0.000 -0.191 0.001
## HF_typeHF_2:sexM -0.175 -0.178 -0.177 -0.514 -0.004 -0.001
## HF_typeHF_3:sexM -0.174 -0.177 -0.176 -0.509 -0.004 -0.001
## HF_typeHF_5:sexM -0.350 -0.176 -0.175 -0.504 -0.004 -0.001
## HF_typeHF_7:sexM -0.176 -0.359 -0.178 -0.518 -0.005 -0.001
## HF_typeHF_8:sexM -0.175 -0.178 -0.355 -0.512 -0.004 -0.003
## HF_typeHF_2:session2 -0.270 -0.275 -0.273 -0.006 -0.634 -0.234
## HF_typeHF_3:session2 -0.268 -0.273 -0.271 -0.003 -0.626 -0.231
## HF_typeHF_5:session2 -0.541 -0.271 -0.269 -0.006 -0.622 -0.229
## HF_typeHF_7:session2 -0.270 -0.537 -0.273 -0.001 -0.633 -0.235
## HF_typeHF_8:session2 -0.270 -0.275 -0.541 -0.006 -0.634 -0.235
## HF_typeHF_2:session3 -0.226 -0.230 -0.229 -0.003 -0.240 -0.617
## HF_typeHF_3:session3 -0.225 -0.229 -0.227 -0.001 -0.237 -0.609
## HF_typeHF_5:session3 -0.456 -0.228 -0.226 -0.003 -0.236 -0.606
## HF_typeHF_7:session3 -0.227 -0.457 -0.230 -0.001 -0.241 -0.620
## HF_typeHF_8:session3 -0.225 -0.229 -0.448 -0.004 -0.240 -0.612
## HF_typeHF_2:session4 -0.245 -0.250 -0.248 -0.004 -0.259 -0.262
## HF_typeHF_3:session4 -0.243 -0.248 -0.246 -0.003 -0.255 -0.258
## HF_typeHF_5:session4 -0.491 -0.247 -0.246 -0.004 -0.254 -0.257
## HF_typeHF_7:session4 -0.246 -0.491 -0.249 -0.002 -0.260 -0.263
## HF_typeHF_8:session4 -0.245 -0.250 -0.493 -0.004 -0.259 -0.262
## day2:activitystanding 0.005 -0.004 0.000 0.005 -0.001 0.004
## day2:activitywalking 0.006 -0.003 0.002 0.007 -0.090 0.005
## day2:sexM 0.006 0.004 0.004 -0.331 -0.036 0.002
## day2:session2 0.010 0.000 0.003 0.009 -0.279 -0.126
## day2:session3 0.010 0.002 0.002 0.008 -0.136 -0.354
## day2:session4 0.008 -0.001 0.002 0.008 -0.146 -0.148
## activitystanding:sexM -0.002 0.002 0.001 -0.362 -0.050 0.000
## activitywalking:sexM -0.003 0.001 0.001 -0.386 -0.108 0.000
## sexM:session2 -0.002 0.002 -0.001 -0.433 -0.346 -0.132
## sexM:session3 -0.002 0.001 -0.006 -0.366 -0.135 -0.344
## sexM:session4 -0.003 0.000 -0.002 -0.396 -0.145 -0.146
## sessn4 d2:HF_HF_2 d2:HF_HF_3 d2:HF_HF_5 d2:HF_HF_7
## day2
## activitystanding
## activitywalking
## HF_typeHF_2
## HF_typeHF_3
## HF_typeHF_5
## HF_typeHF_7
## HF_typeHF_8
## sexM
## session2
## session3
## session4
## day2:HF_typeHF_2 0.003
## day2:HF_typeHF_3 0.002 0.501

```

## day2:HF_typeHF_5	0.003	0.499	0.496		
## day2:HF_typeHF_7	0.003	0.507	0.503	0.501	
## day2:HF_typeHF_8	0.004	0.503	0.500	0.498	0.505
## activitystanding:HF_typeHF_2	0.000	0.084	0.044	0.044	0.044
## activitywalking:HF_typeHF_2	-0.173	0.032	0.017	0.016	0.017
## activitystanding:HF_typeHF_3	0.000	0.044	0.085	0.043	0.044
## activitywalking:HF_typeHF_3	-0.171	0.017	0.033	0.016	0.017
## activitystanding:HF_typeHF_5	0.000	0.044	0.043	0.085	0.044
## activitywalking:HF_typeHF_5	-0.170	0.016	0.016	0.033	0.016
## activitystanding:HF_typeHF_7	0.000	0.044	0.044	0.044	0.081
## activitywalking:HF_typeHF_7	-0.174	0.017	0.017	0.017	0.031
## activitystanding:HF_typeHF_8	0.000	0.044	0.044	0.044	0.044
## activitywalking:HF_typeHF_8	-0.172	0.017	0.016	0.016	0.017
## HF_typeHF_2:sexM	-0.004	-0.024	-0.013	-0.013	-0.013
## HF_typeHF_3:sexM	-0.006	-0.013	-0.011	-0.013	-0.013
## HF_typeHF_5:sexM	-0.004	-0.013	-0.013	-0.025	-0.013
## HF_typeHF_7:sexM	-0.004	-0.013	-0.013	-0.013	-0.015
## HF_typeHF_8:sexM	-0.004	-0.013	-0.013	-0.013	-0.013
## HF_typeHF_2:session2	-0.254	0.111	0.055	0.054	0.055
## HF_typeHF_3:session2	-0.250	0.055	0.112	0.054	0.055
## HF_typeHF_5:session2	-0.248	0.054	0.054	0.112	0.054
## HF_typeHF_7:session2	-0.254	0.055	0.055	0.054	0.112
## HF_typeHF_8:session2	-0.253	0.055	0.055	0.054	0.055
## HF_typeHF_2:session3	-0.262	-0.005	-0.005	-0.005	-0.005
## HF_typeHF_3:session3	-0.259	-0.005	-0.005	-0.005	-0.005
## HF_typeHF_5:session3	-0.258	-0.005	-0.005	-0.005	-0.005
## HF_typeHF_7:session3	-0.264	-0.005	-0.005	-0.005	-0.005
## HF_typeHF_8:session3	-0.262	-0.005	-0.005	-0.005	-0.005
## HF_typeHF_2:session4	-0.615	-0.003	-0.003	-0.003	-0.003
## HF_typeHF_3:session4	-0.607	-0.003	0.001	-0.003	-0.003
## HF_typeHF_5:session4	-0.603	-0.003	-0.003	-0.003	-0.003
## HF_typeHF_7:session4	-0.617	-0.003	-0.003	-0.003	-0.001
## HF_typeHF_8:session4	-0.614	-0.003	-0.003	-0.003	-0.003
## day2:activitystanding	-0.001	0.001	0.000	0.000	0.001
## day2:activitywalking	-0.097	0.003	0.003	0.003	0.003
## day2:sexM	0.001	0.001	-0.004	0.002	-0.007
## day2:session2	-0.136	-0.002	-0.002	-0.003	0.001
## day2:session3	-0.148	-0.001	-0.001	-0.001	-0.001
## day2:session4	-0.354	0.002	0.004	0.002	0.003
## activitystanding:sexM	0.000	0.001	0.000	0.001	0.002
## activitywalking:sexM	-0.096	-0.002	-0.002	-0.001	-0.001
## sexM:session2	-0.142	-0.003	-0.004	-0.003	-0.005
## sexM:session3	-0.147	-0.001	-0.001	-0.001	-0.001
## sexM:session4	-0.345	-0.002	0.000	-0.002	-0.003
##					
##	d2:HF_HF_8	actvtys:HF_HF_2	actvtyw:HF_HF_2		
## day2					
## activitystanding					
## activitywalking					
## HF_typeHF_2					
## HF_typeHF_3					
## HF_typeHF_5					
## HF_typeHF_7					
## HF_typeHF_8					
## sexM					

```

## session2
## session3
## session4
## day2:HF_typeHF_2
## day2:HF_typeHF_3
## day2:HF_typeHF_5
## day2:HF_typeHF_7
## day2:HF_typeHF_8
## activitystanding:HF_typeHF_2 0.044
## activitywalking:HF_typeHF_2 0.017 0.396
## activitystanding:HF_typeHF_3 0.044 0.502 0.199
## activitywalking:HF_typeHF_3 0.016 0.200 0.500
## activitystanding:HF_typeHF_5 0.044 0.500 0.198
## activitywalking:HF_typeHF_5 0.016 0.198 0.498
## activitystanding:HF_typeHF_7 0.044 0.508 0.202
## activitywalking:HF_typeHF_7 0.017 0.202 0.506
## activitystanding:HF_typeHF_8 0.084 0.504 0.200
## activitywalking:HF_typeHF_8 0.035 0.200 0.502
## HF_typeHF_2:sexM -0.013 -0.005 0.001
## HF_typeHF_3:sexM -0.013 -0.005 0.001
## HF_typeHF_5:sexM -0.013 -0.005 0.001
## HF_typeHF_7:sexM -0.013 -0.005 0.001
## HF_typeHF_8:sexM -0.023 -0.005 0.001
## HF_typeHF_2:session2 0.055 0.129 0.302
## HF_typeHF_3:session2 0.054 0.065 0.152
## HF_typeHF_5:session2 0.054 0.064 0.151
## HF_typeHF_7:session2 0.055 0.065 0.153
## HF_typeHF_8:session2 0.112 0.065 0.153
## HF_typeHF_2:session3 -0.005 -0.002 -0.002
## HF_typeHF_3:session3 -0.005 -0.002 -0.002
## HF_typeHF_5:session3 -0.005 -0.002 -0.002
## HF_typeHF_7:session3 -0.005 -0.002 -0.002
## HF_typeHF_8:session3 -0.009 -0.002 -0.002
## HF_typeHF_2:session4 -0.003 0.000 0.279
## HF_typeHF_3:session4 -0.003 0.000 0.140
## HF_typeHF_5:session4 -0.003 0.000 0.140
## HF_typeHF_7:session4 -0.003 0.000 0.142
## HF_typeHF_8:session4 -0.002 0.000 0.141
## day2:activitystanding 0.000 0.000 0.002
## day2:activitywalking 0.000 0.002 0.001
## day2:sexM -0.001 0.001 -0.001
## day2:session2 -0.003 0.001 0.000
## day2:session3 0.003 -0.003 -0.003
## day2:session4 0.002 0.001 -0.001
## activitystanding:sexM -0.001 0.003 0.001
## activitywalking:sexM -0.003 0.000 0.001
## sexM:session2 -0.003 0.004 0.003
## sexM:session3 0.000 0.001 0.001
## sexM:session4 -0.002 0.000 0.001
## actvtys:HF_HF_3 actvtyw:HF_HF_3 actvtys:HF_HF_5
## day2
## activitystanding
## activitywalking
## HF_typeHF_2

```

```

## HF_typeHF_3
## HF_typeHF_5
## HF_typeHF_7
## HF_typeHF_8
## sexM
## session2
## session3
## session4
## day2:HF_typeHF_2
## day2:HF_typeHF_3
## day2:HF_typeHF_5
## day2:HF_typeHF_7
## day2:HF_typeHF_8
## activitystanding:HF_typeHF_2
## activitywalking:HF_typeHF_2
## activitystanding:HF_typeHF_3
## activitywalking:HF_typeHF_3    0.397
## activitystanding:HF_typeHF_5    0.496          0.197
## activitywalking:HF_typeHF_5    0.197          0.495          0.396
## activitystanding:HF_typeHF_7    0.504          0.200          0.501
## activitywalking:HF_typeHF_7    0.200          0.503          0.199
## activitystanding:HF_typeHF_8    0.500          0.199          0.498
## activitywalking:HF_typeHF_8    0.199          0.499          0.198
## HF_typeHF_2:sexM          -0.005          0.001          -0.005
## HF_typeHF_3:sexM          -0.005          0.001          -0.005
## HF_typeHF_5:sexM          -0.005          0.001          -0.005
## HF_typeHF_7:sexM          -0.005          0.001          -0.005
## HF_typeHF_8:sexM          -0.005          0.001          -0.005
## HF_typeHF_2:session2          0.065          0.152          0.065
## HF_typeHF_3:session2          0.131          0.302          0.064
## HF_typeHF_5:session2          0.064          0.150          0.132
## HF_typeHF_7:session2          0.065          0.152          0.065
## HF_typeHF_8:session2          0.065          0.152          0.065
## HF_typeHF_2:session3          -0.002          -0.002          -0.002
## HF_typeHF_3:session3          -0.002          -0.002          -0.002
## HF_typeHF_5:session3          -0.002          -0.002          -0.002
## HF_typeHF_7:session3          -0.002          -0.002          -0.002
## HF_typeHF_8:session3          -0.002          -0.002          -0.002
## HF_typeHF_2:session4          0.000          0.140          0.000
## HF_typeHF_3:session4          0.000          0.278          0.000
## HF_typeHF_5:session4          0.000          0.139          0.000
## HF_typeHF_7:session4          0.000          0.141          0.000
## HF_typeHF_8:session4          0.000          0.140          0.000
## day2:activitystanding          -0.004          0.000          -0.013
## day2:activitywalking          0.001          -0.005          -0.003
## day2:sexM          0.000          -0.002          0.001
## day2:session2          0.001          -0.002          0.000
## day2:session3          -0.003          -0.003          -0.003
## day2:session4          0.001          -0.003          0.001
## activitystanding:sexM          -0.002          -0.002          0.004
## activitywalking:sexM          -0.002          -0.005          0.001
## sexM:session2          0.002          0.001          0.004
## sexM:session3          0.001          0.001          0.001
## sexM:session4          0.000          0.000          0.000

```

```

##                                activtyw:HF_HF_5 activtys:HF_HF_7 activtyw:HF_HF_7
## day2
## activitystanding
## activitywalking
## HF_typeHF_2
## HF_typeHF_3
## HF_typeHF_5
## HF_typeHF_7
## HF_typeHF_8
## sexM
## session2
## session3
## session4
## day2:HF_typeHF_2
## day2:HF_typeHF_3
## day2:HF_typeHF_5
## day2:HF_typeHF_7
## day2:HF_typeHF_8
## activitystanding:HF_typeHF_2
## activitywalking:HF_typeHF_2
## activitystanding:HF_typeHF_3
## activitywalking:HF_typeHF_3
## activitystanding:HF_typeHF_5
## activitywalking:HF_typeHF_5
## activitystanding:HF_typeHF_7 0.199
## activitywalking:HF_typeHF_7 0.500          0.396
## activitystanding:HF_typeHF_8 0.198          0.506          0.201
## activitywalking:HF_typeHF_8 0.496          0.201          0.504
## HF_typeHF_2:sexM          0.001          -0.005          0.001
## HF_typeHF_3:sexM          0.001          -0.005          0.001
## HF_typeHF_5:sexM          0.001          -0.005          0.001
## HF_typeHF_7:sexM          0.001          -0.005          0.001
## HF_typeHF_8:sexM          0.001          -0.005          0.001
## HF_typeHF_2:session2      0.151          0.066          0.154
## HF_typeHF_3:session2      0.150          0.065          0.153
## HF_typeHF_5:session2      0.303          0.065          0.151
## HF_typeHF_7:session2      0.151          0.128          0.301
## HF_typeHF_8:session2      0.151          0.066          0.153
## HF_typeHF_2:session3     -0.002          -0.002          -0.002
## HF_typeHF_3:session3     -0.002          -0.002          -0.002
## HF_typeHF_5:session3     -0.002          -0.002          -0.002
## HF_typeHF_7:session3     -0.002          -0.002          -0.002
## HF_typeHF_8:session3     -0.002          -0.002          -0.002
## HF_typeHF_2:session4      0.140          0.000          0.142
## HF_typeHF_3:session4      0.138          0.000          0.141
## HF_typeHF_5:session4      0.279          0.000          0.140
## HF_typeHF_7:session4      0.140          -0.002          0.278
## HF_typeHF_8:session4      0.140          0.000          0.142
## day2:activitystanding     -0.003          0.006          0.004
## day2:activitywalking     -0.012          0.005          0.007
## day2:sexM                 -0.001          0.002          -0.001
## day2:session2             -0.004          0.000          0.001
## day2:session3             -0.003          -0.003          -0.003
## day2:session4             -0.005          0.000          0.000

```


## activitystanding:sexM	0.001	-0.006	-0.003
## activitywalking:sexM	0.002	-0.003	-0.006
## sexM:session2	0.003	0.001	0.000
## sexM:session3	0.001	0.001	0.001
## sexM:session4	0.002	0.001	0.000
##	actvtys:HF_HF_8	actvtyw:HF_HF_8	HF_HF_2:M
## day2			
## activitystanding			
## activitywalking			
## HF_typeHF_2			
## HF_typeHF_3			
## HF_typeHF_5			
## HF_typeHF_7			
## HF_typeHF_8			
## sexM			
## session2			
## session3			
## session4			
## day2:HF_typeHF_2			
## day2:HF_typeHF_3			
## day2:HF_typeHF_5			
## day2:HF_typeHF_7			
## day2:HF_typeHF_8			
## activitystanding:HF_typeHF_2			
## activitywalking:HF_typeHF_2			
## activitystanding:HF_typeHF_3			
## activitywalking:HF_typeHF_3			
## activitystanding:HF_typeHF_5			
## activitywalking:HF_typeHF_5			
## activitystanding:HF_typeHF_7			
## activitywalking:HF_typeHF_7			
## activitystanding:HF_typeHF_8			
## activitywalking:HF_typeHF_8	0.397		
## HF_typeHF_2:sexM	-0.005	0.001	
## HF_typeHF_3:sexM	-0.005	0.001	0.501
## HF_typeHF_5:sexM	-0.005	0.001	0.499
## HF_typeHF_7:sexM	-0.005	0.001	0.507
## HF_typeHF_8:sexM	-0.001	0.005	0.503
## HF_typeHF_2:session2	0.065	0.152	0.005
## HF_typeHF_3:session2	0.065	0.151	0.005
## HF_typeHF_5:session2	0.064	0.150	0.005
## HF_typeHF_7:session2	0.065	0.153	0.005
## HF_typeHF_8:session2	0.130	0.303	0.005
## HF_typeHF_2:session3	-0.002	-0.002	0.002
## HF_typeHF_3:session3	-0.002	-0.002	0.002
## HF_typeHF_5:session3	-0.002	-0.002	0.002
## HF_typeHF_7:session3	-0.002	-0.002	0.002
## HF_typeHF_8:session3	-0.004	-0.009	0.002
## HF_typeHF_2:session4	0.000	0.141	0.003
## HF_typeHF_3:session4	0.000	0.140	0.003
## HF_typeHF_5:session4	0.000	0.139	0.003
## HF_typeHF_7:session4	0.000	0.141	0.003
## HF_typeHF_8:session4	0.001	0.280	0.003
## day2:activitystanding	-0.002	0.002	-0.001

## day2:activitywalking	0.002	0.000	-0.002		
## day2:sexM	-0.001	-0.003	-0.001		
## day2:session2	0.002	0.000	-0.003		
## day2:session3	-0.004	-0.001	-0.001		
## day2:session4	0.001	-0.001	-0.002		
## activitystanding:sexM	0.001	-0.002	0.001		
## activitywalking:sexM	-0.002	-0.002	0.003		
## sexM:session2	0.003	0.002	-0.002		
## sexM:session3	0.004	0.004	-0.001		
## sexM:session4	-0.001	0.001	0.002		
##	HF_HF_3:M	HF_HF_5:M	HF_HF_7:M	HF_HF_8:M	HF_HF_2:2
## day2					
## activitystanding					
## activitywalking					
## HF_typeHF_2					
## HF_typeHF_3					
## HF_typeHF_5					
## HF_typeHF_7					
## HF_typeHF_8					
## sexM					
## session2					
## session3					
## session4					
## day2:HF_typeHF_2					
## day2:HF_typeHF_3					
## day2:HF_typeHF_5					
## day2:HF_typeHF_7					
## day2:HF_typeHF_8					
## activitystanding:HF_typeHF_2					
## activitywalking:HF_typeHF_2					
## activitystanding:HF_typeHF_3					
## activitywalking:HF_typeHF_3					
## activitystanding:HF_typeHF_5					
## activitywalking:HF_typeHF_5					
## activitystanding:HF_typeHF_7					
## activitywalking:HF_typeHF_7					
## activitystanding:HF_typeHF_8					
## activitywalking:HF_typeHF_8					
## HF_typeHF_2:sexM					
## HF_typeHF_3:sexM					
## HF_typeHF_5:sexM	0.496				
## HF_typeHF_7:sexM	0.503	0.501			
## HF_typeHF_8:sexM	0.500	0.497	0.505		
## HF_typeHF_2:session2	0.005	0.005	0.005	0.005	
## HF_typeHF_3:session2	0.005	0.005	0.005	0.005	0.505
## HF_typeHF_5:session2	0.005	0.005	0.005	0.005	0.501
## HF_typeHF_7:session2	0.005	0.005	0.000	0.005	0.509
## HF_typeHF_8:session2	0.005	0.005	0.005	0.006	0.508
## HF_typeHF_2:session3	0.002	0.002	0.002	0.002	0.383
## HF_typeHF_3:session3	0.002	0.002	0.002	0.002	0.191
## HF_typeHF_5:session3	0.002	0.002	0.002	0.002	0.190
## HF_typeHF_7:session3	0.002	0.002	0.002	0.002	0.193
## HF_typeHF_8:session3	0.002	0.002	0.002	-0.003	0.191
## HF_typeHF_2:session4	0.003	0.003	0.003	0.003	0.411

## HF_typeHF_3:session4	0.007	0.003	0.003	0.003	0.206
## HF_typeHF_5:session4	0.003	0.003	0.003	0.003	0.205
## HF_typeHF_7:session4	0.003	0.003	0.002	0.003	0.208
## HF_typeHF_8:session4	0.003	0.003	0.003	0.004	0.207
## day2:activitystanding	-0.001	-0.001	0.001	-0.001	0.001
## day2:activitywalking	-0.002	-0.002	-0.001	-0.003	0.000
## day2:sexM	-0.005	-0.013	0.007	-0.003	-0.003
## day2:session2	-0.003	-0.003	-0.004	-0.003	-0.002
## day2:session3	-0.001	-0.001	-0.001	0.000	-0.003
## day2:session4	0.000	-0.002	-0.003	-0.002	-0.002
## activitystanding:sexM	0.001	0.001	0.001	0.000	0.004
## activitywalking:sexM	0.003	0.003	0.003	0.000	0.003
## sexM:session2	-0.002	-0.002	0.001	-0.003	-0.001
## sexM:session3	-0.001	-0.001	-0.001	0.003	0.002
## sexM:session4	0.004	0.002	0.003	0.001	0.002
##	HF_HF_3:2	HF_HF_5:2	HF_HF_7:2	HF_HF_8:2	HF_HF_2:3
## day2					
## activitystanding					
## activitywalking					
## HF_typeHF_2					
## HF_typeHF_3					
## HF_typeHF_5					
## HF_typeHF_7					
## HF_typeHF_8					
## sexM					
## session2					
## session3					
## session4					
## day2:HF_typeHF_2					
## day2:HF_typeHF_3					
## day2:HF_typeHF_5					
## day2:HF_typeHF_7					
## day2:HF_typeHF_8					
## activitystanding:HF_typeHF_2					
## activitywalking:HF_typeHF_2					
## activitystanding:HF_typeHF_3					
## activitywalking:HF_typeHF_3					
## activitystanding:HF_typeHF_5					
## activitywalking:HF_typeHF_5					
## activitystanding:HF_typeHF_7					
## activitywalking:HF_typeHF_7					
## activitystanding:HF_typeHF_8					
## activitywalking:HF_typeHF_8					
## HF_typeHF_2:sexM					
## HF_typeHF_3:sexM					
## HF_typeHF_5:sexM					
## HF_typeHF_7:sexM					
## HF_typeHF_8:sexM					
## HF_typeHF_2:session2					
## HF_typeHF_3:session2					
## HF_typeHF_5:session2	0.498				
## HF_typeHF_7:session2	0.506	0.502			
## HF_typeHF_8:session2	0.505	0.501	0.509		
## HF_typeHF_2:session3	0.191	0.190	0.193	0.192	

## HF_typeHF_3:session3	0.382	0.189	0.192	0.191	0.502
## HF_typeHF_5:session3	0.189	0.382	0.190	0.190	0.499
## HF_typeHF_7:session3	0.192	0.191	0.381	0.193	0.507
## HF_typeHF_8:session3	0.190	0.189	0.192	0.378	0.502
## HF_typeHF_2:session4	0.206	0.204	0.208	0.207	0.426
## HF_typeHF_3:session4	0.409	0.203	0.206	0.205	0.213
## HF_typeHF_5:session4	0.204	0.410	0.205	0.205	0.212
## HF_typeHF_7:session4	0.207	0.205	0.409	0.208	0.215
## HF_typeHF_8:session4	0.206	0.204	0.208	0.411	0.214
## day2:activitystanding	0.000	-0.001	0.000	0.000	-0.003
## day2:activitywalking	-0.002	-0.004	0.001	-0.001	-0.003
## day2:sexM	-0.004	-0.003	-0.005	-0.004	-0.001
## day2:session2	-0.007	-0.014	0.006	-0.002	-0.002
## day2:session3	-0.005	-0.008	0.000	-0.002	-0.001
## day2:session4	-0.004	-0.007	0.001	-0.002	-0.002
## activitystanding:sexM	0.002	0.004	0.001	0.003	0.001
## activitywalking:sexM	0.001	0.003	0.000	0.002	0.001
## sexM:session2	-0.008	0.000	-0.014	-0.001	0.002
## sexM:session3	0.000	0.002	-0.001	0.003	0.003
## sexM:session4	0.000	0.002	-0.001	0.002	0.002
##	HF_HF_3:3	HF_HF_5:3	HF_HF_7:3	HF_HF_8:3	HF_HF_2:4
## day2					
## activitystanding					
## activitywalking					
## HF_typeHF_2					
## HF_typeHF_3					
## HF_typeHF_5					
## HF_typeHF_7					
## HF_typeHF_8					
## sexM					
## session2					
## session3					
## session4					
## day2:HF_typeHF_2					
## day2:HF_typeHF_3					
## day2:HF_typeHF_5					
## day2:HF_typeHF_7					
## day2:HF_typeHF_8					
## activitystanding:HF_typeHF_2					
## activitywalking:HF_typeHF_2					
## activitystanding:HF_typeHF_3					
## activitywalking:HF_typeHF_3					
## activitystanding:HF_typeHF_5					
## activitywalking:HF_typeHF_5					
## activitystanding:HF_typeHF_7					
## activitywalking:HF_typeHF_7					
## activitystanding:HF_typeHF_8					
## activitywalking:HF_typeHF_8					
## HF_typeHF_2:sexM					
## HF_typeHF_3:sexM					
## HF_typeHF_5:sexM					
## HF_typeHF_7:sexM					
## HF_typeHF_8:sexM					
## HF_typeHF_2:session2					

```

## HF_typeHF_3:session2
## HF_typeHF_5:session2
## HF_typeHF_7:session2
## HF_typeHF_8:session2
## HF_typeHF_2:session3
## HF_typeHF_3:session3
## HF_typeHF_5:session3      0.496
## HF_typeHF_7:session3      0.504      0.501
## HF_typeHF_8:session3      0.500      0.497      0.505
## HF_typeHF_2:session4      0.213      0.212      0.216      0.213
## HF_typeHF_3:session4      0.425      0.210      0.214      0.212      0.498
## HF_typeHF_5:session4      0.211      0.426      0.213      0.211      0.497
## HF_typeHF_7:session4      0.214      0.213      0.426      0.214      0.504
## HF_typeHF_8:session4      0.213      0.212      0.216      0.422      0.502
## day2:activitystanding     -0.003     -0.003     -0.003     -0.004      0.001
## day2:activitywalking     -0.003     -0.003     -0.003     -0.002     -0.001
## day2:sexM                 -0.001     -0.001     -0.001      0.000     -0.002
## day2:session2             -0.005     -0.007      0.000     -0.002     -0.002
## day2:session3             -0.007     -0.014      0.005     -0.004     -0.002
## day2:session4             -0.005     -0.008      0.000     -0.002     -0.002
## activitystanding:sexM      0.001      0.001      0.001      0.004      0.000
## activitywalking:sexM      0.001      0.001      0.001      0.004      0.001
## sexM:session2             0.000      0.002     -0.001      0.003      0.002
## sexM:session3            -0.003      0.003     -0.003      0.000      0.002
## sexM:session4            -0.001      0.002     -0.001      0.003      0.002
##                          HF_HF_3:4 HF_HF_5:4 HF_HF_7:4 HF_HF_8:4 dy2:ctvtys
## day2
## activitystanding
## activitywalking
## HF_typeHF_2
## HF_typeHF_3
## HF_typeHF_5
## HF_typeHF_7
## HF_typeHF_8
## sexM
## session2
## session3
## session4
## day2:HF_typeHF_2
## day2:HF_typeHF_3
## day2:HF_typeHF_5
## day2:HF_typeHF_7
## day2:HF_typeHF_8
## activitystanding:HF_typeHF_2
## activitywalking:HF_typeHF_2
## activitystanding:HF_typeHF_3
## activitywalking:HF_typeHF_3
## activitystanding:HF_typeHF_5
## activitywalking:HF_typeHF_5
## activitystanding:HF_typeHF_7
## activitywalking:HF_typeHF_7
## activitystanding:HF_typeHF_8
## activitywalking:HF_typeHF_8
## HF_typeHF_2:sexM

```

```

## HF_typeHF_3:sexM
## HF_typeHF_5:sexM
## HF_typeHF_7:sexM
## HF_typeHF_8:sexM
## HF_typeHF_2:session2
## HF_typeHF_3:session2
## HF_typeHF_5:session2
## HF_typeHF_7:session2
## HF_typeHF_8:session2
## HF_typeHF_2:session3
## HF_typeHF_3:session3
## HF_typeHF_5:session3
## HF_typeHF_7:session3
## HF_typeHF_8:session3
## HF_typeHF_2:session4
## HF_typeHF_3:session4
## HF_typeHF_5:session4      0.493
## HF_typeHF_7:session4      0.500      0.498
## HF_typeHF_8:session4      0.498      0.497      0.504
## day2:activitystanding      0.001      0.001      -0.001      0.001
## day2:activitywalking      -0.003      -0.005      0.000      -0.001      0.404
## day2:sexM                  0.000      -0.002      -0.003      -0.002      -0.001
## day2:session2              -0.004      -0.007      0.000      -0.002      0.191
## day2:session3              -0.005      -0.008      0.000      -0.002      -0.001
## day2:session4              -0.005      -0.015      0.006      -0.002      -0.001
## activitystanding:sexM      0.000      0.000      0.001      -0.001      -0.014
## activitywalking:sexM      0.000      0.002      0.000      0.001      -0.005
## sexM:session2              0.000      0.002      -0.001      0.002      0.001
## sexM:session3              -0.001      0.002      -0.001      0.003      -0.001
## sexM:session4              0.000      0.003      -0.006      0.002      0.001
##                             dy2:ctvtw dy2:sM dy2:s2 dy2:s3 dy2:s4 actvtys:M
## day2
## activitystanding
## activitywalking
## HF_typeHF_2
## HF_typeHF_3
## HF_typeHF_5
## HF_typeHF_7
## HF_typeHF_8
## sexM
## session2
## session3
## session4
## day2:HF_typeHF_2
## day2:HF_typeHF_3
## day2:HF_typeHF_5
## day2:HF_typeHF_7
## day2:HF_typeHF_8
## activitystanding:HF_typeHF_2
## activitywalking:HF_typeHF_2
## activitystanding:HF_typeHF_3
## activitywalking:HF_typeHF_3
## activitystanding:HF_typeHF_5
## activitywalking:HF_typeHF_5

```

```

## activitystanding:HF_typeHF_7
## activitywalking:HF_typeHF_7
## activitystanding:HF_typeHF_8
## activitywalking:HF_typeHF_8
## HF_typeHF_2:sexM
## HF_typeHF_3:sexM
## HF_typeHF_5:sexM
## HF_typeHF_7:sexM
## HF_typeHF_8:sexM
## HF_typeHF_2:session2
## HF_typeHF_3:session2
## HF_typeHF_5:session2
## HF_typeHF_7:session2
## HF_typeHF_8:session2
## HF_typeHF_2:session3
## HF_typeHF_3:session3
## HF_typeHF_5:session3
## HF_typeHF_7:session3
## HF_typeHF_8:session3
## HF_typeHF_2:session4
## HF_typeHF_3:session4
## HF_typeHF_5:session4
## HF_typeHF_7:session4
## HF_typeHF_8:session4
## day2:activitystanding
## day2:activitywalking
## day2:sexM                0.002
## day2:session2            0.314    0.000
## day2:session3           -0.003    -0.001    0.364
## day2:session4            0.276    0.002    0.389    0.424
## activitystanding:sexM    -0.005    0.081    0.000 -0.001    0.001
## activitywalking:sexM    -0.015    0.033 -0.004 -0.001 -0.004    0.396
## sexM:session2           -0.004    0.114 -0.014 -0.007 -0.007    0.131
## sexM:session3           -0.001    -0.003 -0.006 -0.016 -0.008 -0.001
## sexM:session4           -0.004    0.001 -0.007 -0.008 -0.015    0.000
##                          actvtyw:M sxM:s2 sxM:s3
## day2
## activitystanding
## activitywalking
## HF_typeHF_2
## HF_typeHF_3
## HF_typeHF_5
## HF_typeHF_7
## HF_typeHF_8
## sexM
## session2
## session3
## session4
## day2:HF_typeHF_2
## day2:HF_typeHF_3
## day2:HF_typeHF_5
## day2:HF_typeHF_7
## day2:HF_typeHF_8
## activitystanding:HF_typeHF_2

```

```

## activitywalking:HF_typeHF_2
## activitystanding:HF_typeHF_3
## activitywalking:HF_typeHF_3
## activitystanding:HF_typeHF_5
## activitywalking:HF_typeHF_5
## activitystanding:HF_typeHF_7
## activitywalking:HF_typeHF_7
## activitystanding:HF_typeHF_8
## activitywalking:HF_typeHF_8
## HF_typeHF_2:sexM
## HF_typeHF_3:sexM
## HF_typeHF_5:sexM
## HF_typeHF_7:sexM
## HF_typeHF_8:sexM
## HF_typeHF_2:session2
## HF_typeHF_3:session2
## HF_typeHF_5:session2
## HF_typeHF_7:session2
## HF_typeHF_8:session2
## HF_typeHF_2:session3
## HF_typeHF_3:session3
## HF_typeHF_5:session3
## HF_typeHF_7:session3
## HF_typeHF_8:session3
## HF_typeHF_2:session4
## HF_typeHF_3:session4
## HF_typeHF_5:session4
## HF_typeHF_7:session4
## HF_typeHF_8:session4
## day2:activitystanding
## day2:activitywalking
## day2:sexM
## day2:session2
## day2:session3
## day2:session4
## activitystanding:sexM
## activitywalking:sexM
## sexM:session2          0.302
## sexM:session3        -0.003    0.382
## sexM:session4          0.277    0.410    0.424
##
## Standardized residuals:
##      Min      Q1      Med      Q3      Max
## -6.6879660 -0.4692056  0.1175483  0.6343131  3.1302677
##
## Residual standard error: 0.291916
## Degrees of freedom: 2638 total; 2579 residual

```

#now: try running just the model with fixed effects and this interactive effect
#same issue: gls fit singular, rank 18...

```

# mdl2_log_no_session_interact_activity_session <- gls(
#   log(HF_value) ~ (day) + (activity) + (HF_type) + (sex) + (activity * session), # - df issue (too
#   data = df_clean,
#   method = "ML"

```



```
# )
```

We find that : - p-values are too high for all coefficients (over 0.05) for the following; note that 0.05 is not a strict cutoff, but a good starting point to determine which interaction effects to examine: *#sexsession*
#daysex *#day*activity*

*#not enough degrees freedom: #activity*session*

0.6 Section 5.2: Run model with interaction terms

Now, try adding in each interactive term one by one (of the ones not automatically excluded). Test with an ANOVA.

```
#sex*session
mdl2_log_no_session_interact_sex_session<- gls(
  log(HF_value) ~ (day) + (activity) + (HF_type) + (sex) + (sex * session) ,
  data = df_clean,
  method = "ML"
)

#day*sex
dl2_log_no_session_interact_sex_day<- gls(
  log(HF_value) ~ (day) + (activity) + (HF_type) + (sex) + (sex * day) ,
  data = df_clean,
  method = "ML"
)

#day*activity
dl2_log_no_session_interact_day_activity<- gls(
  log(HF_value) ~ (day) + (activity) + (HF_type) + (sex) + (activity * day) ,
  data = df_clean,
  method = "ML"
)

#HF_type*day
mdl2_log_no_session_interact_sensor_day <- gls(
  log(HF_value) ~ (day) + (activity) + (HF_type) + (sex) + (HF_type * day) ,
  data = df_clean,
  method = "ML"
)
# print(anova(mdl1_log_no_session, mdl2_log_no_session_interact_sensor_day))

#HF_type*activity
mdl2_log_no_session_interact_sensor_activity <- gls(
  log(HF_value) ~ (day) + (activity) + (HF_type) + (sex) + (HF_type * activity) ,
  data = df_clean,
  method = "ML"
)
# print(anova(mdl1_log_no_session, mdl2_log_no_session_interact_sensor_activity))

#HF_type*sex
mdl2_log_no_session_interact_sensor_sex <- gls(
  log(HF_value) ~ (day) + (activity) + (HF_type) + (sex) + (HF_type * sex) ,
  data = df_clean,
```

```

    method = "ML"
  )
  # print(anova mdl1_log_no_session, mdl2_log_no_session_interact_sensor_sex))

#HF_type*session
mdl2_log_no_session_interact_sensor_session <- gls(
  log(HF_value) ~ (day) + (activity) + (HF_type) + (sex) + (HF_type * session) ,
  data = df_clean,
  method = "ML"
)
# print(anova mdl1_log_no_session, mdl2_log_no_session_interact_sensor_session))

#activity*sex
mdl2_log_no_session_interact_activity_sex <- gls(
  log(HF_value) ~ (day) + (activity) + (HF_type) + (sex) + (activity * sex) ,
  data = df_clean,
  method = "ML"
)
# print(anova mdl1_log_no_session, mdl2_log_no_session_interact_activity_sex))

#sex*session
mdl2_log_no_session_interact_session_sex <- gls(
  log(HF_value) ~ (day) + (activity) + (HF_type) + (sex) + (session * sex) ,
  data = df_clean,
  method = "ML"
)
# print(anova mdl1_log_no_session, mdl2_log_no_session_interact_session_sex))

#day*session
mdl2_log_no_session_interact_session_day <- gls(
  log(HF_value) ~ (day) + (activity) + (HF_type) + (sex) + (session * day) ,
  data = df_clean,
  method = "ML"
)

anova_summary_manual_mdl1_nosession_interact1 <- bind_rows(
  get_baseline_metrics(mdl1_log_no_session, "Current Model (No Session)"),
  get_anova_outputs(mdl1_log_no_session, mdl2_log_no_session_interact_sensor_day, "Interact Sensor & Day"),
  get_anova_outputs(mdl1_log_no_session, mdl2_log_no_session_interact_sensor_activity, "Interact Sensor & Activity"),
  get_anova_outputs(mdl1_log_no_session, mdl2_log_no_session_interact_sensor_sex, "Interact Sensor & Sex"),
  get_anova_outputs(mdl1_log_no_session, mdl2_log_no_session_interact_sensor_session, "Interact Sensor & Session"),
  get_anova_outputs(mdl1_log_no_session, mdl2_log_no_session_interact_day_activity, "Interact Activity & Day"),
  get_anova_outputs(mdl1_log_no_session, mdl2_log_no_session_interact_activity_sex, "Interact Activity & Sex"),
  get_anova_outputs(mdl1_log_no_session, mdl2_log_no_session_interact_sex_session, "Interact Session & Sex"),
  get_anova_outputs(mdl1_log_no_session, mdl2_log_no_session_interact_sex_day, "Interact Day & Sex"),
  get_anova_outputs(mdl1_log_no_session, mdl2_log_no_session_interact_session_day, "Interact Day & Session")
)

#day*session
#activity*session

```

```
print(anova_summary_manual_md11_nosession_interact1)
```

##		Model	Df	AIC	BIC	LogLik	LRT_Stat	p_value
## 1	Current Model (No Session)	11	1498.179	1562.835	-738.0895	NA	NA	
## 2	Interact Sensor & Day	16	1311.286	1405.331	-639.6431	196.8927	0.0000	
## 3	Interact Sensor & Activity	21	1339.514	1462.947	-648.7569	178.6653	0.0000	
## 4	Interact Sensor & Sex	16	1482.395	1576.439	-725.1973	25.7845	0.0001	
## 5	Interact Sensor & Session	29	1454.597	1625.052	-698.2983	79.5823	0.0000	
## 6	Interact Activity & Day	13	1501.803	1578.214	-737.9015	0.3761	0.8286	
## 7	Interact Activity & Sex	13	1498.749	1575.160	-736.3744	3.4303	0.1799	
## 8	Interact Session & Sex	17	1503.649	1603.572	-734.8247	6.5296	0.3665	
## 9	Interact Day & Sex	12	1499.547	1570.080	-737.7735	0.6320	0.4266	
## 10	Interact Day & Session	17	1493.782	1593.704	-729.8910	16.3970	0.0118	

This suggests we should include the below. Both their AIC and BIC decreases, and they have significant p-values from the ANOVA test.

- HF_type * day
- HF_type * activity

We should potentially include the below. Although their AIC decreases and they have significant p-values from the ANOVA test, their BIC increased.

- HF_type * sex
- HF_type * session
- Day * session

And not include the below, as both their AIC and BIC increase, and they do not have significant p-values from the ANOVA test.

- activity * day
- activity * sex
- session * sex
- day * sex

And we do not include the following due to issues with degrees of freedom:

- activity * session

0.7 Section 5.3

Now, try adding all five of the suggested interactive variables, and then compare that to only having four interactive variables

```
#HF_type*day, HF_type * activity, HF_type * sex, HF_type * session, day * session
mdl3 <- gls(
  log(HF_value) ~ (day) + (activity) + (HF_type) + (sex) + (HF_type * day) + (HF_type * activity) +
  data = df_clean,
  method = "ML"
)

#remove hf_type * day
mdl3a <- gls(
  log(HF_value) ~ (day) + (activity) + (HF_type) + (sex) + (HF_type * activity) + (HF_type * sex) +
  data = df_clean,
```

```

    method = "ML"
  )

  #remove hf_type * activity
  mdl3b <- gls(
    log(HF_value) ~ (day) + (activity) + (HF_type) + (sex) + (HF_type * day) + (HF_type * sex) + (HF_
    data = df_clean,
    method = "ML"
  )

  #remove hf_type * sex
  mdl3c <- gls(
    log(HF_value) ~ (day) + (activity) + (HF_type) + (sex) + (HF_type * day) + (HF_type * activity) +
    data = df_clean,
    method = "ML"
  )

  #remove hf_type * session
  mdl3d <- gls(
    log(HF_value) ~ (day) + (activity) + (HF_type) + (sex) + (HF_type * day) + (HF_type * activity) +
    data = df_clean,
    method = "ML"
  )

  #remove day * session
  mdl3e <- gls(
    log(HF_value) ~ (day) + (activity) + (HF_type) + (sex) + (HF_type * day) + (HF_type * activity) +
    data = df_clean,
    method = "ML"
  )

anova_summary_manual_md11_nosession_interact2 <- bind_rows(
  get_baseline_metrics(mdl3, "Current Model"),
  get_anova_outputs(mdl3, mdl3a, "Drop Sensor & Day"),
  get_anova_outputs(mdl3, mdl3b, "Drop Sensor & Activity"),
  get_anova_outputs(mdl3, mdl3c, "Drop Sensor & Sex"),
  get_anova_outputs(mdl3, mdl3d, "Drop Sensor & Session"),
  get_anova_outputs(mdl3, mdl3e, "Drop Day & Session")
)
print(anova_summary_manual_md11_nosession_interact2)

```

##	Model	Df	AIC	BIC	LogLik	LRT_Stat	p_value
## 1	Current Model	52	1099.586	1405.230	-497.7930	NA	NA
## 2	Drop Sensor & Day	47	1307.264	1583.519	-606.6318	217.6775	0.0000
## 3	Drop Sensor & Activity	42	1240.924	1487.791	-578.4621	161.3382	0.0000
## 4	Drop Sensor & Sex	47	1119.786	1396.042	-512.8932	30.2003	0.0000
## 5	Drop Sensor & Session	49	1107.169	1395.180	-504.5847	13.5834	0.0035
## 6	Drop Day & Session	37	1105.828	1323.306	-515.9139	36.2419	0.0016

Now, we see a similar story.

Both AIC and BIC decrease (and p-value is stat. sig) when we include the following:

- Sensor & Day
- Sensor & Activity

However, BIC increases when we include the following:

- Sensor & Sex
- Sensor & Session
- Day & Session

For now, we will take that to mean to include all interactive variables! Note that including sensor & Session + Day & Session adds session back to the model, in general. We will remove variables later, after looking at random intercepts.

0.8 Section 6: Run model with random intercepts

0.9 Section 6.1: Does adding the person random intercept do anything (with interaction variables)?

Now, I will test if there's inter-person variation.

```
#final model before random effects
# log(HF_value) ~ (day) + (activity) + (HF_type) + (sex)
#               + (HF_type * day) + (HF_type * activity) + (HF_type * sex) + (HF_type * session),

mdl4_no_random_with_interact_reml <- gls(
  log(HF_value) ~ (day) + (activity) + (HF_type) + (sex) + (HF_type * day) + (HF_type * activity) +
  data = df_clean,
  method = "REML"
)

mdl4_random_person_with_interact_reml <- lme(
  fixed = log(HF_value) ~ (day) + (activity) + (HF_type) + (sex) + (HF_type * day) + (HF_type * act.
  random = ~ 1 | person,
  data = df_clean,
  method = "REML"
)

#anova to compare
print(anova(mdl4_no_random_with_interact_reml, mdl4_random_person_with_interact_reml))

##                                Model df      AIC      BIC    logLik
## mdl4_no_random_with_interact_reml      1 52 1363.879 1668.508 -629.9394
## mdl4_random_person_with_interact_reml    2 53 1215.207 1525.694 -554.6032
##                                Test  L.Ratio p-value
## mdl4_no_random_with_interact_reml
## mdl4_random_person_with_interact_reml 1 vs 2 150.6722 <.0001

#now see if the variance of the random effect is 0.
print(summary(mdl4_random_person_with_interact_reml))

## Linear mixed-effects model fit by REML
##   Data: df_clean
##       AIC      BIC    logLik
##  1215.206 1525.694 -554.6032
##
## Random effects:
## Formula: ~1 | person
##      (Intercept) Residual
## StdDev:  0.08332656 0.2837776
```

```

##
## Fixed effects: log(HF_value) ~ (day) + (activity) + (HF_type) + (sex) + (HF_type *
##
## Value Std.Error DF t-value p-value
## (Intercept) 4.452016 0.04479621 2565 99.38376 0.0000
## day2 -0.108628 0.03185325 2565 -3.41027 0.0007
## activitystanding -0.088607 0.03118034 2565 -2.84175 0.0045
## activitywalking 0.037399 0.03927511 2565 0.95222 0.3411
## HF_typeHF_2 -0.712535 0.05206180 2565 -13.68632 0.0000
## HF_typeHF_3 -0.642538 0.05266376 2565 -12.20077 0.0000
## HF_typeHF_5 -0.412219 0.05287316 2565 -7.79638 0.0000
## HF_typeHF_7 -0.156199 0.05184797 2565 -3.01264 0.0026
## HF_typeHF_8 -0.893326 0.05223878 2565 -17.10082 0.0000
## sexM 0.003851 0.04353336 22 0.08845 0.9303
## session2 -0.035486 0.04661032 2565 -0.76133 0.4465
## session3 -0.032443 0.03692475 2565 -0.87863 0.3797
## session4 -0.144673 0.04280785 2565 -3.37960 0.0007
## day2:HF_typeHF_2 0.492350 0.03867158 2565 12.73158 0.0000
## day2:HF_typeHF_3 0.500757 0.03895440 2565 12.85496 0.0000
## day2:HF_typeHF_5 0.427280 0.03915474 2565 10.91259 0.0000
## day2:HF_typeHF_7 0.342409 0.03850939 2565 8.89157 0.0000
## day2:HF_typeHF_8 0.479066 0.03878476 2565 12.35191 0.0000
## activitystanding:HF_typeHF_2 -0.091681 0.04369535 2565 -2.09818 0.0360
## activitywalking:HF_typeHF_2 -0.377406 0.05533355 2565 -6.82056 0.0000
## activitystanding:HF_typeHF_3 0.189726 0.04402757 2565 4.30926 0.0000
## activitywalking:HF_typeHF_3 0.232380 0.05565090 2565 4.17567 0.0000
## activitystanding:HF_typeHF_5 0.062779 0.04423849 2565 1.41909 0.1560
## activitywalking:HF_typeHF_5 -0.038559 0.05595404 2565 -0.68913 0.4908
## activitystanding:HF_typeHF_7 -0.043026 0.04353023 2565 -0.98842 0.3230
## activitywalking:HF_typeHF_7 -0.119747 0.05505787 2565 -2.17494 0.0297
## activitystanding:HF_typeHF_8 0.182487 0.04385365 2565 4.16127 0.0000
## activitywalking:HF_typeHF_8 0.166283 0.05550516 2565 2.99582 0.0028
## HF_typeHF_2:sexM -0.073982 0.03824705 2565 -1.93431 0.0532
## HF_typeHF_3:sexM 0.083197 0.03850601 2565 2.16063 0.0308
## HF_typeHF_5:sexM 0.013619 0.03873016 2565 0.35165 0.7251
## HF_typeHF_7:sexM 0.113927 0.03806730 2565 2.99278 0.0028
## HF_typeHF_8:sexM 0.074473 0.03835389 2565 1.94172 0.0523
## HF_typeHF_2:session2 0.126562 0.06244310 2565 2.02684 0.0428
## HF_typeHF_3:session2 0.047576 0.06286785 2565 0.75676 0.4493
## HF_typeHF_5:session2 0.088155 0.06331171 2565 1.39239 0.1639
## HF_typeHF_7:session2 0.066791 0.06232700 2565 1.07163 0.2840
## HF_typeHF_8:session2 0.093582 0.06246893 2565 1.49806 0.1342
## HF_typeHF_2:session3 0.101822 0.04801480 2565 2.12064 0.0340
## HF_typeHF_3:session3 0.043753 0.04827395 2565 0.90635 0.3648
## HF_typeHF_5:session3 0.063755 0.04854524 2565 1.31330 0.1892
## HF_typeHF_7:session3 0.022956 0.04776700 2565 0.48059 0.6308
## HF_typeHF_8:session3 0.129210 0.04823949 2565 2.67851 0.0074
## HF_typeHF_2:session4 0.283852 0.05617238 2565 5.05323 0.0000
## HF_typeHF_3:session4 0.185957 0.05665228 2565 3.28242 0.0010
## HF_typeHF_5:session4 0.201682 0.05679608 2565 3.55098 0.0004
## HF_typeHF_7:session4 0.089382 0.05596397 2565 1.59714 0.1104
## HF_typeHF_8:session4 0.232961 0.05618452 2565 4.14635 0.0000
## day2:session2 -0.072200 0.03596208 2565 -2.00767 0.0448
## day2:session3 -0.070164 0.02779766 2565 -2.52408 0.0117
## day2:session4 -0.111670 0.03104562 2565 -3.59697 0.0003

```

```

## Correlation:
##
## (Intr) day2   actvtys actvtyw HF_tHF_2 HF_tHF_3
## day2        -0.372
## activitystanding -0.368  0.063
## activitywalking -0.397  0.022  0.399
## HF_typeHF_2    -0.586  0.243  0.319   0.343
## HF_typeHF_3    -0.578  0.239  0.315   0.338   0.499
## HF_typeHF_5    -0.576  0.236  0.314   0.337   0.497   0.491
## HF_typeHF_7    -0.589  0.246  0.320   0.344   0.507   0.501
## HF_typeHF_8    -0.584  0.243  0.318   0.341   0.503   0.497
## sexM          -0.477 -0.016 -0.006   0.001   0.156   0.154
## session2      -0.460  0.204  0.103   0.280   0.365   0.360
## session3      -0.405  0.158 -0.005  -0.005   0.297   0.292
## session4      -0.431  0.139  0.000   0.260   0.326   0.322
## day2:HF_typeHF_2  0.235 -0.612 -0.062  -0.023  -0.403  -0.200
## day2:HF_typeHF_3  0.233 -0.609 -0.062  -0.023  -0.201  -0.411
## day2:HF_typeHF_5  0.232 -0.605 -0.061  -0.023  -0.200  -0.198
## day2:HF_typeHF_7  0.236 -0.616 -0.062  -0.023  -0.203  -0.201
## day2:HF_typeHF_8  0.234 -0.612 -0.062  -0.023  -0.202  -0.200
## activitystanding:HF_typeHF_2 0.264 -0.053 -0.709  -0.283  -0.450  -0.225
## activitywalking:HF_typeHF_2  0.282 -0.019 -0.281  -0.709  -0.480  -0.240
## activitystanding:HF_typeHF_3 0.262 -0.053 -0.703  -0.281  -0.226  -0.449
## activitywalking:HF_typeHF_3  0.281 -0.019 -0.280  -0.705  -0.242  -0.477
## activitystanding:HF_typeHF_5 0.261 -0.053 -0.700  -0.279  -0.224  -0.222
## activitywalking:HF_typeHF_5  0.279 -0.019 -0.278  -0.701  -0.240  -0.237
## activitystanding:HF_typeHF_7 0.265 -0.053 -0.712  -0.284  -0.228  -0.226
## activitywalking:HF_typeHF_7  0.283 -0.019 -0.283  -0.713  -0.244  -0.241
## activitystanding:HF_typeHF_8 0.263 -0.053 -0.706  -0.282  -0.226  -0.224
## activitywalking:HF_typeHF_8  0.281 -0.019 -0.280  -0.707  -0.242  -0.239
## HF_typeHF_2:sexM    0.206  0.018  0.007  -0.001  -0.350  -0.174
## HF_typeHF_3:sexM    0.204  0.017  0.007  -0.001  -0.175  -0.358
## HF_typeHF_5:sexM    0.203  0.018  0.007  -0.001  -0.175  -0.172
## HF_typeHF_7:sexM    0.207  0.019  0.007  -0.001  -0.178  -0.176
## HF_typeHF_8:sexM    0.205  0.018  0.007  -0.001  -0.176  -0.174
## HF_typeHF_2:session2 0.318 -0.065 -0.092  -0.215  -0.541  -0.270
## HF_typeHF_3:session2 0.315 -0.063 -0.091  -0.214  -0.272  -0.539
## HF_typeHF_5:session2 0.312 -0.060 -0.091  -0.213  -0.270  -0.267
## HF_typeHF_7:session2 0.319 -0.068 -0.091  -0.216  -0.274  -0.271
## HF_typeHF_8:session2 0.318 -0.066 -0.092  -0.215  -0.273  -0.270
## HF_typeHF_2:session3 0.266  0.007  0.003   0.003  -0.458  -0.227
## HF_typeHF_3:session3 0.264  0.009  0.003   0.003  -0.228  -0.455
## HF_typeHF_5:session3 0.261  0.012  0.003   0.003  -0.227  -0.224
## HF_typeHF_7:session3 0.269  0.004  0.003   0.003  -0.231  -0.228
## HF_typeHF_8:session3 0.265  0.007  0.003   0.003  -0.228  -0.226
## HF_typeHF_2:session4 0.289  0.005  0.000  -0.199  -0.493  -0.246
## HF_typeHF_3:session4 0.286  0.006  0.000  -0.197  -0.247  -0.491
## HF_typeHF_5:session4 0.284  0.010  0.000  -0.197  -0.246  -0.243
## HF_typeHF_7:session4 0.291  0.002  0.000  -0.200  -0.250  -0.247
## HF_typeHF_8:session4 0.289  0.005  0.000  -0.199  -0.249  -0.246
## day2:session2      0.092 -0.333  0.073   0.031   0.004   0.006
## day2:session3      0.153 -0.434  0.004   0.005   0.005   0.007
## day2:session4      0.139 -0.391  0.000   0.002   0.002   0.003
##
## HF_tHF_5 HF_tHF_7 HF_tHF_8 sexM   sessn2 sessn3
## day2

```

```

## activitystanding
## activitywalking
## HF_typeHF_2
## HF_typeHF_3
## HF_typeHF_5
## HF_typeHF_7      0.499
## HF_typeHF_8      0.495      0.505
## sexM              0.153      0.156      0.155
## session2          0.358      0.368      0.364      0.005
## session3          0.290      0.299      0.296      0.002      0.391
## session4          0.319      0.328      0.325      0.004      0.413      0.427
## day2:HF_typeHF_2 -0.199 -0.203 -0.202      0.013 -0.073      0.006
## day2:HF_typeHF_3 -0.198 -0.201 -0.200      0.013 -0.072      0.006
## day2:HF_typeHF_5 -0.411 -0.200 -0.199      0.013 -0.071      0.006
## day2:HF_typeHF_7 -0.200 -0.401 -0.202      0.013 -0.074      0.007
## day2:HF_typeHF_8 -0.199 -0.202 -0.404      0.013 -0.073      0.005
## activitystanding:HF_typeHF_2 -0.224 -0.228 -0.227      0.005 -0.088      0.004
## activitywalking:HF_typeHF_2 -0.239 -0.244 -0.242 -0.001 -0.204      0.004
## activitystanding:HF_typeHF_3 -0.222 -0.227 -0.225      0.004 -0.087      0.004
## activitywalking:HF_typeHF_3 -0.238 -0.243 -0.241 -0.001 -0.203      0.004
## activitystanding:HF_typeHF_5 -0.449 -0.225 -0.224      0.004 -0.087      0.004
## activitywalking:HF_typeHF_5 -0.478 -0.241 -0.240 -0.001 -0.202      0.004
## activitystanding:HF_typeHF_7 -0.225 -0.448 -0.227      0.005 -0.087      0.004
## activitywalking:HF_typeHF_7 -0.241 -0.479 -0.244 -0.001 -0.205      0.004
## activitystanding:HF_typeHF_8 -0.223 -0.227 -0.452      0.004 -0.088      0.004
## activitywalking:HF_typeHF_8 -0.239 -0.243 -0.483 -0.001 -0.204      0.003
## HF_typeHF_2:sexM -0.174 -0.178 -0.176 -0.443 -0.006 -0.002
## HF_typeHF_3:sexM -0.173 -0.177 -0.175 -0.440 -0.005 -0.001
## HF_typeHF_5:sexM -0.349 -0.176 -0.174 -0.438 -0.006 -0.002
## HF_typeHF_7:sexM -0.175 -0.358 -0.177 -0.445 -0.005 -0.002
## HF_typeHF_8:sexM -0.174 -0.178 -0.354 -0.442 -0.006 -0.002
## HF_typeHF_2:session2 -0.269 -0.275 -0.273 -0.004 -0.681 -0.249
## HF_typeHF_3:session2 -0.268 -0.273 -0.271 -0.004 -0.674 -0.246
## HF_typeHF_5:session2 -0.540 -0.271 -0.269 -0.004 -0.668 -0.244
## HF_typeHF_7:session2 -0.270 -0.537 -0.273 -0.004 -0.684 -0.250
## HF_typeHF_8:session2 -0.269 -0.275 -0.541 -0.004 -0.680 -0.249
## HF_typeHF_2:session3 -0.226 -0.230 -0.229 -0.001 -0.257 -0.656
## HF_typeHF_3:session3 -0.225 -0.229 -0.227 -0.001 -0.255 -0.650
## HF_typeHF_5:session3 -0.456 -0.228 -0.226 -0.001 -0.252 -0.644
## HF_typeHF_7:session3 -0.227 -0.457 -0.230 -0.001 -0.259 -0.661
## HF_typeHF_8:session3 -0.225 -0.229 -0.448 -0.001 -0.256 -0.651
## HF_typeHF_2:session4 -0.245 -0.250 -0.248 -0.003 -0.277 -0.278
## HF_typeHF_3:session4 -0.243 -0.248 -0.246 -0.003 -0.274 -0.275
## HF_typeHF_5:session4 -0.491 -0.247 -0.246 -0.003 -0.272 -0.273
## HF_typeHF_7:session4 -0.246 -0.491 -0.249 -0.003 -0.279 -0.280
## HF_typeHF_8:session4 -0.245 -0.250 -0.493 -0.003 -0.277 -0.278
## day2:session2      0.008      0.002      0.003      0.002 -0.292 -0.146
## day2:session3      0.010      0.002      0.002      0.001 -0.148 -0.383
## day2:session4      0.007      0.000      0.002      0.001 -0.134 -0.169
## sessn4 d2:HF_HF_2 d2:HF_HF_3 d2:HF_HF_5 d2:HF_HF_7
## day2
## activitystanding
## activitywalking
## HF_typeHF_2

```



```

## HF_typeHF_3
## HF_typeHF_5
## HF_typeHF_7
## HF_typeHF_8
## sexM
## session2
## session3
## session4
## day2:HF_typeHF_2          0.003
## day2:HF_typeHF_3          0.003  0.502
## day2:HF_typeHF_5          0.003  0.500      0.496
## day2:HF_typeHF_7          0.003  0.507      0.503      0.501
## day2:HF_typeHF_8          0.003  0.504      0.501      0.498      0.505
## activitystanding:HF_typeHF_2 0.000  0.084      0.044      0.044      0.044
## activitywalking:HF_typeHF_2 -0.185  0.032      0.016      0.016      0.017
## activitystanding:HF_typeHF_3 0.000  0.044      0.085      0.043      0.044
## activitywalking:HF_typeHF_3 -0.184  0.016      0.032      0.016      0.017
## activitystanding:HF_typeHF_5 0.000  0.044      0.043      0.085      0.044
## activitywalking:HF_typeHF_5 -0.183  0.016      0.016      0.033      0.016
## activitystanding:HF_typeHF_7 0.000  0.044      0.044      0.044      0.081
## activitywalking:HF_typeHF_7 -0.186  0.017      0.016      0.016      0.031
## activitystanding:HF_typeHF_8 0.000  0.044      0.044      0.043      0.044
## activitywalking:HF_typeHF_8 -0.184  0.016      0.016      0.016      0.017
## HF_typeHF_2:sexM          -0.004 -0.026     -0.016     -0.015     -0.014
## HF_typeHF_3:sexM          -0.005 -0.015     -0.012     -0.015     -0.014
## HF_typeHF_5:sexM          -0.004 -0.015     -0.016     -0.027     -0.014
## HF_typeHF_7:sexM          -0.004 -0.014     -0.014     -0.014     -0.016
## HF_typeHF_8:sexM          -0.004 -0.016     -0.016     -0.015     -0.014
## HF_typeHF_2:session2      -0.271  0.111      0.054      0.054      0.055
## HF_typeHF_3:session2      -0.269  0.054      0.111      0.054      0.055
## HF_typeHF_5:session2      -0.266  0.054      0.053      0.112      0.054
## HF_typeHF_7:session2      -0.273  0.055      0.054      0.054      0.112
## HF_typeHF_8:session2      -0.271  0.055      0.054      0.054      0.055
## HF_typeHF_2:session3      -0.281 -0.005     -0.005     -0.004     -0.005
## HF_typeHF_3:session3      -0.278 -0.004     -0.005     -0.004     -0.005
## HF_typeHF_5:session3      -0.276 -0.004     -0.005     -0.004     -0.005
## HF_typeHF_7:session3      -0.283 -0.005     -0.005     -0.004     -0.005
## HF_typeHF_8:session3      -0.279 -0.004     -0.005     -0.004     -0.005
## HF_typeHF_2:session4      -0.658 -0.003     -0.003     -0.003     -0.003
## HF_typeHF_3:session4      -0.652 -0.003      0.001     -0.003     -0.003
## HF_typeHF_5:session4      -0.647 -0.003     -0.003     -0.003     -0.003
## HF_typeHF_7:session4      -0.664 -0.003     -0.003     -0.003     -0.001
## HF_typeHF_8:session4      -0.658 -0.003     -0.003     -0.003     -0.003
## day2:session2             -0.125 -0.003     -0.003     -0.004      0.000
## day2:session3             -0.162 -0.001     -0.001     -0.001     -0.001
## day2:session4             -0.368  0.002      0.004      0.002      0.003
##
## d2:HF_HF_8 actvtys:HF_HF_2 actvtyw:HF_HF_2
## day2
## activitystanding
## activitywalking
## HF_typeHF_2
## HF_typeHF_3
## HF_typeHF_5
## HF_typeHF_7

```

```

## HF_typeHF_8
## sexM
## session2
## session3
## session4
## day2:HF_typeHF_2
## day2:HF_typeHF_3
## day2:HF_typeHF_5
## day2:HF_typeHF_7
## day2:HF_typeHF_8
## activitystanding:HF_typeHF_2  0.044
## activitywalking:HF_typeHF_2   0.016      0.396
## activitystanding:HF_typeHF_3  0.044      0.502      0.199
## activitywalking:HF_typeHF_3   0.016      0.200      0.500
## activitystanding:HF_typeHF_5  0.043      0.500      0.198
## activitywalking:HF_typeHF_5   0.016      0.199      0.498
## activitystanding:HF_typeHF_7  0.044      0.508      0.202
## activitywalking:HF_typeHF_7   0.016      0.202      0.506
## activitystanding:HF_typeHF_8  0.083      0.504      0.200
## activitywalking:HF_typeHF_8   0.035      0.200      0.502
## HF_typeHF_2:sexM              -0.016     -0.005      0.001
## HF_typeHF_3:sexM              -0.015     -0.005      0.001
## HF_typeHF_5:sexM              -0.015     -0.005      0.001
## HF_typeHF_7:sexM              -0.014     -0.005      0.001
## HF_typeHF_8:sexM              -0.025     -0.005      0.001
## HF_typeHF_2:session2          0.055      0.129      0.302
## HF_typeHF_3:session2          0.054      0.065      0.152
## HF_typeHF_5:session2          0.054      0.064      0.151
## HF_typeHF_7:session2          0.055      0.066      0.153
## HF_typeHF_8:session2          0.112      0.065      0.153
## HF_typeHF_2:session3         -0.005     -0.002     -0.002
## HF_typeHF_3:session3         -0.005     -0.002     -0.002
## HF_typeHF_5:session3         -0.005     -0.002     -0.002
## HF_typeHF_7:session3         -0.005     -0.002     -0.002
## HF_typeHF_8:session3         -0.009     -0.002     -0.002
## HF_typeHF_2:session4         -0.003      0.000      0.279
## HF_typeHF_3:session4         -0.003      0.000      0.140
## HF_typeHF_5:session4         -0.003      0.000      0.140
## HF_typeHF_7:session4         -0.003      0.000      0.142
## HF_typeHF_8:session4         -0.002      0.000      0.141
## day2:session2                 -0.003      0.001     -0.001
## day2:session3                  0.003     -0.003     -0.003
## day2:session4                  0.002      0.000     -0.001
##                                actvtys:HF_HF_3 actvtyw:HF_HF_3 actvtys:HF_HF_5
## day2
## activitystanding
## activitywalking
## HF_typeHF_2
## HF_typeHF_3
## HF_typeHF_5
## HF_typeHF_7
## HF_typeHF_8
## sexM
## session2

```

```

## session3
## session4
## day2:HF_typeHF_2
## day2:HF_typeHF_3
## day2:HF_typeHF_5
## day2:HF_typeHF_7
## day2:HF_typeHF_8
## activitystanding:HF_typeHF_2
## activitywalking:HF_typeHF_2
## activitystanding:HF_typeHF_3
## activitywalking:HF_typeHF_3    0.397
## activitystanding:HF_typeHF_5    0.496          0.197
## activitywalking:HF_typeHF_5    0.197          0.495          0.396
## activitystanding:HF_typeHF_7    0.504          0.200          0.502
## activitywalking:HF_typeHF_7    0.200          0.503          0.199
## activitystanding:HF_typeHF_8    0.500          0.199          0.498
## activitywalking:HF_typeHF_8    0.199          0.499          0.198
## HF_typeHF_2:sexM                -0.005          0.001          -0.005
## HF_typeHF_3:sexM                -0.005          0.001          -0.005
## HF_typeHF_5:sexM                -0.005          0.001          -0.005
## HF_typeHF_7:sexM                -0.005          0.001          -0.005
## HF_typeHF_8:sexM                -0.005          0.001          -0.005
## HF_typeHF_2:session2            0.065          0.152          0.065
## HF_typeHF_3:session2            0.131          0.302          0.064
## HF_typeHF_5:session2            0.064          0.150          0.132
## HF_typeHF_7:session2            0.065          0.152          0.065
## HF_typeHF_8:session2            0.065          0.152          0.065
## HF_typeHF_2:session3            -0.002          -0.002          -0.002
## HF_typeHF_3:session3            -0.002          -0.002          -0.002
## HF_typeHF_5:session3            -0.002          -0.002          -0.002
## HF_typeHF_7:session3            -0.002          -0.002          -0.002
## HF_typeHF_8:session3            -0.002          -0.002          -0.002
## HF_typeHF_2:session4            0.000          0.140          0.000
## HF_typeHF_3:session4            0.000          0.278          0.000
## HF_typeHF_5:session4            0.000          0.139          0.000
## HF_typeHF_7:session4            0.000          0.141          0.000
## HF_typeHF_8:session4            0.000          0.140          0.000
## day2:session2                   0.002          -0.001          0.002
## day2:session3                   -0.003          -0.003          -0.003
## day2:session4                   0.000          -0.001          0.000
##                                actvtyw:HF_HF_5 actvtys:HF_HF_7 actvtyw:HF_HF_7
## day2
## activitystanding
## activitywalking
## HF_typeHF_2
## HF_typeHF_3
## HF_typeHF_5
## HF_typeHF_7
## HF_typeHF_8
## sexM
## session2
## session3
## session4
## day2:HF_typeHF_2

```

```

## day2:HF_typeHF_3
## day2:HF_typeHF_5
## day2:HF_typeHF_7
## day2:HF_typeHF_8
## activitystanding:HF_typeHF_2
## activitywalking:HF_typeHF_2
## activitystanding:HF_typeHF_3
## activitywalking:HF_typeHF_3
## activitystanding:HF_typeHF_5
## activitywalking:HF_typeHF_5
## activitystanding:HF_typeHF_7 0.199
## activitywalking:HF_typeHF_7 0.500          0.396
## activitystanding:HF_typeHF_8 0.198          0.506          0.201
## activitywalking:HF_typeHF_8 0.496          0.201          0.504
## HF_typeHF_2:sexM 0.001          -0.005          0.001
## HF_typeHF_3:sexM 0.001          -0.005          0.001
## HF_typeHF_5:sexM 0.001          -0.005          0.001
## HF_typeHF_7:sexM 0.001          -0.005          0.001
## HF_typeHF_8:sexM 0.001          -0.005          0.001
## HF_typeHF_2:session2 0.151          0.066          0.154
## HF_typeHF_3:session2 0.150          0.065          0.153
## HF_typeHF_5:session2 0.303          0.065          0.152
## HF_typeHF_7:session2 0.151          0.128          0.301
## HF_typeHF_8:session2 0.151          0.066          0.154
## HF_typeHF_2:session3 -0.002          -0.002          -0.002
## HF_typeHF_3:session3 -0.002          -0.002          -0.002
## HF_typeHF_5:session3 -0.002          -0.002          -0.002
## HF_typeHF_7:session3 -0.002          -0.002          -0.002
## HF_typeHF_8:session3 -0.002          -0.002          -0.002
## HF_typeHF_2:session4 0.140          0.000          0.142
## HF_typeHF_3:session4 0.138          0.000          0.141
## HF_typeHF_5:session4 0.279          0.000          0.140
## HF_typeHF_7:session4 0.140          -0.002          0.278
## HF_typeHF_8:session4 0.140          0.000          0.142
## day2:session2 -0.001          -0.002          -0.002
## day2:session3 -0.003          -0.003          -0.003
## day2:session4 -0.001          -0.001          -0.002
##
##          actvtys:HF_HF_8 actvtyw:HF_HF_8 HF_HF_2:M
## day2
## activitystanding
## activitywalking
## HF_typeHF_2
## HF_typeHF_3
## HF_typeHF_5
## HF_typeHF_7
## HF_typeHF_8
## sexM
## session2
## session3
## session4
## day2:HF_typeHF_2
## day2:HF_typeHF_3
## day2:HF_typeHF_5
## day2:HF_typeHF_7

```

```

## day2:HF_typeHF_8
## activitystanding:HF_typeHF_2
## activitywalking:HF_typeHF_2
## activitystanding:HF_typeHF_3
## activitywalking:HF_typeHF_3
## activitystanding:HF_typeHF_5
## activitywalking:HF_typeHF_5
## activitystanding:HF_typeHF_7
## activitywalking:HF_typeHF_7
## activitystanding:HF_typeHF_8
## activitywalking:HF_typeHF_8 0.397
## HF_typeHF_2:sexM -0.005 0.001
## HF_typeHF_3:sexM -0.005 0.001 0.502
## HF_typeHF_5:sexM -0.005 0.001 0.500
## HF_typeHF_7:sexM -0.005 0.001 0.507
## HF_typeHF_8:sexM -0.001 0.005 0.504
## HF_typeHF_2:session2 0.065 0.152 0.005
## HF_typeHF_3:session2 0.065 0.151 0.005
## HF_typeHF_5:session2 0.064 0.150 0.005
## HF_typeHF_7:session2 0.065 0.153 0.005
## HF_typeHF_8:session2 0.130 0.303 0.005
## HF_typeHF_2:session3 -0.002 -0.002 0.001
## HF_typeHF_3:session3 -0.002 -0.002 0.001
## HF_typeHF_5:session3 -0.002 -0.002 0.001
## HF_typeHF_7:session3 -0.002 -0.002 0.001
## HF_typeHF_8:session3 -0.004 -0.009 0.001
## HF_typeHF_2:session4 0.000 0.141 0.004
## HF_typeHF_3:session4 0.000 0.140 0.003
## HF_typeHF_5:session4 0.000 0.139 0.004
## HF_typeHF_7:session4 0.000 0.141 0.004
## HF_typeHF_8:session4 0.001 0.280 0.004
## day2:session2 0.002 0.000 -0.003
## day2:session3 -0.004 -0.001 -0.001
## day2:session4 0.000 -0.001 -0.002
## HF_HF_3:M HF_HF_5:M HF_HF_7:M HF_HF_8:M HF_HF_2:2
## day2
## activitystanding
## activitywalking
## HF_typeHF_2
## HF_typeHF_3
## HF_typeHF_5
## HF_typeHF_7
## HF_typeHF_8
## sexM
## session2
## session3
## session4
## day2:HF_typeHF_2
## day2:HF_typeHF_3
## day2:HF_typeHF_5
## day2:HF_typeHF_7
## day2:HF_typeHF_8
## activitystanding:HF_typeHF_2
## activitywalking:HF_typeHF_2

```

```

## activitystanding:HF_typeHF_3
## activitywalking:HF_typeHF_3
## activitystanding:HF_typeHF_5
## activitywalking:HF_typeHF_5
## activitystanding:HF_typeHF_7
## activitywalking:HF_typeHF_7
## activitystanding:HF_typeHF_8
## activitywalking:HF_typeHF_8
## HF_typeHF_2:sexM
## HF_typeHF_3:sexM
## HF_typeHF_5:sexM          0.496
## HF_typeHF_7:sexM          0.503      0.501
## HF_typeHF_8:sexM          0.501      0.498      0.505
## HF_typeHF_2:session2      0.005      0.005      0.005      0.005
## HF_typeHF_3:session2      0.005      0.005      0.005      0.005      0.505
## HF_typeHF_5:session2      0.005      0.005      0.005      0.005      0.501
## HF_typeHF_7:session2      0.005      0.005      0.000      0.005      0.509
## HF_typeHF_8:session2      0.005      0.005      0.005      0.006      0.508
## HF_typeHF_2:session3      0.001      0.001      0.002      0.001      0.383
## HF_typeHF_3:session3      0.001      0.001      0.002      0.001      0.191
## HF_typeHF_5:session3      0.001      0.001      0.002      0.001      0.190
## HF_typeHF_7:session3      0.001      0.001      0.002      0.001      0.193
## HF_typeHF_8:session3      0.001      0.002      0.002      -0.003      0.191
## HF_typeHF_2:session4      0.003      0.004      0.003      0.004      0.411
## HF_typeHF_3:session4      0.007      0.003      0.003      0.003      0.206
## HF_typeHF_5:session4      0.003      0.003      0.003      0.004      0.205
## HF_typeHF_7:session4      0.003      0.004      0.002      0.004      0.208
## HF_typeHF_8:session4      0.003      0.003      0.003      0.004      0.207
## day2:session2              -0.003      -0.003      -0.004      -0.002      -0.002
## day2:session3              -0.001      -0.001      -0.001      0.000      -0.003
## day2:session4              0.001      -0.002      -0.003      -0.002      -0.002
## HF_HF_3:2 HF_HF_5:2 HF_HF_7:2 HF_HF_8:2 HF_HF_2:3
## day2
## activitystanding
## activitywalking
## HF_typeHF_2
## HF_typeHF_3
## HF_typeHF_5
## HF_typeHF_7
## HF_typeHF_8
## sexM
## session2
## session3
## session4
## day2:HF_typeHF_2
## day2:HF_typeHF_3
## day2:HF_typeHF_5
## day2:HF_typeHF_7
## day2:HF_typeHF_8
## activitystanding:HF_typeHF_2
## activitywalking:HF_typeHF_2
## activitystanding:HF_typeHF_3
## activitywalking:HF_typeHF_3
## activitystanding:HF_typeHF_5

```

```

## activitywalking:HF_typeHF_5
## activitystanding:HF_typeHF_7
## activitywalking:HF_typeHF_7
## activitystanding:HF_typeHF_8
## activitywalking:HF_typeHF_8
## HF_typeHF_2:sexM
## HF_typeHF_3:sexM
## HF_typeHF_5:sexM
## HF_typeHF_7:sexM
## HF_typeHF_8:sexM
## HF_typeHF_2:session2
## HF_typeHF_3:session2
## HF_typeHF_5:session2      0.498
## HF_typeHF_7:session2      0.506      0.502
## HF_typeHF_8:session2      0.505      0.501      0.509
## HF_typeHF_2:session3      0.191      0.190      0.193      0.192
## HF_typeHF_3:session3      0.382      0.189      0.192      0.191      0.502
## HF_typeHF_5:session3      0.189      0.382      0.190      0.190      0.499
## HF_typeHF_7:session3      0.192      0.191      0.381      0.193      0.507
## HF_typeHF_8:session3      0.190      0.189      0.192      0.378      0.502
## HF_typeHF_2:session4      0.206      0.204      0.208      0.207      0.426
## HF_typeHF_3:session4      0.409      0.203      0.206      0.205      0.213
## HF_typeHF_5:session4      0.204      0.410      0.205      0.205      0.212
## HF_typeHF_7:session4      0.207      0.205      0.409      0.208      0.215
## HF_typeHF_8:session4      0.206      0.204      0.208      0.411      0.214
## day2:session2              -0.007      -0.013      0.006      -0.002      -0.001
## day2:session3              -0.005      -0.008      0.000      -0.002      -0.001
## day2:session4              -0.004      -0.006      0.000      -0.002      -0.002
##                            HF_HF_3:3 HF_HF_5:3 HF_HF_7:3 HF_HF_8:3 HF_HF_2:4
## day2
## activitystanding
## activitywalking
## HF_typeHF_2
## HF_typeHF_3
## HF_typeHF_5
## HF_typeHF_7
## HF_typeHF_8
## sexM
## session2
## session3
## session4
## day2:HF_typeHF_2
## day2:HF_typeHF_3
## day2:HF_typeHF_5
## day2:HF_typeHF_7
## day2:HF_typeHF_8
## activitystanding:HF_typeHF_2
## activitywalking:HF_typeHF_2
## activitystanding:HF_typeHF_3
## activitywalking:HF_typeHF_3
## activitystanding:HF_typeHF_5
## activitywalking:HF_typeHF_5
## activitystanding:HF_typeHF_7
## activitywalking:HF_typeHF_7

```

```

## activitystanding:HF_typeHF_8
## activitywalking:HF_typeHF_8
## HF_typeHF_2:sexM
## HF_typeHF_3:sexM
## HF_typeHF_5:sexM
## HF_typeHF_7:sexM
## HF_typeHF_8:sexM
## HF_typeHF_2:session2
## HF_typeHF_3:session2
## HF_typeHF_5:session2
## HF_typeHF_7:session2
## HF_typeHF_8:session2
## HF_typeHF_2:session3
## HF_typeHF_3:session3
## HF_typeHF_5:session3      0.496
## HF_typeHF_7:session3      0.504      0.502
## HF_typeHF_8:session3      0.500      0.497      0.505
## HF_typeHF_2:session4      0.213      0.212      0.216      0.213
## HF_typeHF_3:session4      0.425      0.210      0.214      0.212      0.498
## HF_typeHF_5:session4      0.211      0.426      0.213      0.211      0.497
## HF_typeHF_7:session4      0.214      0.213      0.426      0.214      0.504
## HF_typeHF_8:session4      0.213      0.212      0.216      0.422      0.502
## day2:session2             -0.004      -0.006      0.001      -0.002      -0.002
## day2:session3             -0.007      -0.014      0.005      -0.004      -0.002
## day2:session4             -0.005      -0.008      0.001      -0.002      -0.002
##                           HF_HF_3:4 HF_HF_5:4 HF_HF_7:4 HF_HF_8:4 dy2:s2
## day2
## activitystanding
## activitywalking
## HF_typeHF_2
## HF_typeHF_3
## HF_typeHF_5
## HF_typeHF_7
## HF_typeHF_8
## sexM
## session2
## session3
## session4
## day2:HF_typeHF_2
## day2:HF_typeHF_3
## day2:HF_typeHF_5
## day2:HF_typeHF_7
## day2:HF_typeHF_8
## activitystanding:HF_typeHF_2
## activitywalking:HF_typeHF_2
## activitystanding:HF_typeHF_3
## activitywalking:HF_typeHF_3
## activitystanding:HF_typeHF_5
## activitywalking:HF_typeHF_5
## activitystanding:HF_typeHF_7
## activitywalking:HF_typeHF_7
## activitystanding:HF_typeHF_8
## activitywalking:HF_typeHF_8
## HF_typeHF_2:sexM

```



```

## HF_typeHF_3:sexM
## HF_typeHF_5:sexM
## HF_typeHF_7:sexM
## HF_typeHF_8:sexM
## HF_typeHF_2:session2
## HF_typeHF_3:session2
## HF_typeHF_5:session2
## HF_typeHF_7:session2
## HF_typeHF_8:session2
## HF_typeHF_2:session3
## HF_typeHF_3:session3
## HF_typeHF_5:session3
## HF_typeHF_7:session3
## HF_typeHF_8:session3
## HF_typeHF_2:session4
## HF_typeHF_3:session4
## HF_typeHF_5:session4      0.493
## HF_typeHF_7:session4      0.500      0.499
## HF_typeHF_8:session4      0.498      0.497      0.504
## day2:session2             -0.004      -0.006      0.000      -0.002
## day2:session3             -0.005      -0.008      0.000      -0.002      0.385
## day2:session4             -0.004      -0.014      0.006      -0.002      0.345
##                           dy2:s3
## day2
## activitystanding
## activitywalking
## HF_typeHF_2
## HF_typeHF_3
## HF_typeHF_5
## HF_typeHF_7
## HF_typeHF_8
## sexM
## session2
## session3
## session4
## day2:HF_typeHF_2
## day2:HF_typeHF_3
## day2:HF_typeHF_5
## day2:HF_typeHF_7
## day2:HF_typeHF_8
## activitystanding:HF_typeHF_2
## activitywalking:HF_typeHF_2
## activitystanding:HF_typeHF_3
## activitywalking:HF_typeHF_3
## activitystanding:HF_typeHF_5
## activitywalking:HF_typeHF_5
## activitystanding:HF_typeHF_7
## activitywalking:HF_typeHF_7
## activitystanding:HF_typeHF_8
## activitywalking:HF_typeHF_8
## HF_typeHF_2:sexM
## HF_typeHF_3:sexM
## HF_typeHF_5:sexM
## HF_typeHF_7:sexM

```

```
## HF_typeHF_8:sexM
## HF_typeHF_2:session2
## HF_typeHF_3:session2
## HF_typeHF_5:session2
## HF_typeHF_7:session2
## HF_typeHF_8:session2
## HF_typeHF_2:session3
## HF_typeHF_3:session3
## HF_typeHF_5:session3
## HF_typeHF_7:session3
## HF_typeHF_8:session3
## HF_typeHF_2:session4
## HF_typeHF_3:session4
## HF_typeHF_5:session4
## HF_typeHF_7:session4
## HF_typeHF_8:session4
## day2:session2
## day2:session3
## day2:session4          0.446
##
## Standardized Within-Group Residuals:
##      Min      Q1      Med      Q3      Max
## -6.468812 -0.4840518  0.1015446  0.6370308  3.7508645
##
## Number of Observations: 2638
## Number of Groups: 24
```

#yes - it stdev = 0.0795, so diff levels. however there is still leftover noise (0.2819). ICC is .073 (

Yes, the adding person as the random intercept does decrease AIC

0.9.1 Section 6.2: Does adding HF_type as random intercept help?

#hat about comparing the HF_type intercept with interaction effects, to the baseline person random inte

```
mdl4_random_person_sensor_with_interact_reml <- lme(
  fixed = log(HF_value) ~ (day) + (activity) + (HF_type) + (sex) + (session) + (HF_type * day) + (HF
  random = ~ 1 | person/HF_type,
  data = df_clean,
  method = "REML"
)
print(anova(mdl4_random_person_sensor_with_interact_reml, mdl4_random_person_with_interact_reml))
```

```
##              Model df          AIC          BIC
## mdl4_random_person_sensor_with_interact_reml      1 54  487.1302  803.4759
## mdl4_random_person_with_interact_reml            2 53 1215.2065 1525.6940
##              logLik    Test  L.Ratio p-value
## mdl4_random_person_sensor_with_interact_reml -189.5651
## mdl4_random_person_with_interact_reml        -554.6032 1 vs 2 730.0763  <.0001
```

#now see if the variance of the random effect is 0.

```
print(summary(mdl4_random_person_sensor_with_interact_reml))
```

```
## Linear mixed-effects model fit by REML
##   Data: df_clean
##       AIC       BIC    logLik
##  487.1302 803.4759 -189.5651
```

```

##
## Random effects:
## Formula: ~1 | person
## (Intercept)
## StdDev: 0.04079463
##
## Formula: ~1 | HF_type %in% person
## (Intercept) Residual
## StdDev: 0.1848808 0.2331964
##
## Fixed effects: log(HF_value) ~ (day) + (activity) + (HF_type) + (sex) + (session) + (HF_type *
## Value Std.Error DF t-value p-value
## (Intercept) 4.448996 0.06288118 2455 70.75243 0.0000
## day2 -0.119320 0.02635793 2455 -4.52692 0.0000
## activitystanding -0.092245 0.02563196 2455 -3.59882 0.0003
## activitywalking 0.042070 0.03228582 2455 1.30305 0.1927
## HF_typeHF_2 -0.715719 0.08678543 110 -8.24699 0.0000
## HF_typeHF_3 -0.630258 0.08715436 110 -7.23151 0.0000
## HF_typeHF_5 -0.404686 0.08724754 110 -4.63837 0.0000
## HF_typeHF_7 -0.153461 0.08668834 110 -1.77026 0.0795
## HF_typeHF_8 -0.895648 0.08686065 110 -10.31132 0.0000
## sexM 0.014990 0.08051215 22 0.18618 0.8540
## session2 -0.031633 0.03832987 2455 -0.82529 0.4093
## session3 -0.032927 0.03035050 2455 -1.08490 0.2781
## session4 -0.137739 0.03518841 2455 -3.91433 0.0001
## day2:HF_typeHF_2 0.516308 0.03205330 2455 16.10779 0.0000
## day2:HF_typeHF_3 0.506994 0.03242861 2455 15.63416 0.0000
## day2:HF_typeHF_5 0.432679 0.03274203 2455 13.21479 0.0000
## day2:HF_typeHF_7 0.349343 0.03179691 2455 10.98671 0.0000
## day2:HF_typeHF_8 0.494847 0.03215307 2455 15.39036 0.0000
## activitystanding:HF_typeHF_2 -0.088084 0.03591381 2455 -2.45265 0.0142
## activitywalking:HF_typeHF_2 -0.382098 0.04547871 2455 -8.40169 0.0000
## activitystanding:HF_typeHF_3 0.193346 0.03618677 2455 5.34299 0.0000
## activitywalking:HF_typeHF_3 0.227698 0.04573945 2455 4.97816 0.0000
## activitystanding:HF_typeHF_5 0.066375 0.03636007 2455 1.82548 0.0681
## activitywalking:HF_typeHF_5 -0.043252 0.04598851 2455 -0.94050 0.3471
## activitystanding:HF_typeHF_7 -0.038171 0.03578076 2455 -1.06679 0.2862
## activitywalking:HF_typeHF_7 -0.123810 0.04525319 2455 -2.73595 0.0063
## activitystanding:HF_typeHF_8 0.185465 0.03604682 2455 5.14511 0.0000
## activitywalking:HF_typeHF_8 0.165942 0.04562472 2455 3.63711 0.0003
## HF_typeHF_2:sexM -0.100771 0.11135545 110 -0.90495 0.3675
## HF_typeHF_3:sexM 0.061173 0.11145460 110 0.54886 0.5842
## HF_typeHF_5:sexM 0.002587 0.11154132 110 0.02319 0.9815
## HF_typeHF_7:sexM 0.105892 0.11127467 110 0.95162 0.3434
## HF_typeHF_8:sexM 0.065078 0.11138090 110 0.58428 0.5602
## HF_typeHF_2:session2 0.122040 0.05133469 2455 2.37734 0.0175
## HF_typeHF_3:session2 0.043086 0.05168385 2455 0.83365 0.4046
## HF_typeHF_5:session2 0.083605 0.05204825 2455 1.60630 0.1083
## HF_typeHF_7:session2 0.056986 0.05124801 2455 1.11197 0.2663
## HF_typeHF_8:session2 0.091408 0.05135613 2455 1.77989 0.0752
## HF_typeHF_2:session3 0.100253 0.03946126 2455 2.54055 0.0111
## HF_typeHF_3:session3 0.042140 0.03967419 2455 1.06215 0.2883
## HF_typeHF_5:session3 0.062095 0.03989708 2455 1.55637 0.1197
## HF_typeHF_7:session3 0.021430 0.03925766 2455 0.54589 0.5852

```

## HF_typeHF_8:session3	0.121427	0.03965542	2455	3.06206	0.0022	
## HF_typeHF_2:session4	0.276577	0.04617054	2455	5.99033	0.0000	
## HF_typeHF_3:session4	0.179282	0.04657722	2455	3.84913	0.0001	
## HF_typeHF_5:session4	0.194392	0.04668308	2455	4.16407	0.0000	
## HF_typeHF_7:session4	0.078603	0.04599982	2455	1.70877	0.0876	
## HF_typeHF_8:session4	0.227239	0.04618088	2455	4.92064	0.0000	
## day2:session2	-0.071051	0.02955827	2455	-2.40377	0.0163	
## day2:session3	-0.066144	0.02284600	2455	-2.89520	0.0038	
## day2:session4	-0.111003	0.02551616	2455	-4.35030	0.0000	
## Correlation:						
##	(Intr)	day2	actvtys	actvtyw	HF_tHF_2	HF_tHF_3
## day2	-0.218					
## activitystanding	-0.215	0.063				
## activitywalking	-0.233	0.021	0.399			
## HF_typeHF_2	-0.691	0.121	0.157	0.169		
## HF_typeHF_3	-0.688	0.119	0.156	0.168	0.499	
## HF_typeHF_5	-0.687	0.118	0.156	0.168	0.499	0.496
## HF_typeHF_7	-0.692	0.121	0.157	0.169	0.502	0.500
## HF_typeHF_8	-0.691	0.121	0.157	0.169	0.501	0.499
## sexM	-0.636	-0.012	-0.003	0.001	0.441	0.439
## session2	-0.269	0.202	0.103	0.280	0.180	0.179
## session3	-0.237	0.157	-0.006	-0.006	0.146	0.145
## session4	-0.252	0.137	0.000	0.261	0.161	0.160
## day2:HF_typeHF_2	0.138	-0.614	-0.062	-0.023	-0.200	-0.100
## day2:HF_typeHF_3	0.137	-0.607	-0.061	-0.023	-0.099	-0.209
## day2:HF_typeHF_5	0.135	-0.601	-0.061	-0.022	-0.098	-0.098
## day2:HF_typeHF_7	0.139	-0.620	-0.062	-0.023	-0.101	-0.100
## day2:HF_typeHF_8	0.138	-0.613	-0.062	-0.023	-0.100	-0.099
## activitystanding:HF_typeHF_2	0.154	-0.053	-0.709	-0.283	-0.222	-0.112
## activitywalking:HF_typeHF_2	0.165	-0.018	-0.281	-0.709	-0.237	-0.120
## activitystanding:HF_typeHF_3	0.153	-0.053	-0.704	-0.281	-0.111	-0.223
## activitywalking:HF_typeHF_3	0.164	-0.018	-0.280	-0.705	-0.119	-0.237
## activitystanding:HF_typeHF_5	0.153	-0.052	-0.700	-0.279	-0.111	-0.110
## activitywalking:HF_typeHF_5	0.164	-0.018	-0.278	-0.701	-0.119	-0.118
## activitystanding:HF_typeHF_7	0.155	-0.053	-0.712	-0.284	-0.112	-0.112
## activitywalking:HF_typeHF_7	0.166	-0.018	-0.283	-0.713	-0.121	-0.120
## activitystanding:HF_typeHF_8	0.154	-0.053	-0.706	-0.282	-0.112	-0.111
## activitywalking:HF_typeHF_8	0.165	-0.019	-0.280	-0.707	-0.120	-0.119
## HF_typeHF_2:sexM	0.440	0.008	0.002	0.000	-0.637	-0.318
## HF_typeHF_3:sexM	0.440	0.008	0.002	0.000	-0.319	-0.638
## HF_typeHF_5:sexM	0.440	0.008	0.002	0.000	-0.318	-0.317
## HF_typeHF_7:sexM	0.441	0.008	0.002	0.000	-0.319	-0.318
## HF_typeHF_8:sexM	0.440	0.008	0.002	0.000	-0.319	-0.318
## HF_typeHF_2:session2	0.186	-0.064	-0.092	-0.216	-0.267	-0.134
## HF_typeHF_3:session2	0.185	-0.062	-0.091	-0.214	-0.134	-0.268
## HF_typeHF_5:session2	0.183	-0.059	-0.091	-0.213	-0.133	-0.133
## HF_typeHF_7:session2	0.187	-0.067	-0.091	-0.216	-0.135	-0.135
## HF_typeHF_8:session2	0.186	-0.065	-0.092	-0.216	-0.135	-0.134
## HF_typeHF_2:session3	0.156	0.007	0.003	0.003	-0.226	-0.113
## HF_typeHF_3:session3	0.154	0.010	0.003	0.003	-0.112	-0.226
## HF_typeHF_5:session3	0.153	0.013	0.003	0.003	-0.112	-0.111
## HF_typeHF_7:session3	0.157	0.004	0.003	0.003	-0.114	-0.113
## HF_typeHF_8:session3	0.155	0.008	0.003	0.003	-0.113	-0.112
## HF_typeHF_2:session4	0.169	0.006	0.000	-0.199	-0.243	-0.122

## HF_typeHF_3:session4	0.168	0.007	0.000	-0.198	-0.122	-0.244
## HF_typeHF_5:session4	0.167	0.010	0.000	-0.197	-0.122	-0.121
## HF_typeHF_7:session4	0.171	0.003	0.000	-0.200	-0.123	-0.123
## HF_typeHF_8:session4	0.169	0.005	0.000	-0.199	-0.123	-0.122
## day2:session2	0.054	-0.331	0.073	0.031	0.002	0.003
## day2:session3	0.089	-0.431	0.004	0.005	0.002	0.004
## day2:session4	0.081	-0.389	0.000	0.002	0.001	0.001
##	HF_tHF_5	HF_tHF_7	HF_tHF_8	sexM	sessn2	sessn3
## day2						
## activitystanding						
## activitywalking						
## HF_typeHF_2						
## HF_typeHF_3						
## HF_typeHF_5						
## HF_typeHF_7	0.499					
## HF_typeHF_8	0.498	0.501				
## sexM	0.439	0.442	0.441			
## session2	0.178	0.181	0.180	0.002		
## session3	0.144	0.147	0.146	0.001	0.391	
## session4	0.159	0.161	0.161	0.002	0.413	0.427
## day2:HF_typeHF_2	-0.100	-0.100	-0.100	0.009	-0.072	0.007
## day2:HF_typeHF_3	-0.098	-0.099	-0.099	0.009	-0.071	0.007
## day2:HF_typeHF_5	-0.210	-0.098	-0.098	0.009	-0.070	0.007
## day2:HF_typeHF_7	-0.100	-0.197	-0.101	0.009	-0.073	0.007
## day2:HF_typeHF_8	-0.099	-0.100	-0.201	0.009	-0.072	0.005
## activitystanding:HF_typeHF_2	-0.111	-0.112	-0.112	0.002	-0.088	0.004
## activitywalking:HF_typeHF_2	-0.119	-0.120	-0.120	0.000	-0.205	0.004
## activitystanding:HF_typeHF_3	-0.111	-0.111	-0.111	0.002	-0.087	0.004
## activitywalking:HF_typeHF_3	-0.119	-0.119	-0.119	0.000	-0.204	0.004
## activitystanding:HF_typeHF_5	-0.224	-0.111	-0.111	0.002	-0.087	0.004
## activitywalking:HF_typeHF_5	-0.238	-0.119	-0.119	0.000	-0.202	0.004
## activitystanding:HF_typeHF_7	-0.112	-0.220	-0.112	0.002	-0.087	0.004
## activitywalking:HF_typeHF_7	-0.120	-0.236	-0.121	0.000	-0.205	0.004
## activitystanding:HF_typeHF_8	-0.111	-0.112	-0.223	0.002	-0.087	0.004
## activitywalking:HF_typeHF_8	-0.119	-0.120	-0.239	0.000	-0.204	0.003
## HF_typeHF_2:sexM	-0.317	-0.319	-0.319	-0.692	-0.002	-0.001
## HF_typeHF_3:sexM	-0.317	-0.319	-0.318	-0.691	-0.002	-0.001
## HF_typeHF_5:sexM	-0.635	-0.319	-0.318	-0.691	-0.002	-0.001
## HF_typeHF_7:sexM	-0.318	-0.640	-0.319	-0.693	-0.002	-0.001
## HF_typeHF_8:sexM	-0.317	-0.319	-0.638	-0.692	-0.002	-0.001
## HF_typeHF_2:session2	-0.134	-0.135	-0.135	-0.002	-0.681	-0.249
## HF_typeHF_3:session2	-0.133	-0.134	-0.134	-0.002	-0.675	-0.246
## HF_typeHF_5:session2	-0.269	-0.133	-0.133	-0.002	-0.668	-0.243
## HF_typeHF_7:session2	-0.135	-0.264	-0.135	-0.002	-0.684	-0.250
## HF_typeHF_8:session2	-0.134	-0.135	-0.268	-0.002	-0.681	-0.249
## HF_typeHF_2:session3	-0.112	-0.113	-0.113	-0.001	-0.257	-0.656
## HF_typeHF_3:session3	-0.112	-0.113	-0.112	-0.001	-0.255	-0.650
## HF_typeHF_5:session3	-0.227	-0.112	-0.112	-0.001	-0.252	-0.644
## HF_typeHF_7:session3	-0.113	-0.225	-0.114	-0.001	-0.259	-0.662
## HF_typeHF_8:session3	-0.112	-0.113	-0.221	-0.001	-0.255	-0.651
## HF_typeHF_2:session4	-0.122	-0.123	-0.123	-0.002	-0.277	-0.278
## HF_typeHF_3:session4	-0.121	-0.122	-0.122	-0.002	-0.274	-0.275
## HF_typeHF_5:session4	-0.245	-0.122	-0.122	-0.002	-0.273	-0.273
## HF_typeHF_7:session4	-0.123	-0.242	-0.123	-0.002	-0.279	-0.280

## HF_typeHF_8:session4	-0.122	-0.123	-0.244	-0.002	-0.277	-0.278
## day2:session2	0.004	0.001	0.002	0.001	-0.292	-0.146
## day2:session3	0.005	0.001	0.001	0.000	-0.148	-0.383
## day2:session4	0.004	0.000	0.001	0.001	-0.133	-0.169
##	sessn4	d2:HF_HF_2	d2:HF_HF_3	d2:HF_HF_5	d2:HF_HF_7	
## day2						
## activitystanding						
## activitywalking						
## HF_typeHF_2						
## HF_typeHF_3						
## HF_typeHF_5						
## HF_typeHF_7						
## HF_typeHF_8						
## sexM						
## session2						
## session3						
## session4						
## day2:HF_typeHF_2	0.004					
## day2:HF_typeHF_3	0.003	0.499				
## day2:HF_typeHF_5	0.004	0.495	0.489			
## day2:HF_typeHF_7	0.003	0.509	0.503	0.499		
## day2:HF_typeHF_8	0.004	0.504	0.498	0.493	0.508	
## activitystanding:HF_typeHF_2	0.000	0.084	0.044	0.043	0.044	
## activitywalking:HF_typeHF_2	-0.185	0.032	0.016	0.016	0.016	
## activitystanding:HF_typeHF_3	0.000	0.044	0.084	0.043	0.044	
## activitywalking:HF_typeHF_3	-0.184	0.016	0.032	0.016	0.016	
## activitystanding:HF_typeHF_5	0.000	0.044	0.043	0.084	0.044	
## activitywalking:HF_typeHF_5	-0.183	0.016	0.016	0.032	0.016	
## activitystanding:HF_typeHF_7	0.001	0.044	0.044	0.043	0.081	
## activitywalking:HF_typeHF_7	-0.186	0.016	0.016	0.016	0.031	
## activitystanding:HF_typeHF_8	0.000	0.044	0.043	0.043	0.044	
## activitywalking:HF_typeHF_8	-0.185	0.016	0.016	0.016	0.016	
## HF_typeHF_2:sexM	-0.001	-0.012	-0.007	-0.006	-0.007	
## HF_typeHF_3:sexM	-0.002	-0.007	-0.006	-0.006	-0.007	
## HF_typeHF_5:sexM	-0.001	-0.007	-0.007	-0.013	-0.007	
## HF_typeHF_7:sexM	-0.001	-0.007	-0.007	-0.006	-0.007	
## HF_typeHF_8:sexM	-0.001	-0.007	-0.007	-0.006	-0.007	
## HF_typeHF_2:session2	-0.272	0.110	0.053	0.053	0.055	
## HF_typeHF_3:session2	-0.269	0.054	0.110	0.053	0.054	
## HF_typeHF_5:session2	-0.266	0.053	0.053	0.109	0.054	
## HF_typeHF_7:session2	-0.273	0.054	0.054	0.053	0.111	
## HF_typeHF_8:session2	-0.272	0.054	0.053	0.053	0.055	
## HF_typeHF_2:session3	-0.281	-0.005	-0.005	-0.005	-0.005	
## HF_typeHF_3:session3	-0.278	-0.005	-0.005	-0.005	-0.005	
## HF_typeHF_5:session3	-0.276	-0.005	-0.005	-0.005	-0.005	
## HF_typeHF_7:session3	-0.283	-0.005	-0.005	-0.005	-0.005	
## HF_typeHF_8:session3	-0.279	-0.005	-0.005	-0.005	-0.005	
## HF_typeHF_2:session4	-0.658	-0.003	-0.003	-0.003	-0.003	
## HF_typeHF_3:session4	-0.652	-0.003	0.001	-0.003	-0.003	
## HF_typeHF_5:session4	-0.647	-0.003	-0.003	-0.003	-0.003	
## HF_typeHF_7:session4	-0.664	-0.003	-0.003	-0.003	-0.002	
## HF_typeHF_8:session4	-0.658	-0.003	-0.003	-0.003	-0.003	
## day2:session2	-0.125	-0.003	-0.004	-0.004	0.001	
## day2:session3	-0.162	-0.001	-0.001	-0.001	-0.001	

```

## day2:session4          -0.367  0.002      0.004      0.002      0.003
##                        d2:HF_HF_8 actvtys:HF_HF_2 actvtyw:HF_HF_2
## day2
## activitystanding
## activitywalking
## HF_typeHF_2
## HF_typeHF_3
## HF_typeHF_5
## HF_typeHF_7
## HF_typeHF_8
## sexM
## session2
## session3
## session4
## day2:HF_typeHF_2
## day2:HF_typeHF_3
## day2:HF_typeHF_5
## day2:HF_typeHF_7
## day2:HF_typeHF_8
## activitystanding:HF_typeHF_2  0.044
## activitywalking:HF_typeHF_2   0.016      0.396
## activitystanding:HF_typeHF_3  0.044      0.502      0.199
## activitywalking:HF_typeHF_3   0.016      0.200      0.501
## activitystanding:HF_typeHF_5  0.043      0.500      0.198
## activitywalking:HF_typeHF_5   0.016      0.199      0.498
## activitystanding:HF_typeHF_7  0.044      0.508      0.202
## activitywalking:HF_typeHF_7   0.016      0.202      0.506
## activitystanding:HF_typeHF_8  0.083      0.504      0.200
## activitywalking:HF_typeHF_8   0.035      0.200      0.502
## HF_typeHF_2:sexM             -0.007      -0.002      0.000
## HF_typeHF_3:sexM             -0.007      -0.002      0.000
## HF_typeHF_5:sexM             -0.007      -0.002      0.000
## HF_typeHF_7:sexM             -0.007      -0.002      0.000
## HF_typeHF_8:sexM             -0.012      -0.002      0.000
## HF_typeHF_2:session2         0.054      0.129      0.302
## HF_typeHF_3:session2         0.054      0.065      0.152
## HF_typeHF_5:session2         0.053      0.064      0.151
## HF_typeHF_7:session2         0.054      0.065      0.153
## HF_typeHF_8:session2         0.111      0.065      0.153
## HF_typeHF_2:session3        -0.005      -0.002      -0.002
## HF_typeHF_3:session3        -0.005      -0.002      -0.002
## HF_typeHF_5:session3        -0.005      -0.002      -0.002
## HF_typeHF_7:session3        -0.005      -0.002      -0.002
## HF_typeHF_8:session3        -0.010      -0.002      -0.002
## HF_typeHF_2:session4        -0.003      0.000      0.279
## HF_typeHF_3:session4        -0.003      0.000      0.140
## HF_typeHF_5:session4        -0.003      0.000      0.140
## HF_typeHF_7:session4        -0.003      0.000      0.142
## HF_typeHF_8:session4        -0.002      0.000      0.141
## day2:session2               -0.003      0.001      -0.001
## day2:session3                0.003      -0.003      -0.003
## day2:session4                0.002      0.000      -0.001
##                        actvtys:HF_HF_3 actvtyw:HF_HF_3 actvtys:HF_HF_5
## day2

```

```

## activitystanding
## activitywalking
## HF_typeHF_2
## HF_typeHF_3
## HF_typeHF_5
## HF_typeHF_7
## HF_typeHF_8
## sexM
## session2
## session3
## session4
## day2:HF_typeHF_2
## day2:HF_typeHF_3
## day2:HF_typeHF_5
## day2:HF_typeHF_7
## day2:HF_typeHF_8
## activitystanding:HF_typeHF_2
## activitywalking:HF_typeHF_2
## activitystanding:HF_typeHF_3
## activitywalking:HF_typeHF_3    0.396
## activitystanding:HF_typeHF_5    0.496          0.197
## activitywalking:HF_typeHF_5    0.197          0.495          0.396
## activitystanding:HF_typeHF_7    0.504          0.200          0.502
## activitywalking:HF_typeHF_7    0.200          0.503          0.199
## activitystanding:HF_typeHF_8    0.500          0.199          0.498
## activitywalking:HF_typeHF_8    0.199          0.499          0.198
## HF_typeHF_2:sexM          -0.002          0.000          -0.002
## HF_typeHF_3:sexM          -0.001          0.000          -0.001
## HF_typeHF_5:sexM          -0.002          0.000          -0.002
## HF_typeHF_7:sexM          -0.002          0.000          -0.002
## HF_typeHF_8:sexM          -0.002          0.000          -0.002
## HF_typeHF_2:session2          0.065          0.152          0.064
## HF_typeHF_3:session2          0.131          0.302          0.064
## HF_typeHF_5:session2          0.064          0.150          0.132
## HF_typeHF_7:session2          0.065          0.152          0.065
## HF_typeHF_8:session2          0.065          0.152          0.064
## HF_typeHF_2:session3          -0.002          -0.002          -0.002
## HF_typeHF_3:session3          -0.002          -0.002          -0.002
## HF_typeHF_5:session3          -0.002          -0.002          -0.002
## HF_typeHF_7:session3          -0.002          -0.002          -0.002
## HF_typeHF_8:session3          -0.002          -0.002          -0.002
## HF_typeHF_2:session4          0.000          0.141          0.000
## HF_typeHF_3:session4          0.000          0.278          0.000
## HF_typeHF_5:session4          0.000          0.139          0.000
## HF_typeHF_7:session4          0.000          0.141          0.000
## HF_typeHF_8:session4          0.000          0.141          0.000
## day2:session2          0.002          -0.001          0.002
## day2:session3          -0.003          -0.003          -0.003
## day2:session4          0.000          -0.001          0.000
##
##          actvttyw:HF_HF_5 actvttyw:HF_HF_7 actvttyw:HF_HF_7
## day2
## activitystanding
## activitywalking
## HF_typeHF_2

```



```

## HF_typeHF_3
## HF_typeHF_5
## HF_typeHF_7
## HF_typeHF_8
## sexM
## session2
## session3
## session4
## day2:HF_typeHF_2
## day2:HF_typeHF_3
## day2:HF_typeHF_5
## day2:HF_typeHF_7
## day2:HF_typeHF_8
## activitystanding:HF_typeHF_2
## activitywalking:HF_typeHF_2
## activitystanding:HF_typeHF_3
## activitywalking:HF_typeHF_3
## activitystanding:HF_typeHF_5
## activitywalking:HF_typeHF_5
## activitystanding:HF_typeHF_7 0.199
## activitywalking:HF_typeHF_7 0.500          0.396
## activitystanding:HF_typeHF_8 0.198          0.506          0.201
## activitywalking:HF_typeHF_8 0.496          0.201          0.504
## HF_typeHF_2:sexM          0.000          -0.002          0.000
## HF_typeHF_3:sexM          0.000          -0.002          0.000
## HF_typeHF_5:sexM          0.000          -0.002          0.000
## HF_typeHF_7:sexM          0.000          -0.001          0.000
## HF_typeHF_8:sexM          0.000          -0.002          0.000
## HF_typeHF_2:session2      0.151          0.066          0.154
## HF_typeHF_3:session2      0.150          0.065          0.153
## HF_typeHF_5:session2      0.303          0.065          0.152
## HF_typeHF_7:session2      0.152          0.128          0.301
## HF_typeHF_8:session2      0.151          0.066          0.154
## HF_typeHF_2:session3     -0.002          -0.002          -0.002
## HF_typeHF_3:session3     -0.002          -0.002          -0.002
## HF_typeHF_5:session3     -0.002          -0.002          -0.002
## HF_typeHF_7:session3     -0.002          -0.002          -0.002
## HF_typeHF_8:session3     -0.002          -0.002          -0.002
## HF_typeHF_2:session4      0.140          0.000          0.142
## HF_typeHF_3:session4      0.139          0.000          0.141
## HF_typeHF_5:session4      0.279          0.000          0.141
## HF_typeHF_7:session4      0.140          -0.002          0.278
## HF_typeHF_8:session4      0.140          0.000          0.142
## day2:session2             -0.001          -0.002          -0.002
## day2:session3             -0.003          -0.003          -0.003
## day2:session4             -0.001          -0.001          -0.002
##
##                                actvtys:HF_HF_8 actvtyw:HF_HF_8 HF_HF_2:M
## day2
## activitystanding
## activitywalking
## HF_typeHF_2
## HF_typeHF_3
## HF_typeHF_5
## HF_typeHF_7

```

```

## HF_typeHF_8
## sexM
## session2
## session3
## session4
## day2:HF_typeHF_2
## day2:HF_typeHF_3
## day2:HF_typeHF_5
## day2:HF_typeHF_7
## day2:HF_typeHF_8
## activitystanding:HF_typeHF_2
## activitywalking:HF_typeHF_2
## activitystanding:HF_typeHF_3
## activitywalking:HF_typeHF_3
## activitystanding:HF_typeHF_5
## activitywalking:HF_typeHF_5
## activitystanding:HF_typeHF_7
## activitywalking:HF_typeHF_7
## activitystanding:HF_typeHF_8
## activitywalking:HF_typeHF_8 0.397
## HF_typeHF_2:sexM -0.002 0.000
## HF_typeHF_3:sexM -0.002 0.000 0.500
## HF_typeHF_5:sexM -0.002 0.000 0.500
## HF_typeHF_7:sexM -0.002 0.000 0.501
## HF_typeHF_8:sexM 0.000 0.002 0.500
## HF_typeHF_2:session2 0.065 0.153 0.001
## HF_typeHF_3:session2 0.065 0.152 0.001
## HF_typeHF_5:session2 0.064 0.151 0.001
## HF_typeHF_7:session2 0.065 0.153 0.001
## HF_typeHF_8:session2 0.129 0.303 0.001
## HF_typeHF_2:session3 -0.002 -0.002 0.001
## HF_typeHF_3:session3 -0.002 -0.002 0.001
## HF_typeHF_5:session3 -0.002 -0.002 0.001
## HF_typeHF_7:session3 -0.002 -0.002 0.001
## HF_typeHF_8:session3 -0.004 -0.009 0.001
## HF_typeHF_2:session4 0.000 0.141 0.001
## HF_typeHF_3:session4 0.000 0.140 0.001
## HF_typeHF_5:session4 0.000 0.139 0.001
## HF_typeHF_7:session4 0.000 0.142 0.001
## HF_typeHF_8:session4 0.000 0.280 0.001
## day2:session2 0.002 day2:session2 0.000 -0.001
## day2:session3 -0.004 -0.001 0.000
## day2:session4 0.000 -0.001 0.000
## HF_HF_3:M HF_HF_5:M HF_HF_7:M HF_HF_8:M HF_HF_2:2
## day2
## activitystanding
## activitywalking
## HF_typeHF_2
## HF_typeHF_3
## HF_typeHF_5
## HF_typeHF_7
## HF_typeHF_8
## sexM
## session2

```

```

## session3
## session4
## day2:HF_typeHF_2
## day2:HF_typeHF_3
## day2:HF_typeHF_5
## day2:HF_typeHF_7
## day2:HF_typeHF_8
## activitystanding:HF_typeHF_2
## activitywalking:HF_typeHF_2
## activitystanding:HF_typeHF_3
## activitywalking:HF_typeHF_3
## activitystanding:HF_typeHF_5
## activitywalking:HF_typeHF_5
## activitystanding:HF_typeHF_7
## activitywalking:HF_typeHF_7
## activitystanding:HF_typeHF_8
## activitywalking:HF_typeHF_8
## HF_typeHF_2:sexM
## HF_typeHF_3:sexM
## HF_typeHF_5:sexM          0.499
## HF_typeHF_7:sexM          0.500      0.500
## HF_typeHF_8:sexM          0.500      0.499      0.501
## HF_typeHF_2:session2      0.001      0.001      0.001      0.001
## HF_typeHF_3:session2      0.002      0.001      0.001      0.001      0.505
## HF_typeHF_5:session2      0.001      0.001      0.001      0.001      0.502
## HF_typeHF_7:session2      0.001      0.001      0.000      0.001      0.510
## HF_typeHF_8:session2      0.001      0.001      0.001      0.002      0.509
## HF_typeHF_2:session3      0.001      0.001      0.001      0.001      0.383
## HF_typeHF_3:session3      0.001      0.001      0.001      0.001      0.191
## HF_typeHF_5:session3      0.001      0.001      0.001      0.001      0.190
## HF_typeHF_7:session3      0.001      0.001      0.001      0.001      0.193
## HF_typeHF_8:session3      0.001      0.001      0.001      -0.001      0.191
## HF_typeHF_2:session4      0.001      0.001      0.001      0.001      0.411
## HF_typeHF_3:session4      0.002      0.001      0.001      0.001      0.206
## HF_typeHF_5:session4      0.001      0.001      0.001      0.001      0.205
## HF_typeHF_7:session4      0.001      0.001      0.001      0.001      0.208
## HF_typeHF_8:session4      0.001      0.001      0.001      0.001      0.207
## day2:session2              -0.001      -0.001      -0.001      -0.001      -0.002
## day2:session3              0.000      0.000      0.000      0.000      -0.003
## day2:session4              0.000      0.000      -0.001      0.000      -0.002
##                            HF_HF_3:2 HF_HF_5:2 HF_HF_7:2 HF_HF_8:2 HF_HF_2:3
## day2
## activitystanding
## activitywalking
## HF_typeHF_2
## HF_typeHF_3
## HF_typeHF_5
## HF_typeHF_7
## HF_typeHF_8
## sexM
## session2
## session3
## session4
## day2:HF_typeHF_2

```

```

## day2:HF_typeHF_3
## day2:HF_typeHF_5
## day2:HF_typeHF_7
## day2:HF_typeHF_8
## activitystanding:HF_typeHF_2
## activitywalking:HF_typeHF_2
## activitystanding:HF_typeHF_3
## activitywalking:HF_typeHF_3
## activitystanding:HF_typeHF_5
## activitywalking:HF_typeHF_5
## activitystanding:HF_typeHF_7
## activitywalking:HF_typeHF_7
## activitystanding:HF_typeHF_8
## activitywalking:HF_typeHF_8
## HF_typeHF_2:sexM
## HF_typeHF_3:sexM
## HF_typeHF_5:sexM
## HF_typeHF_7:sexM
## HF_typeHF_8:sexM
## HF_typeHF_2:session2
## HF_typeHF_3:session2
## HF_typeHF_5:session2      0.499
## HF_typeHF_7:session2      0.506      0.503
## HF_typeHF_8:session2      0.505      0.502      0.509
## HF_typeHF_2:session3      0.191      0.190      0.192      0.192
## HF_typeHF_3:session3      0.382      0.189      0.191      0.191      0.502
## HF_typeHF_5:session3      0.189      0.382      0.190      0.190      0.499
## HF_typeHF_7:session3      0.192      0.190      0.381      0.193      0.507
## HF_typeHF_8:session3      0.190      0.189      0.192      0.378      0.502
## HF_typeHF_2:session4      0.206      0.205      0.208      0.207      0.426
## HF_typeHF_3:session4      0.409      0.203      0.206      0.206      0.213
## HF_typeHF_5:session4      0.204      0.410      0.206      0.205      0.212
## HF_typeHF_7:session4      0.207      0.205      0.409      0.208      0.215
## HF_typeHF_8:session4      0.206      0.205      0.208      0.411      0.215
## day2:session2      -0.007      -0.013      0.006      -0.002      -0.001
## day2:session3      -0.005      -0.008      0.000      -0.002      -0.001
## day2:session4      -0.004      -0.006      0.000      -0.002      -0.002
##      HF_HF_3:3 HF_HF_5:3 HF_HF_7:3 HF_HF_8:3 HF_HF_2:4
## day2
## activitystanding
## activitywalking
## HF_typeHF_2
## HF_typeHF_3
## HF_typeHF_5
## HF_typeHF_7
## HF_typeHF_8
## sexM
## session2
## session3
## session4
## day2:HF_typeHF_2
## day2:HF_typeHF_3
## day2:HF_typeHF_5
## day2:HF_typeHF_7

```

```

## day2:HF_typeHF_8
## activitystanding:HF_typeHF_2
## activitywalking:HF_typeHF_2
## activitystanding:HF_typeHF_3
## activitywalking:HF_typeHF_3
## activitystanding:HF_typeHF_5
## activitywalking:HF_typeHF_5
## activitystanding:HF_typeHF_7
## activitywalking:HF_typeHF_7
## activitystanding:HF_typeHF_8
## activitywalking:HF_typeHF_8
## HF_typeHF_2:sexM
## HF_typeHF_3:sexM
## HF_typeHF_5:sexM
## HF_typeHF_7:sexM
## HF_typeHF_8:sexM
## HF_typeHF_2:session2
## HF_typeHF_3:session2
## HF_typeHF_5:session2
## HF_typeHF_7:session2
## HF_typeHF_8:session2
## HF_typeHF_2:session3
## HF_typeHF_3:session3
## HF_typeHF_5:session3      0.497
## HF_typeHF_7:session3      0.505      0.502
## HF_typeHF_8:session3      0.500      0.497      0.505
## HF_typeHF_2:session4      0.213      0.212      0.216      0.214
## HF_typeHF_3:session4      0.425      0.210      0.214      0.212      0.498
## HF_typeHF_5:session4      0.211      0.426      0.213      0.211      0.497
## HF_typeHF_7:session4      0.214      0.213      0.426      0.214      0.504
## HF_typeHF_8:session4      0.213      0.212      0.216      0.422      0.502
## day2:session2      -0.004      -0.006      0.001      -0.002      -0.002
## day2:session3      -0.007      -0.014      0.005      -0.004      -0.003
## day2:session4      -0.005      -0.008      0.001      -0.002      -0.002
##      HF_HF_3:4 HF_HF_5:4 HF_HF_7:4 HF_HF_8:4 dy2:s2
## day2
## activitystanding
## activitywalking
## HF_typeHF_2
## HF_typeHF_3
## HF_typeHF_5
## HF_typeHF_7
## HF_typeHF_8
## sexM
## session2
## session3
## session4
## day2:HF_typeHF_2
## day2:HF_typeHF_3
## day2:HF_typeHF_5
## day2:HF_typeHF_7
## day2:HF_typeHF_8
## activitystanding:HF_typeHF_2
## activitywalking:HF_typeHF_2

```

```

## activitystanding:HF_typeHF_3
## activitywalking:HF_typeHF_3
## activitystanding:HF_typeHF_5
## activitywalking:HF_typeHF_5
## activitystanding:HF_typeHF_7
## activitywalking:HF_typeHF_7
## activitystanding:HF_typeHF_8
## activitywalking:HF_typeHF_8
## HF_typeHF_2:sexM
## HF_typeHF_3:sexM
## HF_typeHF_5:sexM
## HF_typeHF_7:sexM
## HF_typeHF_8:sexM
## HF_typeHF_2:session2
## HF_typeHF_3:session2
## HF_typeHF_5:session2
## HF_typeHF_7:session2
## HF_typeHF_8:session2
## HF_typeHF_2:session3
## HF_typeHF_3:session3
## HF_typeHF_5:session3
## HF_typeHF_7:session3
## HF_typeHF_8:session3
## HF_typeHF_2:session4
## HF_typeHF_3:session4
## HF_typeHF_5:session4      0.493
## HF_typeHF_7:session4      0.500      0.499
## HF_typeHF_8:session4      0.498      0.497      0.504
## day2:session2             -0.004      -0.006      0.000      -0.002
## day2:session3             -0.005      -0.008      0.000      -0.002      0.385
## day2:session4             -0.004      -0.015      0.006      -0.002      0.345
##                           dy2:s3
## day2
## activitystanding
## activitywalking
## HF_typeHF_2
## HF_typeHF_3
## HF_typeHF_5
## HF_typeHF_7
## HF_typeHF_8
## sexM
## session2
## session3
## session4
## day2:HF_typeHF_2
## day2:HF_typeHF_3
## day2:HF_typeHF_5
## day2:HF_typeHF_7
## day2:HF_typeHF_8
## activitystanding:HF_typeHF_2
## activitywalking:HF_typeHF_2
## activitystanding:HF_typeHF_3
## activitywalking:HF_typeHF_3
## activitystanding:HF_typeHF_5

```

```

## activitywalking:HF_typeHF_5
## activitystanding:HF_typeHF_7
## activitywalking:HF_typeHF_7
## activitystanding:HF_typeHF_8
## activitywalking:HF_typeHF_8
## HF_typeHF_2:sexM
## HF_typeHF_3:sexM
## HF_typeHF_5:sexM
## HF_typeHF_7:sexM
## HF_typeHF_8:sexM
## HF_typeHF_2:session2
## HF_typeHF_3:session2
## HF_typeHF_5:session2
## HF_typeHF_7:session2
## HF_typeHF_8:session2
## HF_typeHF_2:session3
## HF_typeHF_3:session3
## HF_typeHF_5:session3
## HF_typeHF_7:session3
## HF_typeHF_8:session3
## HF_typeHF_2:session4
## HF_typeHF_3:session4
## HF_typeHF_5:session4
## HF_typeHF_7:session4
## HF_typeHF_8:session4
## day2:session2
## day2:session3
## day2:session4          0.446
##
## Standardized Within-Group Residuals:
##      Min      Q1      Med      Q3      Max
## -6.45127993 -0.42906957  0.05405537  0.52729493  4.00105074
##
## Number of Observations: 2638
## Number of Groups:
##           person HF_type %in% person
##           24          144

```

#it is not! actually, sensor is more important than person.

0.10 Section 6.5: Check other random intercepts.

```

mdl4_random_person_sensor_activity_with_interact_reml <- lme(
  fixed = log(HF_value) ~ (day) + (activity) + (HF_type) + (sex) + (session) + (HF_type * day) + (HF_type * activity),
  random = ~ 1 | person/HF_type/activity,
  data = df_clean,
  method = "REML"
)

#anova to compare
print(anova(mdl4_random_person_sensor_with_interact_reml, mdl4_random_person_sensor_activity_with_interact_reml))

##           Model df      AIC
## mdl4_random_person_sensor_with_interact_reml      1 54 487.1302

```

```

## mdl4_random_person_sensor_activity_with_interact_reml      2 55 471.4760
##                                     BIC      logLik      Test
## mdl4_random_person_sensor_with_interact_reml              803.4759 -189.5651
## mdl4_random_person_sensor_activity_with_interact_reml      793.6800 -180.7380 1 vs 2
##                                     L.Ratio p-value
## mdl4_random_person_sensor_with_interact_reml
## mdl4_random_person_sensor_activity_with_interact_reml      17.65421 <.0001

mdl4_random_person_sensor_day_with_interact_reml<- lme(
  fixed = log(HF_value) ~ (day) + (activity) + (HF_type) + (sex) + (session) + (HF_type * day) + (HF_type *
  random = ~ 1 | person/HF_type/day,
  data = df_clean,
  method = "REML"
)

#how much of variance does this explain
print(summary(mdl4_random_person_sensor_day_with_interact_reml))

## Linear mixed-effects model fit by REML
##   Data: df_clean
##       AIC      BIC    logLik
##   -602.3829 -280.1789 356.1914
##
## Random effects:
## Formula: ~1 | person
##      (Intercept)
## StdDev:  0.04454982
##
## Formula: ~1 | HF_type %in% person
##      (Intercept)
## StdDev:   0.1002826
##
## Formula: ~1 | day %in% HF_type %in% person
##      (Intercept) Residual
## StdDev:    0.221534 0.1733527
##
## Fixed effects: log(HF_value) ~ (day) + (activity) + (HF_type) + (sex) + (session) + (HF_type *
##                                     Value Std.Error   DF   t-value p-value
## (Intercept)                4.454162 0.06805956 2325   65.44505  0.0000
## day2                -0.110969 0.06765631  130   -1.64018  0.1034
## activitystanding    -0.093504 0.01906490 2325   -4.90452  0.0000
## activitywalking      0.036583 0.02401608 2325    1.52327  0.1278
## HF_typeHF_2         -0.716179 0.09422299  110   -7.60089  0.0000
## HF_typeHF_3         -0.634013 0.09564929  110   -6.62852  0.0000
## HF_typeHF_5         -0.414878 0.09587380  110   -4.32734  0.0000
## HF_typeHF_7         -0.164328 0.09403604  110   -1.74750  0.0833
## HF_typeHF_8         -0.904035 0.09426315  110   -9.59054  0.0000
## sexM                  0.002089 0.08050778   22    0.02595  0.9795
## session2             -0.038156 0.02852481 2325   -1.33765  0.1811
## session3             -0.032676 0.02256897 2325   -1.44783  0.1478
## session4             -0.139695 0.02617555 2325   -5.33687  0.0000
## day2:HF_typeHF_2      0.497832 0.09462672  130    5.26101  0.0000
## day2:HF_typeHF_3      0.496711 0.09526732  130    5.21387  0.0000
## day2:HF_typeHF_5      0.429309 0.09597158  130    4.47330  0.0000
## day2:HF_typeHF_7      0.339162 0.09403119  130    3.60691  0.0004

```



```

## day2:HF_typeHF_8      0.488259 0.09464642 130 5.15877 0.0000
## activitystanding:HF_typeHF_2 -0.086842 0.02670547 2325 -3.25185 0.0012
## activitywalking:HF_typeHF_2 -0.376620 0.03381880 2325 -11.13640 0.0000
## activitystanding:HF_typeHF_3 0.194346 0.02690835 2325 7.22250 0.0000
## activitywalking:HF_typeHF_3 0.233056 0.03401257 2325 6.85205 0.0000
## activitystanding:HF_typeHF_5 0.067616 0.02703711 2325 2.50087 0.0125
## activitywalking:HF_typeHF_5 -0.037774 0.03419766 2325 -1.10457 0.2695
## activitystanding:HF_typeHF_7 -0.033888 0.02660941 2325 -1.27355 0.2030
## activitywalking:HF_typeHF_7 -0.116812 0.03365178 2325 -3.47119 0.0005
## activitystanding:HF_typeHF_8 0.186278 0.02680763 2325 6.94868 0.0000
## activitywalking:HF_typeHF_8 0.172892 0.03393485 2325 5.09481 0.0000
## HF_typeHF_2:sexM      -0.075404 0.11088593 110 -0.68002 0.4979
## HF_typeHF_3:sexM      0.074464 0.11138169 110 0.66854 0.5052
## HF_typeHF_5:sexM      0.017129 0.11192745 110 0.15304 0.8786
## HF_typeHF_7:sexM      0.129002 0.11037640 110 1.16875 0.2450
## HF_typeHF_8:sexM      0.082882 0.11090063 110 0.74736 0.4564
## HF_typeHF_2:session2  0.129464 0.03818680 2325 3.39028 0.0007
## HF_typeHF_3:session2  0.050049 0.03844634 2325 1.30179 0.1931
## HF_typeHF_5:session2  0.091070 0.03871705 2325 2.35220 0.0187
## HF_typeHF_7:session2  0.059399 0.03813281 2325 1.55768 0.1194
## HF_typeHF_8:session2  0.099502 0.03820388 2325 2.60449 0.0093
## HF_typeHF_2:session3  0.100263 0.02933949 2325 3.41734 0.0006
## HF_typeHF_3:session3  0.042155 0.02949774 2325 1.42909 0.1531
## HF_typeHF_5:session3  0.062116 0.02966341 2325 2.09403 0.0364
## HF_typeHF_7:session3  0.021435 0.02918817 2325 0.73436 0.4628
## HF_typeHF_8:session3  0.119189 0.02949510 2325 4.04099 0.0001
## HF_typeHF_2:session4  0.281439 0.03433849 2325 8.19602 0.0000
## HF_typeHF_3:session4  0.177909 0.03465690 2325 5.13345 0.0000
## HF_typeHF_5:session4  0.199383 0.03471950 2325 5.74269 0.0000
## HF_typeHF_7:session4  0.075709 0.03421467 2325 2.21276 0.0270
## HF_typeHF_8:session4  0.232664 0.03434677 2325 6.77396 0.0000
## day2:session2         -0.072883 0.02197988 2325 -3.31591 0.0009
## day2:session3         -0.066654 0.01698627 2325 -3.92401 0.0001
## day2:session4         -0.116694 0.01897343 2325 -6.15037 0.0000
## Correlation:
## (Intr) day2 actvtys actvtyw HF_tHF_2 HF_tHF_3
## day2 -0.499
## activitystanding -0.148 0.019
## activitywalking -0.160 0.006 0.398
## HF_typeHF_2 -0.693 0.353 0.107 0.116
## HF_typeHF_3 -0.683 0.347 0.106 0.114 0.493
## HF_typeHF_5 -0.681 0.346 0.106 0.114 0.492 0.485
## HF_typeHF_7 -0.694 0.354 0.108 0.116 0.502 0.494
## HF_typeHF_8 -0.693 0.353 0.107 0.116 0.500 0.493
## sexM -0.571 -0.021 -0.002 0.001 0.391 0.385
## session2 -0.185 0.058 0.102 0.281 0.124 0.122
## session3 -0.163 0.045 -0.006 -0.006 0.100 0.098
## session4 -0.174 0.039 -0.001 0.262 0.111 0.109
## day2:HF_typeHF_2 0.349 -0.700 -0.016 -0.006 -0.505 -0.249
## day2:HF_typeHF_3 0.347 -0.695 -0.016 -0.006 -0.251 -0.513
## day2:HF_typeHF_5 0.344 -0.690 -0.016 -0.006 -0.249 -0.245
## day2:HF_typeHF_7 0.352 -0.704 -0.016 -0.006 -0.254 -0.250
## day2:HF_typeHF_8 0.349 -0.700 -0.016 -0.006 -0.252 -0.249
## activitystanding:HF_typeHF_2 0.106 -0.016 -0.709 -0.282 -0.152 -0.075

```

```

## activitywalking:HF_typeHF_2  0.114 -0.005 -0.281 -0.709 -0.162 -0.081
## activitystanding:HF_typeHF_3 0.105 -0.015 -0.704 -0.280 -0.076 -0.151
## activitywalking:HF_typeHF_3  0.113 -0.005 -0.279 -0.705 -0.082 -0.161
## activitystanding:HF_typeHF_5 0.105 -0.015 -0.700 -0.279 -0.076 -0.075
## activitywalking:HF_typeHF_5  0.113 -0.005 -0.278 -0.702 -0.082 -0.080
## activitystanding:HF_typeHF_7 0.106 -0.015 -0.712 -0.284 -0.077 -0.076
## activitywalking:HF_typeHF_7  0.114 -0.005 -0.282 -0.713 -0.083 -0.082
## activitystanding:HF_typeHF_8 0.106 -0.015 -0.706 -0.281 -0.076 -0.075
## activitywalking:HF_typeHF_8  0.114 -0.005 -0.280 -0.707 -0.082 -0.081
## HF_typeHF_2:sexM              0.392  0.016  0.002 -0.001 -0.567 -0.279
## HF_typeHF_3:sexM              0.391  0.016  0.002 -0.001 -0.282 -0.572
## HF_typeHF_5:sexM              0.389  0.015  0.002 -0.001 -0.281 -0.277
## HF_typeHF_7:sexM              0.394  0.016  0.002 -0.001 -0.285 -0.281
## HF_typeHF_8:sexM              0.392  0.016  0.002 -0.001 -0.283 -0.279
## HF_typeHF_2:session2          0.128 -0.018 -0.091 -0.216 -0.183 -0.091
## HF_typeHF_3:session2          0.127 -0.018 -0.091 -0.215 -0.092 -0.182
## HF_typeHF_5:session2          0.126 -0.017 -0.091 -0.214 -0.091 -0.090
## HF_typeHF_7:session2          0.129 -0.019 -0.091 -0.216 -0.093 -0.091
## HF_typeHF_8:session2          0.128 -0.018 -0.091 -0.216 -0.093 -0.091
## HF_typeHF_2:session3          0.107  0.002  0.004  0.003 -0.155 -0.076
## HF_typeHF_3:session3          0.106  0.003  0.004  0.003 -0.077 -0.153
## HF_typeHF_5:session3          0.105  0.004  0.004  0.003 -0.077 -0.075
## HF_typeHF_7:session3          0.108  0.001  0.004  0.003 -0.078 -0.077
## HF_typeHF_8:session3          0.106  0.003  0.004  0.003 -0.077 -0.076
## HF_typeHF_2:session4          0.117  0.002  0.001 -0.200 -0.167 -0.083
## HF_typeHF_3:session4          0.115  0.002  0.001 -0.198 -0.084 -0.166
## HF_typeHF_5:session4          0.115  0.003  0.001 -0.198 -0.084 -0.082
## HF_typeHF_7:session4          0.118  0.001  0.001 -0.201 -0.085 -0.083
## HF_typeHF_8:session4          0.117  0.002  0.001 -0.200 -0.084 -0.083
## day2:session2                 0.037 -0.096  0.072  0.031  0.001  0.002
## day2:session3                 0.061 -0.125  0.004  0.005  0.002  0.003
## day2:session4                 0.056 -0.113 -0.001  0.002  0.001  0.001
##                               HF_tHF_5 HF_tHF_7 HF_tHF_8 sexM  sessn2 sessn3
## day2
## activitystanding
## activitywalking
## HF_typeHF_2
## HF_typeHF_3
## HF_typeHF_5
## HF_typeHF_7          0.493
## HF_typeHF_8          0.492    0.501
## sexM                 0.384    0.391    0.390
## session2             0.121    0.124    0.124    0.002
## session3             0.098    0.101    0.100    0.001    0.391
## session4             0.108    0.111    0.110    0.002    0.413    0.427
## day2:HF_typeHF_2     -0.248 -0.253 -0.252    0.015 -0.018    0.002
## day2:HF_typeHF_3     -0.246 -0.251 -0.251    0.015 -0.018    0.002
## day2:HF_typeHF_5     -0.516 -0.249 -0.249    0.015 -0.017    0.002
## day2:HF_typeHF_7     -0.250 -0.502 -0.254    0.015 -0.018    0.002
## day2:HF_typeHF_8     -0.248 -0.253 -0.505    0.015 -0.018    0.001
## activitystanding:HF_typeHF_2 -0.075 -0.077 -0.077    0.002 -0.087    0.005
## activitywalking:HF_typeHF_2 -0.081 -0.083 -0.082 -0.001 -0.205    0.004
## activitystanding:HF_typeHF_3 -0.075 -0.076 -0.076    0.002 -0.087    0.005
## activitywalking:HF_typeHF_3 -0.081 -0.082 -0.082 -0.001 -0.204    0.004

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## activitystanding:HF_typeHF_5 -0.151 -0.076 -0.076 0.002 -0.086 0.005
## activitywalking:HF_typeHF_5 -0.161 -0.082 -0.081 -0.001 -0.203 0.004
## activitystanding:HF_typeHF_7 -0.076 -0.151 -0.077 0.002 -0.087 0.005
## activitywalking:HF_typeHF_7 -0.081 -0.162 -0.083 -0.001 -0.206 0.004
## activitystanding:HF_typeHF_8 -0.075 -0.076 -0.153 0.002 -0.087 0.005
## activitywalking:HF_typeHF_8 -0.081 -0.082 -0.164 -0.001 -0.205 0.003
## HF_typeHF_2:sexM -0.279 -0.284 -0.283 -0.689 -0.002 -0.001
## HF_typeHF_3:sexM -0.277 -0.283 -0.282 -0.686 -0.002 -0.001
## HF_typeHF_5:sexM -0.561 -0.281 -0.281 -0.683 -0.002 -0.001
## HF_typeHF_7:sexM -0.280 -0.576 -0.285 -0.692 -0.001 -0.001
## HF_typeHF_8:sexM -0.278 -0.284 -0.567 -0.689 -0.002 -0.001
## HF_typeHF_2:session2 -0.091 -0.093 -0.093 -0.002 -0.681 -0.249
## HF_typeHF_3:session2 -0.091 -0.092 -0.092 -0.002 -0.675 -0.246
## HF_typeHF_5:session2 -0.182 -0.092 -0.091 -0.002 -0.669 -0.243
## HF_typeHF_7:session2 -0.091 -0.181 -0.093 -0.002 -0.685 -0.250
## HF_typeHF_8:session2 -0.091 -0.093 -0.184 -0.002 -0.681 -0.249
## HF_typeHF_2:session3 -0.076 -0.078 -0.077 -0.001 -0.257 -0.656
## HF_typeHF_3:session3 -0.076 -0.077 -0.077 -0.001 -0.254 -0.650
## HF_typeHF_5:session3 -0.154 -0.077 -0.077 -0.001 -0.252 -0.644
## HF_typeHF_7:session3 -0.077 -0.154 -0.078 -0.001 -0.259 -0.662
## HF_typeHF_8:session3 -0.076 -0.077 -0.151 -0.001 -0.255 -0.651
## HF_typeHF_2:session4 -0.083 -0.085 -0.084 -0.001 -0.278 -0.278
## HF_typeHF_3:session4 -0.082 -0.084 -0.084 -0.001 -0.274 -0.275
## HF_typeHF_5:session4 -0.166 -0.084 -0.084 -0.001 -0.273 -0.273
## HF_typeHF_7:session4 -0.083 -0.166 -0.085 -0.001 -0.279 -0.280
## HF_typeHF_8:session4 -0.083 -0.085 -0.167 -0.001 -0.278 -0.278
## day2:session2 0.003 0.001 0.001 0.001 -0.291 -0.146
## day2:session3 0.004 0.001 0.001 0.000 -0.148 -0.383
## day2:session4 0.003 0.000 0.001 0.001 -0.133 -0.169
## sessn4 d2:HF_HF_2 d2:HF_HF_3 d2:HF_HF_5 d2:HF_HF_7
## day2
## activitystanding
## activitywalking
## HF_typeHF_2
## HF_typeHF_3
## HF_typeHF_5
## HF_typeHF_7
## HF_typeHF_8
## sexM
## session2
## session3
## session4
## day2:HF_typeHF_2 0.001
## day2:HF_typeHF_3 0.001 0.497
## day2:HF_typeHF_5 0.001 0.494 0.490
## day2:HF_typeHF_7 0.001 0.504 0.500 0.496
## day2:HF_typeHF_8 0.001 0.500 0.497 0.494 0.503
## activitystanding:HF_typeHF_2 0.001 0.021 0.011 0.011 0.011
## activitywalking:HF_typeHF_2 -0.186 0.008 0.004 0.004 0.004
## activitystanding:HF_typeHF_3 0.000 0.011 0.021 0.011 0.011
## activitywalking:HF_typeHF_3 -0.185 0.004 0.008 0.004 0.004
## activitystanding:HF_typeHF_5 0.001 0.011 0.011 0.021 0.011
## activitywalking:HF_typeHF_5 -0.184 0.004 0.004 0.008 0.004
## activitystanding:HF_typeHF_7 0.001 0.011 0.011 0.011 0.020

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## activitywalking:HF_typeHF_7 -0.186 0.004 0.004 0.004 0.008
## activitystanding:HF_typeHF_8 0.001 0.011 0.011 0.011 0.011
## activitywalking:HF_typeHF_8 -0.185 0.004 0.004 0.004 0.004
## HF_typeHF_2:sexM -0.001 -0.022 -0.011 -0.011 -0.011
## HF_typeHF_3:sexM -0.001 -0.011 -0.011 -0.011 -0.011
## HF_typeHF_5:sexM -0.001 -0.011 -0.011 -0.023 -0.011
## HF_typeHF_7:sexM -0.001 -0.011 -0.011 -0.011 -0.011
## HF_typeHF_8:sexM -0.001 -0.011 -0.011 -0.011 -0.011
## HF_typeHF_2:session2 -0.272 0.027 0.013 0.013 0.014
## HF_typeHF_3:session2 -0.270 0.013 0.028 0.013 0.013
## HF_typeHF_5:session2 -0.267 0.013 0.013 0.028 0.013
## HF_typeHF_7:session2 -0.273 0.013 0.013 0.013 0.028
## HF_typeHF_8:session2 -0.272 0.013 0.013 0.013 0.014
## HF_typeHF_2:session3 -0.281 -0.001 -0.001 -0.001 -0.001
## HF_typeHF_3:session3 -0.278 -0.001 -0.001 -0.001 -0.001
## HF_typeHF_5:session3 -0.276 -0.001 -0.001 -0.001 -0.001
## HF_typeHF_7:session3 -0.283 -0.001 -0.001 -0.001 -0.001
## HF_typeHF_8:session3 -0.280 -0.001 -0.001 -0.001 -0.001
## HF_typeHF_2:session4 -0.659 -0.001 -0.001 -0.001 -0.001
## HF_typeHF_3:session4 -0.652 -0.001 0.000 -0.001 -0.001
## HF_typeHF_5:session4 -0.647 -0.001 -0.001 -0.001 -0.001
## HF_typeHF_7:session4 -0.664 -0.001 -0.001 -0.001 -0.001
## HF_typeHF_8:session4 -0.659 -0.001 -0.001 -0.001 -0.001
## day2:session2 -0.125 -0.001 -0.001 -0.001 0.000
## day2:session3 -0.161 0.000 0.000 0.000 0.000
## day2:session4 -0.367 0.001 0.001 0.001 0.001
## d2:HF_HF_8 actvtys:HF_HF_2 actvtyw:HF_HF_2
## day2
## activitystanding
## activitywalking
## HF_typeHF_2
## HF_typeHF_3
## HF_typeHF_5
## HF_typeHF_7
## HF_typeHF_8
## sexM
## session2
## session3
## session4
## day2:HF_typeHF_2
## day2:HF_typeHF_3
## day2:HF_typeHF_5
## day2:HF_typeHF_7
## day2:HF_typeHF_8
## activitystanding:HF_typeHF_2 0.011
## activitywalking:HF_typeHF_2 0.004 0.395
## activitystanding:HF_typeHF_3 0.011 0.503 0.199
## activitywalking:HF_typeHF_3 0.004 0.199 0.501
## activitystanding:HF_typeHF_5 0.011 0.500 0.198
## activitywalking:HF_typeHF_5 0.004 0.198 0.498
## activitystanding:HF_typeHF_7 0.011 0.508 0.201
## activitywalking:HF_typeHF_7 0.004 0.202 0.506
## activitystanding:HF_typeHF_8 0.021 0.505 0.200
## activitywalking:HF_typeHF_8 0.009 0.200 0.502

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## HF_typeHF_2:sexM	-0.011	-0.001	0.000
## HF_typeHF_3:sexM	-0.011	-0.001	0.000
## HF_typeHF_5:sexM	-0.011	-0.001	0.000
## HF_typeHF_7:sexM	-0.011	-0.001	0.000
## HF_typeHF_8:sexM	-0.022	-0.001	0.000
## HF_typeHF_2:session2	0.013	0.128	0.302
## HF_typeHF_3:session2	0.013	0.065	0.152
## HF_typeHF_5:session2	0.013	0.064	0.151
## HF_typeHF_7:session2	0.013	0.065	0.154
## HF_typeHF_8:session2	0.028	0.065	0.153
## HF_typeHF_2:session3	-0.001	-0.003	-0.002
## HF_typeHF_3:session3	-0.001	-0.003	-0.002
## HF_typeHF_5:session3	-0.001	-0.003	-0.002
## HF_typeHF_7:session3	-0.001	-0.003	-0.002
## HF_typeHF_8:session3	-0.003	-0.003	-0.002
## HF_typeHF_2:session4	-0.001	-0.001	0.280
## HF_typeHF_3:session4	-0.001	0.000	0.141
## HF_typeHF_5:session4	-0.001	-0.001	0.140
## HF_typeHF_7:session4	-0.001	-0.001	0.143
## HF_typeHF_8:session4	-0.001	-0.001	0.142
## day2:session2	-0.001	0.001	-0.001
## day2:session3	0.001	-0.003	-0.004
## day2:session4	0.001	0.000	-0.002
##	actvtys:HF_HF_3 actvtyw:HF_HF_3 actvtys:HF_HF_5		
## day2			
## activitystanding			
## activitywalking			
## HF_typeHF_2			
## HF_typeHF_3			
## HF_typeHF_5			
## HF_typeHF_7			
## HF_typeHF_8			
## sexM			
## session2			
## session3			
## session4			
## day2:HF_typeHF_2			
## day2:HF_typeHF_3			
## day2:HF_typeHF_5			
## day2:HF_typeHF_7			
## day2:HF_typeHF_8			
## activitystanding:HF_typeHF_2			
## activitywalking:HF_typeHF_2			
## activitystanding:HF_typeHF_3			
## activitywalking:HF_typeHF_3	0.396		
## activitystanding:HF_typeHF_5	0.496	0.197	
## activitywalking:HF_typeHF_5	0.197	0.495	0.396
## activitystanding:HF_typeHF_7	0.504	0.200	0.502
## activitywalking:HF_typeHF_7	0.200	0.503	0.199
## activitystanding:HF_typeHF_8	0.501	0.199	0.498
## activitywalking:HF_typeHF_8	0.198	0.499	0.197
## HF_typeHF_2:sexM	-0.001	0.000	-0.001
## HF_typeHF_3:sexM	-0.001	0.000	-0.001
## HF_typeHF_5:sexM	-0.001	0.000	-0.001

## HF_typeHF_7:sexM	-0.001	0.000	-0.001
## HF_typeHF_8:sexM	-0.001	0.000	-0.001
## HF_typeHF_2:session2	0.065	0.153	0.064
## HF_typeHF_3:session2	0.130	0.303	0.064
## HF_typeHF_5:session2	0.064	0.151	0.132
## HF_typeHF_7:session2	0.065	0.153	0.064
## HF_typeHF_8:session2	0.064	0.153	0.064
## HF_typeHF_2:session3	-0.003	-0.002	-0.003
## HF_typeHF_3:session3	-0.003	-0.002	-0.003
## HF_typeHF_5:session3	-0.003	-0.002	-0.003
## HF_typeHF_7:session3	-0.003	-0.002	-0.003
## HF_typeHF_8:session3	-0.003	-0.002	-0.003
## HF_typeHF_2:session4	-0.001	0.141	-0.001
## HF_typeHF_3:session4	0.000	0.279	0.000
## HF_typeHF_5:session4	-0.001	0.140	-0.001
## HF_typeHF_7:session4	-0.001	0.142	-0.001
## HF_typeHF_8:session4	-0.001	0.141	-0.001
## day2:session2	0.002	-0.001	0.002
## day2:session3	-0.003	-0.004	-0.003
## day2:session4	0.001	-0.001	0.000
##	actvttyw:HF_HF_5 actvtty:HF_HF_7 actvttyw:HF_HF_7		
## day2			
## activitystanding			
## activitywalking			
## HF_typeHF_2			
## HF_typeHF_3			
## HF_typeHF_5			
## HF_typeHF_7			
## HF_typeHF_8			
## sexM			
## session2			
## session3			
## session4			
## day2:HF_typeHF_2			
## day2:HF_typeHF_3			
## day2:HF_typeHF_5			
## day2:HF_typeHF_7			
## day2:HF_typeHF_8			
## activitystanding:HF_typeHF_2			
## activitywalking:HF_typeHF_2			
## activitystanding:HF_typeHF_3			
## activitywalking:HF_typeHF_3			
## activitystanding:HF_typeHF_5			
## activitywalking:HF_typeHF_5			
## activitystanding:HF_typeHF_7	0.199		
## activitywalking:HF_typeHF_7	0.501	0.396	
## activitystanding:HF_typeHF_8	0.198	0.506	0.201
## activitywalking:HF_typeHF_8	0.496	0.201	0.505
## HF_typeHF_2:sexM	0.000	-0.001	0.000
## HF_typeHF_3:sexM	0.000	-0.001	0.000
## HF_typeHF_5:sexM	0.000	-0.001	0.000
## HF_typeHF_7:sexM	0.000	-0.001	0.000
## HF_typeHF_8:sexM	0.000	-0.001	0.000
## HF_typeHF_2:session2	0.152	0.065	0.154

## HF_typeHF_3:session2	0.151	0.065	0.153
## HF_typeHF_5:session2	0.303	0.064	0.152
## HF_typeHF_7:session2	0.152	0.128	0.301
## HF_typeHF_8:session2	0.152	0.065	0.154
## HF_typeHF_2:session3	-0.002	-0.003	-0.002
## HF_typeHF_3:session3	-0.002	-0.003	-0.002
## HF_typeHF_5:session3	-0.002	-0.003	-0.002
## HF_typeHF_7:session3	-0.002	-0.003	-0.002
## HF_typeHF_8:session3	-0.002	-0.003	-0.002
## HF_typeHF_2:session4	0.140	-0.001	0.143
## HF_typeHF_3:session4	0.139	0.000	0.141
## HF_typeHF_5:session4	0.280	0.000	0.141
## HF_typeHF_7:session4	0.141	-0.003	0.278
## HF_typeHF_8:session4	0.140	-0.001	0.143
## day2:session2	-0.001	-0.002	-0.002
## day2:session3	-0.004	-0.003	-0.004
## day2:session4	-0.002	-0.001	-0.002
##	actvtys:HF_HF_8 actvtyw:HF_HF_8 HF_HF_2:M		
## day2			
## activitystanding			
## activitywalking			
## HF_typeHF_2			
## HF_typeHF_3			
## HF_typeHF_5			
## HF_typeHF_7			
## HF_typeHF_8			
## sexM			
## session2			
## session3			
## session4			
## day2:HF_typeHF_2			
## day2:HF_typeHF_3			
## day2:HF_typeHF_5			
## day2:HF_typeHF_7			
## day2:HF_typeHF_8			
## activitystanding:HF_typeHF_2			
## activitywalking:HF_typeHF_2			
## activitystanding:HF_typeHF_3			
## activitywalking:HF_typeHF_3			
## activitystanding:HF_typeHF_5			
## activitywalking:HF_typeHF_5			
## activitystanding:HF_typeHF_7			
## activitywalking:HF_typeHF_7			
## activitystanding:HF_typeHF_8			
## activitywalking:HF_typeHF_8	0.397		
## HF_typeHF_2:sexM	-0.001	0.000	
## HF_typeHF_3:sexM	-0.001	0.000	0.498
## HF_typeHF_5:sexM	-0.001	0.000	0.496
## HF_typeHF_7:sexM	-0.001	0.000	0.503
## HF_typeHF_8:sexM	0.000	0.001	0.500
## HF_typeHF_2:session2	0.065	0.153	0.001
## HF_typeHF_3:session2	0.064	0.152	0.001
## HF_typeHF_5:session2	0.064	0.151	0.001
## HF_typeHF_7:session2	0.065	0.153	0.001

## HF_typeHF_8:session2	0.129	0.304	0.001		
## HF_typeHF_2:session3	-0.003	-0.002	0.000		
## HF_typeHF_3:session3	-0.003	-0.002	0.000		
## HF_typeHF_5:session3	-0.003	-0.002	0.000		
## HF_typeHF_7:session3	-0.003	-0.002	0.000		
## HF_typeHF_8:session3	-0.005	-0.010	0.000		
## HF_typeHF_2:session4	-0.001	0.142	0.001		
## HF_typeHF_3:session4	0.000	0.140	0.001		
## HF_typeHF_5:session4	-0.001	0.140	0.001		
## HF_typeHF_7:session4	-0.001	0.142	0.001		
## HF_typeHF_8:session4	0.000	0.280	0.001		
## day2:session2	0.002	-0.001	-0.001		
## day2:session3	-0.004	-0.001	0.000		
## day2:session4	0.000	-0.002	0.000		
##	HF_HF_3:M HF_HF_5:M HF_HF_7:M HF_HF_8:M HF_HF_2:2				
## day2					
## activitystanding					
## activitywalking					
## HF_typeHF_2					
## HF_typeHF_3					
## HF_typeHF_5					
## HF_typeHF_7					
## HF_typeHF_8					
## sexM					
## session2					
## session3					
## session4					
## day2:HF_typeHF_2					
## day2:HF_typeHF_3					
## day2:HF_typeHF_5					
## day2:HF_typeHF_7					
## day2:HF_typeHF_8					
## activitystanding:HF_typeHF_2					
## activitywalking:HF_typeHF_2					
## activitystanding:HF_typeHF_3					
## activitywalking:HF_typeHF_3					
## activitystanding:HF_typeHF_5					
## activitywalking:HF_typeHF_5					
## activitystanding:HF_typeHF_7					
## activitywalking:HF_typeHF_7					
## activitystanding:HF_typeHF_8					
## activitywalking:HF_typeHF_8					
## HF_typeHF_2:sexM					
## HF_typeHF_3:sexM					
## HF_typeHF_5:sexM	0.494				
## HF_typeHF_7:sexM	0.500	0.498			
## HF_typeHF_8:sexM	0.498	0.496	0.502		
## HF_typeHF_2:session2	0.001	0.001	0.001	0.001	
## HF_typeHF_3:session2	0.001	0.001	0.001	0.001	0.506
## HF_typeHF_5:session2	0.001	0.001	0.001	0.001	0.503
## HF_typeHF_7:session2	0.001	0.001	0.000	0.001	0.510
## HF_typeHF_8:session2	0.001	0.001	0.001	0.002	0.509
## HF_typeHF_2:session3	0.000	0.000	0.000	0.000	0.382
## HF_typeHF_3:session3	0.000	0.000	0.000	0.000	0.191

## HF_typeHF_5:session3	0.000	0.000	0.000	0.000	0.190
## HF_typeHF_7:session3	0.000	0.000	0.000	0.000	0.193
## HF_typeHF_8:session3	0.000	0.000	0.000	-0.001	0.191
## HF_typeHF_2:session4	0.001	0.001	0.001	0.001	0.411
## HF_typeHF_3:session4	0.002	0.001	0.001	0.001	0.206
## HF_typeHF_5:session4	0.001	0.001	0.001	0.001	0.206
## HF_typeHF_7:session4	0.001	0.001	0.001	0.001	0.209
## HF_typeHF_8:session4	0.001	0.001	0.001	0.001	0.208
## day2:session2	-0.001	-0.001	-0.001	-0.001	-0.002
## day2:session3	0.000	0.000	0.000	0.000	-0.003
## day2:session4	0.000	0.000	-0.001	0.000	-0.002
##	HF_HF_3:2	HF_HF_5:2	HF_HF_7:2	HF_HF_8:2	HF_HF_2:3
## day2					
## activitystanding					
## activitywalking					
## HF_typeHF_2					
## HF_typeHF_3					
## HF_typeHF_5					
## HF_typeHF_7					
## HF_typeHF_8					
## sexM					
## session2					
## session3					
## session4					
## day2:HF_typeHF_2					
## day2:HF_typeHF_3					
## day2:HF_typeHF_5					
## day2:HF_typeHF_7					
## day2:HF_typeHF_8					
## activitystanding:HF_typeHF_2					
## activitywalking:HF_typeHF_2					
## activitystanding:HF_typeHF_3					
## activitywalking:HF_typeHF_3					
## activitystanding:HF_typeHF_5					
## activitywalking:HF_typeHF_5					
## activitystanding:HF_typeHF_7					
## activitywalking:HF_typeHF_7					
## activitystanding:HF_typeHF_8					
## activitywalking:HF_typeHF_8					
## HF_typeHF_2:sexM					
## HF_typeHF_3:sexM					
## HF_typeHF_5:sexM					
## HF_typeHF_7:sexM					
## HF_typeHF_8:sexM					
## HF_typeHF_2:session2					
## HF_typeHF_3:session2					
## HF_typeHF_5:session2	0.499				
## HF_typeHF_7:session2	0.507	0.503			
## HF_typeHF_8:session2	0.506	0.502	0.510		
## HF_typeHF_2:session3	0.191	0.190	0.192	0.192	
## HF_typeHF_3:session3	0.382	0.189	0.191	0.191	0.502
## HF_typeHF_5:session3	0.189	0.381	0.190	0.190	0.499
## HF_typeHF_7:session3	0.192	0.191	0.381	0.193	0.507
## HF_typeHF_8:session3	0.190	0.189	0.191	0.377	0.502

## HF_typeHF_2:session4	0.207	0.205	0.208	0.208	0.426
## HF_typeHF_3:session4	0.410	0.203	0.206	0.206	0.213
## HF_typeHF_5:session4	0.204	0.410	0.206	0.206	0.212
## HF_typeHF_7:session4	0.207	0.206	0.409	0.209	0.216
## HF_typeHF_8:session4	0.207	0.205	0.208	0.412	0.215
## day2:session2	-0.008	-0.014	0.006	-0.002	-0.002
## day2:session3	-0.005	-0.008	-0.001	-0.002	-0.001
## day2:session4	-0.004	-0.007	0.000	-0.002	-0.002
##	HF_HF_3:3	HF_HF_5:3	HF_HF_7:3	HF_HF_8:3	HF_HF_2:4
## day2					
## activitystanding					
## activitywalking					
## HF_typeHF_2					
## HF_typeHF_3					
## HF_typeHF_5					
## HF_typeHF_7					
## HF_typeHF_8					
## sexM					
## session2					
## session3					
## session4					
## day2:HF_typeHF_2					
## day2:HF_typeHF_3					
## day2:HF_typeHF_5					
## day2:HF_typeHF_7					
## day2:HF_typeHF_8					
## activitystanding:HF_typeHF_2					
## activitywalking:HF_typeHF_2					
## activitystanding:HF_typeHF_3					
## activitywalking:HF_typeHF_3					
## activitystanding:HF_typeHF_5					
## activitywalking:HF_typeHF_5					
## activitystanding:HF_typeHF_7					
## activitywalking:HF_typeHF_7					
## activitystanding:HF_typeHF_8					
## activitywalking:HF_typeHF_8					
## HF_typeHF_2:sexM					
## HF_typeHF_3:sexM					
## HF_typeHF_5:sexM					
## HF_typeHF_7:sexM					
## HF_typeHF_8:sexM					
## HF_typeHF_2:session2					
## HF_typeHF_3:session2					
## HF_typeHF_5:session2					
## HF_typeHF_7:session2					
## HF_typeHF_8:session2					
## HF_typeHF_2:session3					
## HF_typeHF_3:session3					
## HF_typeHF_5:session3	0.497				
## HF_typeHF_7:session3	0.505	0.502			
## HF_typeHF_8:session3	0.499	0.497	0.505		
## HF_typeHF_2:session4	0.214	0.212	0.216	0.214	
## HF_typeHF_3:session4	0.425	0.211	0.214	0.212	0.498
## HF_typeHF_5:session4	0.211	0.426	0.214	0.211	0.497

## HF_typeHF_7:session4	0.214	0.213	0.426	0.214	0.505
## HF_typeHF_8:session4	0.214	0.212	0.216	0.422	0.503
## day2:session2	-0.004	-0.007	0.001	-0.002	-0.002
## day2:session3	-0.007	-0.014	0.005	-0.005	-0.003
## day2:session4	-0.005	-0.008	0.001	-0.002	-0.003
##	HF_HF_3:4	HF_HF_5:4	HF_HF_7:4	HF_HF_8:4	dy2:s2
## day2					
## activitystanding					
## activitywalking					
## HF_typeHF_2					
## HF_typeHF_3					
## HF_typeHF_5					
## HF_typeHF_7					
## HF_typeHF_8					
## sexM					
## session2					
## session3					
## session4					
## day2:HF_typeHF_2					
## day2:HF_typeHF_3					
## day2:HF_typeHF_5					
## day2:HF_typeHF_7					
## day2:HF_typeHF_8					
## activitystanding:HF_typeHF_2					
## activitywalking:HF_typeHF_2					
## activitystanding:HF_typeHF_3					
## activitywalking:HF_typeHF_3					
## activitystanding:HF_typeHF_5					
## activitywalking:HF_typeHF_5					
## activitystanding:HF_typeHF_7					
## activitywalking:HF_typeHF_7					
## activitystanding:HF_typeHF_8					
## activitywalking:HF_typeHF_8					
## HF_typeHF_2:sexM					
## HF_typeHF_3:sexM					
## HF_typeHF_5:sexM					
## HF_typeHF_7:sexM					
## HF_typeHF_8:sexM					
## HF_typeHF_2:session2					
## HF_typeHF_3:session2					
## HF_typeHF_5:session2					
## HF_typeHF_7:session2					
## HF_typeHF_8:session2					
## HF_typeHF_2:session3					
## HF_typeHF_3:session3					
## HF_typeHF_5:session3					
## HF_typeHF_7:session3					
## HF_typeHF_8:session3					
## HF_typeHF_2:session4					
## HF_typeHF_3:session4					
## HF_typeHF_5:session4	0.493				
## HF_typeHF_7:session4	0.500	0.499			
## HF_typeHF_8:session4	0.498	0.497	0.505		
## day2:session2	-0.004	-0.006	0.000	-0.002	

## day2:session3	-0.006	-0.008	0.000	-0.002	0.385
## day2:session4	-0.004	-0.015	0.005	-0.003	0.344
##	dy2:s3				
## day2					
## activitystanding					
## activitywalking					
## HF_typeHF_2					
## HF_typeHF_3					
## HF_typeHF_5					
## HF_typeHF_7					
## HF_typeHF_8					
## sexM					
## session2					
## session3					
## session4					
## day2:HF_typeHF_2					
## day2:HF_typeHF_3					
## day2:HF_typeHF_5					
## day2:HF_typeHF_7					
## day2:HF_typeHF_8					
## activitystanding:HF_typeHF_2					
## activitywalking:HF_typeHF_2					
## activitystanding:HF_typeHF_3					
## activitywalking:HF_typeHF_3					
## activitystanding:HF_typeHF_5					
## activitywalking:HF_typeHF_5					
## activitystanding:HF_typeHF_7					
## activitywalking:HF_typeHF_7					
## activitystanding:HF_typeHF_8					
## activitywalking:HF_typeHF_8					
## HF_typeHF_2:sexM					
## HF_typeHF_3:sexM					
## HF_typeHF_5:sexM					
## HF_typeHF_7:sexM					
## HF_typeHF_8:sexM					
## HF_typeHF_2:session2					
## HF_typeHF_3:session2					
## HF_typeHF_5:session2					
## HF_typeHF_7:session2					
## HF_typeHF_8:session2					
## HF_typeHF_2:session3					
## HF_typeHF_3:session3					
## HF_typeHF_5:session3					
## HF_typeHF_7:session3					
## HF_typeHF_8:session3					
## HF_typeHF_2:session4					
## HF_typeHF_3:session4					
## HF_typeHF_5:session4					
## HF_typeHF_7:session4					
## HF_typeHF_8:session4					
## day2:session2					
## day2:session3					
## day2:session4	0.446				
##					

```
## Standardized Within-Group Residuals:
##      Min      Q1      Med      Q3      Max
## -5.92654894 -0.39260627  0.01907578  0.45833306  3.67377731
##
## Number of Observations: 2638
## Number of Groups:
##              person      HF_type %in% person
##              24      144
## day %in% HF_type %in% person
##              280

#day explains a lot of the unexplained residuals!! we should keep that.

#anova to compare
print(anova mdl4_random_person_sensor_with_interact_reml, mdl4_random_person_sensor_day_with_interact_reml)

##              Model df      AIC      BIC
## mdl4_random_person_sensor_with_interact_reml      1 54 487.1302 803.4759
## mdl4_random_person_sensor_day_with_interact_reml      2 55 -602.3829 -280.1789
##              logLik      Test  L.Ratio
## mdl4_random_person_sensor_with_interact_reml      -189.5651
## mdl4_random_person_sensor_day_with_interact_reml      356.1914 1 vs 2 1091.513
##              p-value
## mdl4_random_person_sensor_with_interact_reml
## mdl4_random_person_sensor_day_with_interact_reml      <.0001
```

0.11 Section 7: Let's test removing the variables we were initially unsure of (overlaps with high p-values from final model above)

Stuff with high pvalues:

- day
- sex
- session (although session 4 is significant)
- sensor & sex
- sensor & session (some)

```
mdl4_random_person_sensor_day_with_interact<- lme(
  fixed = log(HF_value) ~ (day) + (activity) + (HF_type) + (sex) + (session) + (HF_type * day) + (HF
  random = ~ 1 | person/HF_type/day,
  data = df_clean,
  method = "ML"
)

#what if I remove the interactive variables that I was unsure of?
mdl4_random_person_sensor_day_with_interact_nodaysession <- lme(
  fixed = log(HF_value) ~ (day) + (activity) + (HF_type) + (session) + (sex) + (HF_type * day) + (HF
  random = ~ 1 | person/HF_type/day,
  data = df_clean,
  method = "ML"
)

mdl4_random_person_sensor_day_with_interact_nosensorsession <- lme(
```

```

    fixed = log(HF_value) ~ (day) + (activity) + (HF_type) + (session) + (sex) + (HF_type * day) + (HF_type * activity)
    random = ~ 1 | person/HF_type/day,
    data = df_clean,
    method = "ML"
  )

mdl4_random_person_sensor_day_with_interact_nosensorsex <- lme(
  fixed = log(HF_value) ~ (day) + (activity) + (HF_type) + (session) + (sex) + (HF_type * day) + (HF_type * activity)
  random = ~ 1 | person/HF_type/day,
  data = df_clean,
  method = "ML"
)

#remove sex (and thus no sensor sex)
mdl4_random_person_sensor_day_with_interact_nosex <- lme(
  fixed = log(HF_value) ~ (day) + (activity) + (HF_type) + (session) + (HF_type * day) + (HF_type * activity)
  random = ~ 1 | person/HF_type/day,
  data = df_clean,
  method = "ML"
)

#remove day (and thus interactions)
mdl4_random_person_sensor_day_with_interact_noday <- lme(
  fixed = log(HF_value) ~ (activity) + (HF_type) + (session) + (HF_type * activity) + (HF_type * session)
  random = ~ 1 | person/HF_type/day,
  data = df_clean,
  method = "ML"
)

mdl4_random_person_sensor_day_with_interact_nosession <- lme(
  fixed = log(HF_value) ~ (activity) + (HF_type) + (day) + (HF_type * activity) + (HF_type * day) + (HF_type * session)
  random = ~ 1 | person/HF_type/day,
  data = df_clean,
  method = "ML"
)

mdl5_activity_sensor_day_session_sensoractivity_sensorday_sessionday <- lme(
  fixed = log(HF_value) ~ (activity) + (HF_type) + (day) + (session) + (HF_type * activity) + (HF_type * day) + (HF_type * session)
  random = ~ 1 | person/HF_type/day,
  data = df_clean,
  method = "ML"
)

mdl5_activity_sensor_day_sensoractivity_sensorday <- lme(
  fixed = log(HF_value) ~ (activity) + (HF_type) + (day) + (HF_type * activity) + (HF_type * day) + (HF_type * session)
  random = ~ 1 | person/HF_type/day,
  data = df_clean,
  method = "ML"
)

anova_summary_manual_mdl4_nosession_interact <- bind_rows(
  get_baseline_metrics(mdl4_random_person_sensor_day_with_interact, "Current Model"),
  get_anova_outputs(mdl4_random_person_sensor_day_with_interact, mdl4_random_person_sensor_day_with_interact)
)

```

```

get_anova_outputs(md14_random_person_sensor_day_with_interact, md14_random_person_sensor_day_with_inte
get_anova_outputs(md14_random_person_sensor_day_with_interact, md14_random_person_sensor_day_with_inte
get_anova_outputs(md14_random_person_sensor_day_with_interact, md14_random_person_sensor_day_with_inte
get_anova_outputs(md14_random_person_sensor_day_with_interact, md14_random_person_sensor_day_with_inte
get_anova_outputs(md14_random_person_sensor_day_with_interact, md14_random_person_sensor_day_with_inte
get_anova_outputs(md14_random_person_sensor_day_with_interact, md15_activity_sensor_day_session_sens
get_anova_outputs(md14_random_person_sensor_day_with_interact, md15_activity_sensor_day_sensoractivit
)
print(anova_summary_manual_md14_nosession_interact)

```

##		Model	Df	AIC	BIC	LogLik	LRT_Stat	p_value
## 1		Current Model	55	-866.5290	-543.2513	488.2645	NA	NA
## 2	Drop Day & Session		52	-831.8426	-526.1982	467.9213	40.6865	0.0000
## 3	Drop Sensor & Session		40	-796.5553	-561.4443	438.2777	99.9737	0.0000
## 4	Drop Sensor & Sex		50	-871.9151	-578.0262	485.9575	4.6139	0.4648
## 5	Drop Sex		49	-872.6361	-584.6250	485.3180	5.8929	0.4353
## 6	Drop Day		46	-764.4846	-494.1069	428.2423	120.0444	0.0000
## 7	Drop Session		34	-753.6617	-553.8173	410.8309	154.8673	0.0000
## 8	Final		34	-802.6793	-602.8350	435.3397	105.8497	0.0000
## 9	Final no session		28	-759.7845	-595.2068	407.8922	160.7445	0.0000

We should definitely drop sensor and sex.

Dropping day & session does nothing - AIC+BIC increase, and p-value is 0 for it. So we will keep it.

Dropping sensor & session increases AIC & decreases BIC, so we will drop it.

Just dropping sensor & session is better than dropping session overall, so we will keep session.

Dropping sensor & sex is unequivocally good. Relatedly, dropping sex overall is even better. So we will drop everything sex.

Dropping day overall is not good, so we will keep it.

Propose to drop sex, sensor & sex, sensor & session.

0.12 Takeaway: Final model: fixed = log(HF_value) ~ (activity) + (HF_type) + (day) + (session) + (HF_type * activity) + (HF_type * day) + (day * session), , random = ~ 1 | person/HF_type/day,

0.13 Section 8: Look at random slopes!

Initial hypothesis: people cool down at different rates. Try adding random slope for session. However, there's convergence issues...

```

#hypothesis: people cool down at different rates

md15_activity_sensor_day_session_sensoractivity_sensorday_sessionday_reml<- lme(
  fixed = log(HF_value) ~ (activity) + (HF_type) + (day) + (session) + (HF_type * activity) + (HF_t
  random = ~ 1 | person/HF_type/day,
  data = df_clean,
  method = "REML"
)
print(summary(md15_activity_sensor_day_session_sensoractivity_sensorday_sessionday_reml))

## Linear mixed-effects model fit by REML
## Data: df_clean

```

```

##          AIC          BIC    logLik
##    -640.229 -440.7734 354.1145
##
## Random effects:
## Formula: ~1 | person
##          (Intercept)
## StdDev:  0.04551663
##
## Formula: ~1 | HF_type %in% person
##          (Intercept)
## StdDev:  0.09977507
##
## Formula: ~1 | day %in% HF_type %in% person
##          (Intercept) Residual
## StdDev:   0.2210764 0.1764976
##
## Fixed effects: log(HF_value) ~ (activity) + (HF_type) + (day) + (session) + (HF_type * activity)
##                                     Value Std.Error   DF   t-value p-value
## (Intercept)                        4.383060 0.05431310 2340   80.69986 0.0000
## activitystanding                   -0.093853 0.01923532 2340   -4.87919 0.0000
## activitywalking                     0.081086 0.02276779 2340    3.56145 0.0004
## HF_typeHF_2                       -0.624414 0.07482992  115   -8.34445 0.0000
## HF_typeHF_3                       -0.530402 0.07565765  115   -7.01055 0.0000
## HF_typeHF_5                       -0.318136 0.07654573  115   -4.15616 0.0001
## HF_typeHF_7                       -0.059330 0.07407033  115   -0.80100 0.4248
## HF_typeHF_8                       -0.752442 0.07486219  115  -10.05103 0.0000
## day2                             -0.112904 0.06758632  130   -1.67051 0.0972
## session2                          0.033295 0.01419859 2340    2.34496 0.0191
## session3                          0.024318 0.01241032 2340    1.95950 0.0502
## session4                          0.020349 0.01414880 2340    1.43823 0.1505
## activitystanding:HF_typeHF_2      -0.086896 0.02689669 2340   -3.23073 0.0013
## activitywalking:HF_typeHF_2      -0.454989 0.03161032 2340  -14.39370 0.0000
## activitystanding:HF_typeHF_3      0.197299 0.02709300 2340    7.28229 0.0000
## activitywalking:HF_typeHF_3      0.186045 0.03179749 2340    5.85093 0.0000
## activitystanding:HF_typeHF_5      0.067339 0.02721450 2340    2.47436 0.0134
## activitywalking:HF_typeHF_5     -0.095560 0.03195842 2340   -2.99015 0.0028
## activitystanding:HF_typeHF_7     -0.037081 0.02679964 2340   -1.38363 0.1666
## activitywalking:HF_typeHF_7     -0.146165 0.03147657 2340   -4.64360 0.0000
## activitystanding:HF_typeHF_8      0.188658 0.02699444 2340    6.98878 0.0000
## activitywalking:HF_typeHF_8      0.119304 0.03165735 2340    3.76862 0.0002
## HF_typeHF_2:day2                  0.497243 0.09447863  130    5.26302 0.0000
## HF_typeHF_3:day2                  0.499802 0.09513092  130    5.25383 0.0000
## HF_typeHF_5:day2                  0.430294 0.09581845  130    4.49072 0.0000
## HF_typeHF_7:day2                  0.338114 0.09389586  130    3.60095 0.0004
## HF_typeHF_8:day2                  0.492708 0.09449664  130    5.21403 0.0000
## day2:session2                    -0.072417 0.02237329 2340   -3.23677 0.0012
## day2:session3                    -0.065812 0.01729029 2340   -3.80628 0.0001
## day2:session4                    -0.115000 0.01931311 2340   -5.95450 0.0000
## Correlation:
##                                     (Intr) actvtys actvtyw HF_tHF_2 HF_tHF_3 HF_tHF_5
## activitystanding                   -0.176
## activitywalking                   -0.151  0.398
## HF_typeHF_2                      -0.690  0.127  0.100
## HF_typeHF_3                      -0.682  0.125  0.099  0.496

```


## HF_typeHF_5	-0.674	0.124	0.098	0.490	0.484	
## HF_typeHF_7	-0.697	0.128	0.102	0.506	0.500	0.494
## HF_typeHF_8	-0.689	0.127	0.100	0.501	0.495	0.490
## day2	-0.638	0.015	0.000	0.453	0.448	0.443
## session2	-0.110	0.000	0.086	-0.001	-0.001	-0.001
## session3	-0.112	-0.004	-0.004	-0.001	-0.001	-0.001
## session4	-0.112	0.000	0.088	0.000	0.000	0.000
## activitystanding:HF_typeHF_2	0.125	-0.708	-0.278	-0.178	-0.090	-0.089
## activitywalking:HF_typeHF_2	0.099	-0.280	-0.698	-0.142	-0.072	-0.071
## activitystanding:HF_typeHF_3	0.124	-0.703	-0.276	-0.090	-0.177	-0.088
## activitywalking:HF_typeHF_3	0.099	-0.278	-0.694	-0.072	-0.142	-0.070
## activitystanding:HF_typeHF_5	0.123	-0.699	-0.275	-0.090	-0.089	-0.176
## activitywalking:HF_typeHF_5	0.099	-0.277	-0.690	-0.072	-0.071	-0.141
## activitystanding:HF_typeHF_7	0.125	-0.711	-0.280	-0.091	-0.090	-0.089
## activitywalking:HF_typeHF_7	0.100	-0.282	-0.701	-0.073	-0.072	-0.071
## activitystanding:HF_typeHF_8	0.124	-0.705	-0.277	-0.090	-0.089	-0.088
## activitywalking:HF_typeHF_8	0.099	-0.280	-0.696	-0.072	-0.071	-0.071
## HF_typeHF_2:day2	0.446	-0.013	0.000	-0.647	-0.321	-0.317
## HF_typeHF_3:day2	0.443	-0.013	0.000	-0.322	-0.653	-0.315
## HF_typeHF_5:day2	0.440	-0.013	0.000	-0.320	-0.316	-0.659
## HF_typeHF_7:day2	0.449	-0.013	0.000	-0.326	-0.322	-0.319
## HF_typeHF_8:day2	0.446	-0.013	0.000	-0.324	-0.321	-0.317
## day2:session2	0.049	0.073	0.032	0.001	0.000	0.000
## day2:session3	0.080	0.004	0.004	0.001	0.001	0.001
## day2:session4	0.073	-0.001	0.001	0.000	0.000	0.000
##	HF_tHF_7	HF_tHF_8	day2	sessn2	sessn3	sessn4
## activitystanding						
## activitywalking						
## HF_typeHF_2						
## HF_typeHF_3						
## HF_typeHF_5						
## HF_typeHF_7						
## HF_typeHF_8	0.506					
## day2	0.457	0.453				
## session2	-0.001	-0.001	0.079			
## session3	-0.001	0.000	0.091	0.432		
## session4	0.000	0.000	0.080	0.426	0.434	
## activitystanding:HF_typeHF_2	-0.092	-0.091	-0.013	0.000	0.003	0.000
## activitywalking:HF_typeHF_2	-0.073	-0.072	0.000	0.001	0.002	0.000
## activitystanding:HF_typeHF_3	-0.091	-0.090	-0.013	0.000	0.003	0.000
## activitywalking:HF_typeHF_3	-0.073	-0.072	0.000	0.000	0.002	-0.002
## activitystanding:HF_typeHF_5	-0.090	-0.089	-0.013	0.000	0.003	0.000
## activitywalking:HF_typeHF_5	-0.072	-0.071	0.000	-0.001	0.002	-0.003
## activitystanding:HF_typeHF_7	-0.179	-0.091	-0.012	0.002	0.003	0.000
## activitywalking:HF_typeHF_7	-0.143	-0.073	0.000	0.002	0.002	0.001
## activitystanding:HF_typeHF_8	-0.091	-0.179	-0.013	0.000	0.003	0.000
## activitywalking:HF_typeHF_8	-0.073	-0.144	0.000	0.001	0.000	0.000
## HF_typeHF_2:day2	-0.327	-0.324	-0.699	0.001	0.001	0.000
## HF_typeHF_3:day2	-0.325	-0.322	-0.695	0.001	0.001	0.000
## HF_typeHF_5:day2	-0.323	-0.320	-0.690	0.001	0.001	0.000
## HF_typeHF_7:day2	-0.642	-0.326	-0.704	0.001	0.001	0.000
## HF_typeHF_8:day2	-0.327	-0.647	-0.699	0.001	-0.001	0.000
## day2:session2	0.001	0.000	-0.098	-0.605	-0.275	-0.241
## day2:session3	0.001	0.000	-0.127	-0.310	-0.718	-0.311

```

## day2:session4          0.000    0.000   -0.115 -0.278 -0.319 -0.698
##          actvtys:HF_HF_2 actvttyw:HF_HF_2 actvtys:HF_HF_3
## activitystanding
## activitywalking
## HF_typeHF_2
## HF_typeHF_3
## HF_typeHF_5
## HF_typeHF_7
## HF_typeHF_8
## day2
## session2
## session3
## session4
## activitystanding:HF_typeHF_2
## activitywalking:HF_typeHF_2  0.395
## activitystanding:HF_typeHF_3  0.503          0.199
## activitywalking:HF_typeHF_3  0.199          0.500          0.395
## activitystanding:HF_typeHF_5  0.500          0.198          0.497
## activitywalking:HF_typeHF_5  0.198          0.497          0.197
## activitystanding:HF_typeHF_7  0.508          0.201          0.505
## activitywalking:HF_typeHF_7  0.201          0.505          0.200
## activitystanding:HF_typeHF_8  0.505          0.200          0.501
## activitywalking:HF_typeHF_8  0.200          0.502          0.199
## HF_typeHF_2:day2          0.017          0.000          0.009
## HF_typeHF_3:day2          0.009          0.000          0.017
## HF_typeHF_5:day2          0.009          0.000          0.009
## HF_typeHF_7:day2          0.009          0.000          0.009
## HF_typeHF_8:day2          0.009          0.000          0.009
## day2:session2          0.001          0.000          0.003
## day2:session3         -0.003          -0.003          -0.003
## day2:session4          0.001          -0.001          0.001
##          actvttyw:HF_HF_3 actvtys:HF_HF_5 actvttyw:HF_HF_5
## activitystanding
## activitywalking
## HF_typeHF_2
## HF_typeHF_3
## HF_typeHF_5
## HF_typeHF_7
## HF_typeHF_8
## day2
## session2
## session3
## session4
## activitystanding:HF_typeHF_2
## activitywalking:HF_typeHF_2
## activitystanding:HF_typeHF_3
## activitywalking:HF_typeHF_3
## activitystanding:HF_typeHF_5  0.197
## activitywalking:HF_typeHF_5  0.494          0.395
## activitystanding:HF_typeHF_7  0.200          0.502          0.199
## activitywalking:HF_typeHF_7  0.502          0.199          0.499
## activitystanding:HF_typeHF_8  0.199          0.499          0.198
## activitywalking:HF_typeHF_8  0.499          0.198          0.496
## HF_typeHF_2:day2          0.000          0.009          0.000

```

## HF_typeHF_3:day2	0.000	0.009	0.000
## HF_typeHF_5:day2	0.000	0.017	0.000
## HF_typeHF_7:day2	0.000	0.009	0.000
## HF_typeHF_8:day2	0.000	0.009	0.000
## day2:session2	0.002	0.004	0.004
## day2:session3	-0.003	-0.003	-0.002
## day2:session4	0.000	0.000	0.003
##	actvtys:HF_HF_7 actvtyw:HF_HF_7 actvtys:HF_HF_8		
## activitystanding			
## activitywalking			
## HF_typeHF_2			
## HF_typeHF_3			
## HF_typeHF_5			
## HF_typeHF_7			
## HF_typeHF_8			
## day2			
## session2			
## session3			
## session4			
## activitystanding:HF_typeHF_2			
## activitywalking:HF_typeHF_2			
## activitystanding:HF_typeHF_3			
## activitywalking:HF_typeHF_3			
## activitystanding:HF_typeHF_5			
## activitywalking:HF_typeHF_5			
## activitystanding:HF_typeHF_7			
## activitywalking:HF_typeHF_7	0.397		
## activitystanding:HF_typeHF_8	0.506	0.201	
## activitywalking:HF_typeHF_8	0.201	0.504	0.397
## HF_typeHF_2:day2	0.009	0.000	0.009
## HF_typeHF_3:day2	0.009	0.000	0.009
## HF_typeHF_5:day2	0.009	0.000	0.009
## HF_typeHF_7:day2	0.016	0.000	0.009
## HF_typeHF_8:day2	0.009	0.000	0.017
## day2:session2	-0.003	-0.004	0.002
## day2:session3	-0.003	-0.003	-0.004
## day2:session4	-0.001	-0.004	0.001
##	actvtyw:HF_HF_8 HF_HF_2: HF_HF_3: HF_HF_5:		
## activitystanding			
## activitywalking			
## HF_typeHF_2			
## HF_typeHF_3			
## HF_typeHF_5			
## HF_typeHF_7			
## HF_typeHF_8			
## day2			
## session2			
## session3			
## session4			
## activitystanding:HF_typeHF_2			
## activitywalking:HF_typeHF_2			
## activitystanding:HF_typeHF_3			
## activitywalking:HF_typeHF_3			
## activitystanding:HF_typeHF_5			

```

## activitywalking:HF_typeHF_5
## activitystanding:HF_typeHF_7
## activitywalking:HF_typeHF_7
## activitystanding:HF_typeHF_8
## activitywalking:HF_typeHF_8
## HF_typeHF_2:day2          0.000
## HF_typeHF_3:day2          0.000          0.497
## HF_typeHF_5:day2          0.000          0.494          0.490
## HF_typeHF_7:day2          0.000          0.503          0.500          0.496
## HF_typeHF_8:day2          0.001          0.501          0.497          0.494
## day2:session2             0.000          -0.001          -0.001          -0.001
## day2:session3             -0.002          0.000          0.000          0.000
## day2:session4             -0.001          0.001          0.001          0.001
## HF_HF_7: HF_HF_8: dy2:s2 dy2:s3
## activitystanding
## activitywalking
## HF_typeHF_2
## HF_typeHF_3
## HF_typeHF_5
## HF_typeHF_7
## HF_typeHF_8
## day2
## session2
## session3
## session4
## activitystanding:HF_typeHF_2
## activitywalking:HF_typeHF_2
## activitystanding:HF_typeHF_3
## activitywalking:HF_typeHF_3
## activitystanding:HF_typeHF_5
## activitywalking:HF_typeHF_5
## activitystanding:HF_typeHF_7
## activitywalking:HF_typeHF_7
## activitystanding:HF_typeHF_8
## activitywalking:HF_typeHF_8
## HF_typeHF_2:day2          0.503
## HF_typeHF_3:day2
## HF_typeHF_5:day2
## HF_typeHF_7:day2
## HF_typeHF_8:day2          0.000          -0.001
## day2:session2             0.000          0.001          0.385
## day2:session3             0.001          0.001          0.344          0.446
## day2:session4
##
## Standardized Within-Group Residuals:
##           Min           Q1           Med           Q3           Max
## -5.57989380 -0.38337175  0.02929192  0.44336609  3.43862334
##
## Number of Observations: 2638
## Number of Groups:
##           person           HF_type %in% person
##           24           144
## day %in% HF_type %in% person
##           280

```

```

# mdl6_rs_session_reml<- lme(
#   fixed = log(HF_value) ~ (activity) + (HF_type) + (day) + (session) + (HF_type * activity) + (HF_t
#   random = list(person = ~ session, HF_type = ~ 1, day = ~ 1),
#   data = df_clean,
#   method = "REML",
#   control = lmeControl(maxIter = 200,
#                         msMaxIter = 200,
#                         tolerance = 1e-6,
#                         niterEM = 100,
#                         opt = "nlminb") #basically, increase iterations
# )

# print(anova mdl5_activity_sensor_day_session_sensoractivity_sensorday_sessionday_reml, mdl6_rs_session
# print(summary(mdl6_rs_session_reml))

#for random effects, we care about the variance! --> is the stdev high?
# for fixed effects, just the estimates --> anova, AIC/BIC, pval

```

It fails when we try to do this for REML. Instead, try numerical. (Also works because when ran for ML, correlation between the sessions was high)

```

df_clean$session_num <- as.numeric(as.character(df_clean$session))

mdl6_rs_session_num_reml<- lme(
  fixed = log(HF_value) ~ (activity) + (HF_type) + (day) + (session) + (HF_type * activity) + (HF_t
  random = list(person = ~ session_num, HF_type = ~ 1, day = ~ 1),
  data = df_clean,
  method = "REML",
  control = lmeControl(maxIter = 200,
                      msMaxIter = 200,
                      tolerance = 1e-6,
                      niterEM = 100,
                      opt = "nlminb") #basically, increase iterations
)

print(anova(mdl5_activity_sensor_day_session_sensoractivity_sensorday_sessionday_reml, mdl6_rs_session_

```

	Model	
## mdl5_activity_sensor_day_session_sensoractivity_sensorday_sessionday_reml	1	
## mdl6_rs_session_num_reml	2	
##	df	
## mdl5_activity_sensor_day_session_sensoractivity_sensorday_sessionday_reml	34	
## mdl6_rs_session_num_reml	36	
##	AIC	
## mdl5_activity_sensor_day_session_sensoractivity_sensorday_sessionday_reml	-640.2290	
## mdl6_rs_session_num_reml	-657.8396	
##	BIC	
## mdl5_activity_sensor_day_session_sensoractivity_sensorday_sessionday_reml	-440.7734	
## mdl6_rs_session_num_reml	-446.6514	
##	logLik	
## mdl5_activity_sensor_day_session_sensoractivity_sensorday_sessionday_reml	354.1145	
## mdl6_rs_session_num_reml	364.9198	
##	Test	

```
## mdl5_activity_sensor_day_session_sensoractivity_sensorday_sessionday_reml
## mdl6_rs_session_num_reml                                1 vs 2
##                                                         L.Ratio
## mdl5_activity_sensor_day_session_sensoractivity_sensorday_sessionday_reml
## mdl6_rs_session_num_reml                                21.61066
##                                                         p-value
## mdl5_activity_sensor_day_session_sensoractivity_sensorday_sessionday_reml
## mdl6_rs_session_num_reml                                <.0001
```

```
print(summary(mdl6_rs_session_num_reml))
```

```
## Linear mixed-effects model fit by REML
```

```
## Data: df_clean
```

```
## AIC BIC logLik
```

```
## -657.8396 -446.6514 364.9198
```

```
##
```

```
## Random effects:
```

```
## Formula: ~session_num | person
```

```
## Structure: General positive-definite, Log-Cholesky parametrization
```

```
## StdDev Corr
```

```
## (Intercept) 0.03931192 (Intr)
```

```
## session_num 0.02099203 -0.494
```

```
##
```

```
## Formula: ~1 | HF_type %in% person
```

```
## (Intercept)
```

```
## StdDev: 0.09882548
```

```
##
```

```
## Formula: ~1 | day %in% HF_type %in% person
```

```
## (Intercept) Residual
```

```
## StdDev: 0.2216079 0.1747833
```

```
##
```

```
## Fixed effects: log(HF_value) ~ (activity) + (HF_type) + (day) + (session) + (HF_type * activity)
```

```
## Value Std.Error DF t-value p-value
```

```
## (Intercept) 4.382900 0.05392890 2340 81.27183 0.0000
```

```
## activitystanding -0.093962 0.01904872 2340 -4.93272 0.0000
```

```
## activitywalking 0.080392 0.02254765 2340 3.56544 0.0004
```

```
## HF_typeHF_2 -0.623684 0.07478953 115 -8.33919 0.0000
```

```
## HF_typeHF_3 -0.531475 0.07561473 115 -7.02872 0.0000
```

```
## HF_typeHF_5 -0.317194 0.07649987 115 -4.14633 0.0001
```

```
## HF_typeHF_7 -0.059971 0.07403273 115 -0.81005 0.4196
```

```
## HF_typeHF_8 -0.751566 0.07482131 115 -10.04482 0.0000
```

```
## day2 -0.111444 0.06767518 130 -1.64674 0.1020
```

```
## session2 0.033613 0.01470458 2340 2.28591 0.0223
```

```
## session3 0.025169 0.01500479 2340 1.67738 0.0936
```

```
## session4 0.021432 0.01905225 2340 1.12491 0.2607
```

```
## activitystanding:HF_typeHF_2 -0.086798 0.02663563 2340 -3.25872 0.0011
```

```
## activitywalking:HF_typeHF_2 -0.454592 0.03130395 2340 -14.52187 0.0000
```

```
## activitystanding:HF_typeHF_3 0.197363 0.02683001 2340 7.35605 0.0000
```

```
## activitywalking:HF_typeHF_3 0.187036 0.03149037 2340 5.93946 0.0000
```

```
## activitystanding:HF_typeHF_5 0.067438 0.02695035 2340 2.50231 0.0124
```

```
## activitywalking:HF_typeHF_5 -0.095228 0.03164917 2340 -3.00885 0.0027
```

```
## activitystanding:HF_typeHF_7 -0.036500 0.02653973 2340 -1.37531 0.1692
```

```
## activitywalking:HF_typeHF_7 -0.145023 0.03117209 2340 -4.65233 0.0000
```

```
## activitystanding:HF_typeHF_8 0.188498 0.02673256 2340 7.05125 0.0000
```

```
## activitywalking:HF_typeHF_8 0.119526 0.03135060 2340 3.81255 0.0001
```

```

## HF_typeHF_2:day2      0.495673 0.09462490 130 5.23829 0.0000
## HF_typeHF_3:day2      0.499811 0.09527266 130 5.24611 0.0000
## HF_typeHF_5:day2      0.428526 0.09595788 130 4.46577 0.0000
## HF_typeHF_7:day2      0.337143 0.09404532 130 3.58490 0.0005
## HF_typeHF_8:day2      0.491143 0.09464263 130 5.18945 0.0000
## day2:session2         -0.073273 0.02215961 2340 -3.30659 0.0010
## day2:session3         -0.067083 0.01714139 2340 -3.91353 0.0001
## day2:session4         -0.116943 0.01916344 2340 -6.10242 0.0000
## Correlation:
## (Intr) actvtys actvtyw HF_tHF_2 HF_tHF_3 HF_tHF_5
## activitystanding      -0.176
## activitywalking        -0.151 0.398
## HF_typeHF_2            -0.694 0.125 0.100
## HF_typeHF_3            -0.687 0.124 0.098 0.495
## HF_typeHF_5            -0.679 0.123 0.097 0.490 0.484
## HF_typeHF_7            -0.701 0.127 0.101 0.506 0.500 0.494
## HF_typeHF_8            -0.694 0.125 0.099 0.501 0.495 0.490
## day2                   -0.643 0.015 0.000 0.454 0.449 0.444
## session2               -0.103 0.000 0.082 -0.001 -0.001 -0.001
## session3               -0.088 -0.004 -0.003 -0.001 -0.001 -0.001
## session4               -0.078 0.000 0.065 0.000 0.001 0.001
## activitystanding:HF_typeHF_2 0.124 -0.708 -0.278 -0.176 -0.089 -0.088
## activitywalking:HF_typeHF_2 0.099 -0.280 -0.698 -0.141 -0.071 -0.070
## activitystanding:HF_typeHF_3 0.123 -0.703 -0.276 -0.089 -0.176 -0.087
## activitywalking:HF_typeHF_3 0.099 -0.278 -0.694 -0.071 -0.140 -0.070
## activitystanding:HF_typeHF_5 0.123 -0.699 -0.275 -0.089 -0.088 -0.174
## activitywalking:HF_typeHF_5 0.098 -0.277 -0.690 -0.071 -0.070 -0.139
## activitystanding:HF_typeHF_7 0.125 -0.711 -0.280 -0.090 -0.089 -0.088
## activitywalking:HF_typeHF_7 0.100 -0.282 -0.701 -0.072 -0.071 -0.070
## activitystanding:HF_typeHF_8 0.124 -0.705 -0.277 -0.089 -0.088 -0.087
## activitywalking:HF_typeHF_8 0.099 -0.280 -0.696 -0.072 -0.071 -0.070
## HF_typeHF_2:day2      0.450 -0.013 0.000 -0.648 -0.321 -0.318
## HF_typeHF_3:day2      0.447 -0.013 0.000 -0.323 -0.654 -0.315
## HF_typeHF_5:day2      0.444 -0.012 0.000 -0.320 -0.317 -0.660
## HF_typeHF_7:day2      0.453 -0.013 0.000 -0.326 -0.323 -0.319
## HF_typeHF_8:day2      0.450 -0.013 0.000 -0.325 -0.321 -0.317
## day2:session2          0.049 0.073 0.032 0.000 0.000 0.000
## day2:session3          0.080 0.004 0.004 0.001 0.001 0.001
## day2:session4          0.073 -0.001 0.001 0.000 0.000 -0.001
## HF_tHF_7 HF_tHF_8 day2 sessn2 sessn3 sessn4
## activitystanding
## activitywalking
## HF_typeHF_2
## HF_typeHF_3
## HF_typeHF_5
## HF_typeHF_7
## HF_typeHF_8          0.506
## day2                 0.458 0.454
## session2             -0.001 -0.001 0.074
## session3             -0.001 0.001 0.074 0.507
## session4             0.000 0.000 0.058 0.498 0.650
## activitystanding:HF_typeHF_2 -0.091 -0.090 -0.012 0.000 0.003 0.000
## activitywalking:HF_typeHF_2 -0.072 -0.072 0.000 0.001 0.002 0.000
## activitystanding:HF_typeHF_3 -0.090 -0.089 -0.012 0.000 0.003 0.000

```

```

## activitywalking:HF_typeHF_3 -0.072 -0.071 0.000 0.000 0.002 -0.001
## activitystanding:HF_typeHF_5 -0.090 -0.089 -0.012 0.000 0.003 0.000
## activitywalking:HF_typeHF_5 -0.072 -0.071 0.000 -0.001 0.002 -0.002
## activitystanding:HF_typeHF_7 -0.177 -0.090 -0.012 0.002 0.003 0.000
## activitywalking:HF_typeHF_7 -0.142 -0.072 0.000 0.002 0.002 0.001
## activitystanding:HF_typeHF_8 -0.090 -0.177 -0.012 0.000 0.003 0.000
## activitywalking:HF_typeHF_8 -0.072 -0.142 0.000 0.001 0.000 0.000
## HF_typeHF_2:day2 -0.328 -0.325 -0.700 0.001 0.000 0.000
## HF_typeHF_3:day2 -0.326 -0.322 -0.695 0.001 0.000 0.000
## HF_typeHF_5:day2 -0.323 -0.320 -0.690 0.001 0.000 0.000
## HF_typeHF_7:day2 -0.643 -0.326 -0.704 0.001 0.001 0.000
## HF_typeHF_8:day2 -0.328 -0.648 -0.700 0.001 -0.001 0.000
## day2:session2 0.001 0.000 -0.097 -0.579 -0.226 -0.178
## day2:session3 0.001 0.000 -0.126 -0.298 -0.590 -0.232
## day2:session4 0.001 0.000 -0.114 -0.267 -0.264 -0.516
## actvtys:HF_HF_2 actvttyw:HF_HF_2 actvtys:HF_HF_3
## activitystanding
## activitywalking
## HF_typeHF_2
## HF_typeHF_3
## HF_typeHF_5
## HF_typeHF_7
## HF_typeHF_8
## day2
## session2
## session3
## session4
## activitystanding:HF_typeHF_2
## activitywalking:HF_typeHF_2 0.395
## activitystanding:HF_typeHF_3 0.503 0.199
## activitywalking:HF_typeHF_3 0.199 0.500 0.395
## activitystanding:HF_typeHF_5 0.500 0.198 0.497
## activitywalking:HF_typeHF_5 0.198 0.497 0.197
## activitystanding:HF_typeHF_7 0.508 0.201 0.505
## activitywalking:HF_typeHF_7 0.201 0.505 0.200
## activitystanding:HF_typeHF_8 0.505 0.200 0.501
## activitywalking:HF_typeHF_8 0.200 0.502 0.199
## HF_typeHF_2:day2 0.017 0.000 0.009
## HF_typeHF_3:day2 0.009 0.000 0.017
## HF_typeHF_5:day2 0.009 0.000 0.009
## HF_typeHF_7:day2 0.009 0.000 0.009
## HF_typeHF_8:day2 0.009 0.000 0.009
## day2:session2 0.001 0.000 0.003
## day2:session3 -0.003 -0.002 -0.003
## day2:session4 0.000 -0.001 0.001
## actvttyw:HF_HF_3 actvtys:HF_HF_5 actvttyw:HF_HF_5
## activitystanding
## activitywalking
## HF_typeHF_2
## HF_typeHF_3
## HF_typeHF_5
## HF_typeHF_7
## HF_typeHF_8
## day2

```



```

## session2
## session3
## session4
## activitystanding:HF_typeHF_2
## activitywalking:HF_typeHF_2
## activitystanding:HF_typeHF_3
## activitywalking:HF_typeHF_3
## activitystanding:HF_typeHF_5 0.197
## activitywalking:HF_typeHF_5 0.494 0.395
## activitystanding:HF_typeHF_7 0.200 0.502 0.199
## activitywalking:HF_typeHF_7 0.502 0.199 0.499
## activitystanding:HF_typeHF_8 0.199 0.499 0.198
## activitywalking:HF_typeHF_8 0.499 0.198 0.496
## HF_typeHF_2:day2 0.000 0.009 0.000
## HF_typeHF_3:day2 0.000 0.009 0.000
## HF_typeHF_5:day2 0.000 0.017 0.000
## HF_typeHF_7:day2 0.000 0.009 0.000
## HF_typeHF_8:day2 0.000 0.009 0.000
## day2:session2 0.002 0.004 0.004
## day2:session3 -0.002 -0.003 -0.002
## day2:session4 0.000 0.000 0.003
## actvtys:HF_HF_7 actvtyw:HF_HF_7 actvtys:HF_HF_8
## activitystanding
## activitywalking
## HF_typeHF_2
## HF_typeHF_3
## HF_typeHF_5
## HF_typeHF_7
## HF_typeHF_8
## day2
## session2
## session3
## session4
## activitystanding:HF_typeHF_2
## activitywalking:HF_typeHF_2
## activitystanding:HF_typeHF_3
## activitywalking:HF_typeHF_3
## activitystanding:HF_typeHF_5
## activitywalking:HF_typeHF_5
## activitystanding:HF_typeHF_7
## activitywalking:HF_typeHF_7 0.397
## activitystanding:HF_typeHF_8 0.506 0.201
## activitywalking:HF_typeHF_8 0.201 0.504 0.397
## HF_typeHF_2:day2 0.009 0.000 0.009
## HF_typeHF_3:day2 0.009 0.000 0.009
## HF_typeHF_5:day2 0.009 0.000 0.009
## HF_typeHF_7:day2 0.016 0.000 0.009
## HF_typeHF_8:day2 0.009 0.000 0.017
## day2:session2 -0.003 -0.004 0.002
## day2:session3 -0.003 -0.003 -0.004
## day2:session4 -0.001 -0.004 0.000
## actvtyw:HF_HF_8 HF_HF_2: HF_HF_3: HF_HF_5:
## activitystanding
## activitywalking

```

```

## HF_typeHF_2
## HF_typeHF_3
## HF_typeHF_5
## HF_typeHF_7
## HF_typeHF_8
## day2
## session2
## session3
## session4
## activitystanding:HF_typeHF_2
## activitywalking:HF_typeHF_2
## activitystanding:HF_typeHF_3
## activitywalking:HF_typeHF_3
## activitystanding:HF_typeHF_5
## activitywalking:HF_typeHF_5
## activitystanding:HF_typeHF_7
## activitywalking:HF_typeHF_7
## activitystanding:HF_typeHF_8
## activitywalking:HF_typeHF_8
## HF_typeHF_2:day2          0.000
## HF_typeHF_3:day2          0.000          0.497
## HF_typeHF_5:day2          0.000          0.494          0.490
## HF_typeHF_7:day2          0.000          0.503          0.500          0.496
## HF_typeHF_8:day2          0.001          0.500          0.497          0.494
## day2:session2             0.000          -0.001          -0.001          -0.001
## day2:session3             -0.001          0.000          0.000          0.000
## day2:session4             -0.001          0.001          0.002          0.001
## HF_HF_7: HF_HF_8: dy2:s2 dy2:s3
## activitystanding
## activitywalking
## HF_typeHF_2
## HF_typeHF_3
## HF_typeHF_5
## HF_typeHF_7
## HF_typeHF_8
## day2
## session2
## session3
## session4
## activitystanding:HF_typeHF_2
## activitywalking:HF_typeHF_2
## activitystanding:HF_typeHF_3
## activitywalking:HF_typeHF_3
## activitystanding:HF_typeHF_5
## activitywalking:HF_typeHF_5
## activitystanding:HF_typeHF_7
## activitywalking:HF_typeHF_7
## activitystanding:HF_typeHF_8
## activitywalking:HF_typeHF_8
## HF_typeHF_2:day2
## HF_typeHF_3:day2
## HF_typeHF_5:day2
## HF_typeHF_7:day2
## HF_typeHF_8:day2          0.503

```

```
## day2:session2          0.000   -0.001
## day2:session3          0.000    0.001    0.385
## day2:session4          0.001    0.001    0.345    0.447
##
## Standardized Within-Group Residuals:
##      Min      Q1      Med      Q3      Max
## -5.62020074 -0.38665545  0.02493225  0.45076438  3.41795349
##
## Number of Observations: 2638
## Number of Groups:
##              person      HF_type %in% person
##              24              144
## day %in% HF_type %in% person
##              280
```

Ok now make everything numerical.

```
mdl6_rs_session_num_full_reml <- lme(
  fixed = log(HF_value) ~ (activity) + (HF_type) + (day) + (session_num) + (HF_type * activity) + (
  random = list(person = ~ session_num, HF_type = ~ 1, day = ~ 1),
  data = df_clean,
  method = "REML",
  control = lmeControl(maxIter = 200,
    msMaxIter = 200,
    tolerance = 1e-6,
    niterEM = 100,
    opt = "nlminb") #basically, increase iterations
)
```

```
print(anova(mdl6_rs_session_num_reml, mdl6_rs_session_num_full_reml))
```

```
## Warning in anova.lme(mdl6_rs_session_num_reml, mdl6_rs_session_num_full_reml):
## fitted objects with different fixed effects. REML comparisons are not
## meaningful.
```

```
##              Model df      AIC      BIC   logLik   Test
## mdl6_rs_session_num_reml      1 36 -657.8396 -446.6514 364.9198
## mdl6_rs_session_num_full_reml    2 32 -679.9084 -492.1365 371.9542 1 vs 2
##              L.Ratio p-value
## mdl6_rs_session_num_reml
## mdl6_rs_session_num_full_reml 14.06881  0.0071
```

```
print(summary(mdl6_rs_session_num_full_reml))
```

```
## Linear mixed-effects model fit by REML
##   Data: df_clean
##       AIC      BIC   logLik
## -679.9084 -492.1365 371.9542
##
## Random effects:
##   Formula: ~session_num | person
##   Structure: General positive-definite, Log-Cholesky parametrization
##              StdDev      Corr
## (Intercept) 0.03929064 (Intr)
## session_num 0.02095441 -0.492
##
```

```

## Formula: ~1 | HF_type %in% person
## (Intercept)
## StdDev: 0.09893083
##
## Formula: ~1 | day %in% HF_type %in% person
## (Intercept) Residual
## StdDev: 0.2215259 0.1749754
##
## Fixed effects: log(HF_value) ~ (activity) + (HF_type) + (day) + (session_num) + (HF_type * act.
## Value Std.Error DF t-value p-value
## (Intercept) 4.382898 0.05454672 2344 80.35127 0.0000
## activitystanding -0.092374 0.01897740 2344 -4.86756 0.0000
## activitywalking 0.083221 0.02226965 2344 3.73699 0.0002
## HF_typeHF_2 -0.623506 0.07478724 115 -8.33707 0.0000
## HF_typeHF_3 -0.531331 0.07561230 115 -7.02704 0.0000
## HF_typeHF_5 -0.317097 0.07649726 115 -4.14520 0.0001
## HF_typeHF_7 -0.059702 0.07403058 115 -0.80645 0.4216
## HF_typeHF_8 -0.751504 0.07481908 115 -10.04428 0.0000
## day2 -0.082669 0.06848323 130 -1.20715 0.2296
## session_num 0.007147 0.00607795 2344 1.17587 0.2398
## activitystanding:HF_typeHF_2 -0.086699 0.02666466 2344 -3.25145 0.0012
## activitywalking:HF_typeHF_2 -0.454686 0.03133820 2344 -14.50899 0.0000
## activitystanding:HF_typeHF_3 0.197530 0.02685909 2344 7.35431 0.0000
## activitywalking:HF_typeHF_3 0.187008 0.03152461 2344 5.93213 0.0000
## activitystanding:HF_typeHF_5 0.067610 0.02697942 2344 2.50598 0.0123
## activitywalking:HF_typeHF_5 -0.095027 0.03168322 2344 -2.99927 0.0027
## activitystanding:HF_typeHF_7 -0.036670 0.02656870 2344 -1.38020 0.1677
## activitywalking:HF_typeHF_7 -0.145388 0.03120605 2344 -4.65898 0.0000
## activitystanding:HF_typeHF_8 0.188701 0.02676151 2344 7.05120 0.0000
## activitywalking:HF_typeHF_8 0.119564 0.03138492 2344 3.80959 0.0001
## HF_typeHF_2:day2 0.495498 0.09459890 130 5.23788 0.0000
## HF_typeHF_3:day2 0.499770 0.09524665 130 5.24712 0.0000
## HF_typeHF_5:day2 0.428342 0.09593192 130 4.46506 0.0000
## HF_typeHF_7:day2 0.337146 0.09401933 130 3.58592 0.0005
## HF_typeHF_8:day2 0.490943 0.09461661 130 5.18876 0.0000
## day2:session_num -0.035598 0.00599332 2344 -5.93969 0.0000
## Correlation:
## (Intr) actvtys actvtyw HF_tHF_2 HF_tHF_3 HF_tHF_5
## activitystanding -0.172
## activitywalking -0.148 0.395
## HF_typeHF_2 -0.686 0.126 0.101
## HF_typeHF_3 -0.679 0.125 0.100 0.496
## HF_typeHF_5 -0.671 0.123 0.099 0.490 0.484
## HF_typeHF_7 -0.693 0.128 0.102 0.506 0.500 0.494
## HF_typeHF_8 -0.686 0.126 0.101 0.501 0.495 0.490
## day2 -0.643 0.019 0.000 0.448 0.443 0.438
## session_num -0.188 -0.003 0.034 0.000 0.000 0.000
## activitystanding:HF_typeHF_2 0.123 -0.712 -0.282 -0.177 -0.089 -0.088
## activitywalking:HF_typeHF_2 0.098 -0.282 -0.707 -0.141 -0.071 -0.070
## activitystanding:HF_typeHF_3 0.122 -0.706 -0.280 -0.089 -0.176 -0.087
## activitywalking:HF_typeHF_3 0.098 -0.280 -0.703 -0.071 -0.141 -0.070
## activitystanding:HF_typeHF_5 0.122 -0.703 -0.279 -0.089 -0.088 -0.175
## activitywalking:HF_typeHF_5 0.098 -0.279 -0.700 -0.071 -0.070 -0.140
## activitystanding:HF_typeHF_7 0.123 -0.714 -0.283 -0.090 -0.089 -0.088

```

```

## activitywalking:HF_typeHF_7  0.099 -0.283 -0.710 -0.072 -0.071 -0.070
## activitystanding:HF_typeHF_8 0.122 -0.709 -0.281 -0.090 -0.089 -0.088
## activitywalking:HF_typeHF_8  0.098 -0.281 -0.706 -0.072 -0.071 -0.070
## HF_typeHF_2:day2             0.445 -0.013  0.000 -0.648 -0.321 -0.317
## HF_typeHF_3:day2             0.442 -0.013  0.000 -0.322 -0.654 -0.315
## HF_typeHF_5:day2             0.439 -0.013  0.000 -0.320 -0.317 -0.660
## HF_typeHF_7:day2             0.447 -0.013  0.000 -0.326 -0.323 -0.319
## HF_typeHF_8:day2             0.445 -0.013  0.000 -0.325 -0.321 -0.317
## day2:session_num             0.135 -0.008  0.003  0.000  0.000  0.000
##                               HF_tHF_7 HF_tHF_8 day2    sssn_n actvtys:HF_HF_2
## activitystanding
## activitywalking
## HF_typeHF_2
## HF_typeHF_3
## HF_typeHF_5
## HF_typeHF_7
## HF_typeHF_8             0.506
## day2                     0.453  0.448
## session_num              0.000  0.001  0.105
## activitystanding:HF_typeHF_2 -0.091 -0.090 -0.012  0.001
## activitywalking:HF_typeHF_2  -0.072 -0.072  0.000  0.000  0.395
## activitystanding:HF_typeHF_3 -0.090 -0.089 -0.012  0.001  0.503
## activitywalking:HF_typeHF_3  -0.072 -0.071  0.000  0.000  0.199
## activitystanding:HF_typeHF_5 -0.090 -0.089 -0.012  0.001  0.500
## activitywalking:HF_typeHF_5  -0.072 -0.071  0.000 -0.001  0.198
## activitystanding:HF_typeHF_7 -0.177 -0.090 -0.012  0.001  0.508
## activitywalking:HF_typeHF_7  -0.142 -0.072  0.000  0.001  0.201
## activitystanding:HF_typeHF_8 -0.090 -0.177 -0.012  0.001  0.505
## activitywalking:HF_typeHF_8  -0.072 -0.142  0.000  0.000  0.200
## HF_typeHF_2:day2           -0.328 -0.325 -0.691  0.000  0.017
## HF_typeHF_3:day2           -0.326 -0.322 -0.687  0.000  0.009
## HF_typeHF_5:day2           -0.323 -0.320 -0.682  0.000  0.009
## HF_typeHF_7:day2           -0.643 -0.326 -0.696  0.000  0.009
## HF_typeHF_8:day2           -0.328 -0.648 -0.691 -0.001  0.009
## day2:session_num           0.001 -0.001 -0.213 -0.502 -0.001
##                               actvttyw:HF_HF_2 actvtys:HF_HF_3 actvttyw:HF_HF_3
## activitystanding
## activitywalking
## HF_typeHF_2
## HF_typeHF_3
## HF_typeHF_5
## HF_typeHF_7
## HF_typeHF_8
## day2
## session_num
## activitystanding:HF_typeHF_2
## activitywalking:HF_typeHF_2
## activitystanding:HF_typeHF_3  0.199
## activitywalking:HF_typeHF_3  0.500  0.395
## activitystanding:HF_typeHF_5  0.198  0.497  0.197
## activitywalking:HF_typeHF_5  0.497  0.197  0.494
## activitystanding:HF_typeHF_7  0.201  0.505  0.200
## activitywalking:HF_typeHF_7  0.505  0.200  0.502
## activitystanding:HF_typeHF_8  0.200  0.501  0.199

```

```

## activitywalking:HF_typeHF_8    0.502          0.199          0.499
## HF_typeHF_2:day2              0.000          0.009          0.000
## HF_typeHF_3:day2              0.000          0.017          0.000
## HF_typeHF_5:day2              0.000          0.009          0.000
## HF_typeHF_7:day2              0.000          0.009          0.000
## HF_typeHF_8:day2              0.000          0.009          0.000
## day2:session_num              -0.001         -0.001         -0.001
##                                actvtys:HF_HF_5 actvtyw:HF_HF_5 actvtys:HF_HF_7
## activitystanding
## activitywalking
## HF_typeHF_2
## HF_typeHF_3
## HF_typeHF_5
## HF_typeHF_7
## HF_typeHF_8
## day2
## session_num
## activitystanding:HF_typeHF_2
## activitywalking:HF_typeHF_2
## activitystanding:HF_typeHF_3
## activitywalking:HF_typeHF_3
## activitystanding:HF_typeHF_5
## activitywalking:HF_typeHF_5    0.395
## activitystanding:HF_typeHF_7    0.502          0.199
## activitywalking:HF_typeHF_7    0.199          0.499          0.397
## activitystanding:HF_typeHF_8    0.499          0.198          0.506
## activitywalking:HF_typeHF_8    0.198          0.496          0.201
## HF_typeHF_2:day2              0.009          0.000          0.009
## HF_typeHF_3:day2              0.009          0.000          0.009
## HF_typeHF_5:day2              0.017          0.000          0.009
## HF_typeHF_7:day2              0.009          0.000          0.016
## HF_typeHF_8:day2              0.009          0.000          0.009
## day2:session_num              -0.001          0.001         -0.001
##                                actvtyw:HF_HF_7 actvtys:HF_HF_8 actvtyw:HF_HF_8
## activitystanding
## activitywalking
## HF_typeHF_2
## HF_typeHF_3
## HF_typeHF_5
## HF_typeHF_7
## HF_typeHF_8
## day2
## session_num
## activitystanding:HF_typeHF_2
## activitywalking:HF_typeHF_2
## activitystanding:HF_typeHF_3
## activitywalking:HF_typeHF_3
## activitystanding:HF_typeHF_5
## activitywalking:HF_typeHF_5
## activitystanding:HF_typeHF_7
## activitywalking:HF_typeHF_7
## activitystanding:HF_typeHF_8    0.201
## activitywalking:HF_typeHF_8    0.504          0.397
## HF_typeHF_2:day2              0.000          0.009          0.000

```

```

## HF_typeHF_3:day2          0.000          0.009          0.000
## HF_typeHF_5:day2          0.000          0.009          0.000
## HF_typeHF_7:day2          0.000          0.009          0.000
## HF_typeHF_8:day2          0.000          0.017          0.001
## day2:session_num         -0.003         -0.001         -0.001
## HF_HF_2: HF_HF_3: HF_HF_5: HF_HF_7: HF_HF_8:
## activitystanding
## activitywalking
## HF_typeHF_2
## HF_typeHF_3
## HF_typeHF_5
## HF_typeHF_7
## HF_typeHF_8
## day2
## session_num
## activitystanding:HF_typeHF_2
## activitywalking:HF_typeHF_2
## activitystanding:HF_typeHF_3
## activitywalking:HF_typeHF_3
## activitystanding:HF_typeHF_5
## activitywalking:HF_typeHF_5
## activitystanding:HF_typeHF_7
## activitywalking:HF_typeHF_7
## activitystanding:HF_typeHF_8
## activitywalking:HF_typeHF_8
## HF_typeHF_2:day2
## HF_typeHF_3:day2          0.497
## HF_typeHF_5:day2          0.494          0.490
## HF_typeHF_7:day2          0.503          0.500          0.496
## HF_typeHF_8:day2          0.500          0.497          0.494          0.503
## day2:session_num          0.001          0.001          0.001          0.001          0.001
##
## Standardized Within-Group Residuals:
##      Min      Q1      Med      Q3      Max
## -5.68320234 -0.38797341  0.03565694  0.46058244  3.42854030
##
## Number of Observations: 2638
## Number of Groups:
##           person      HF_type %in% person
##           24           144
## day %in% HF_type %in% person
##           280

```

0.14 Section 9: Look at different covariance structures

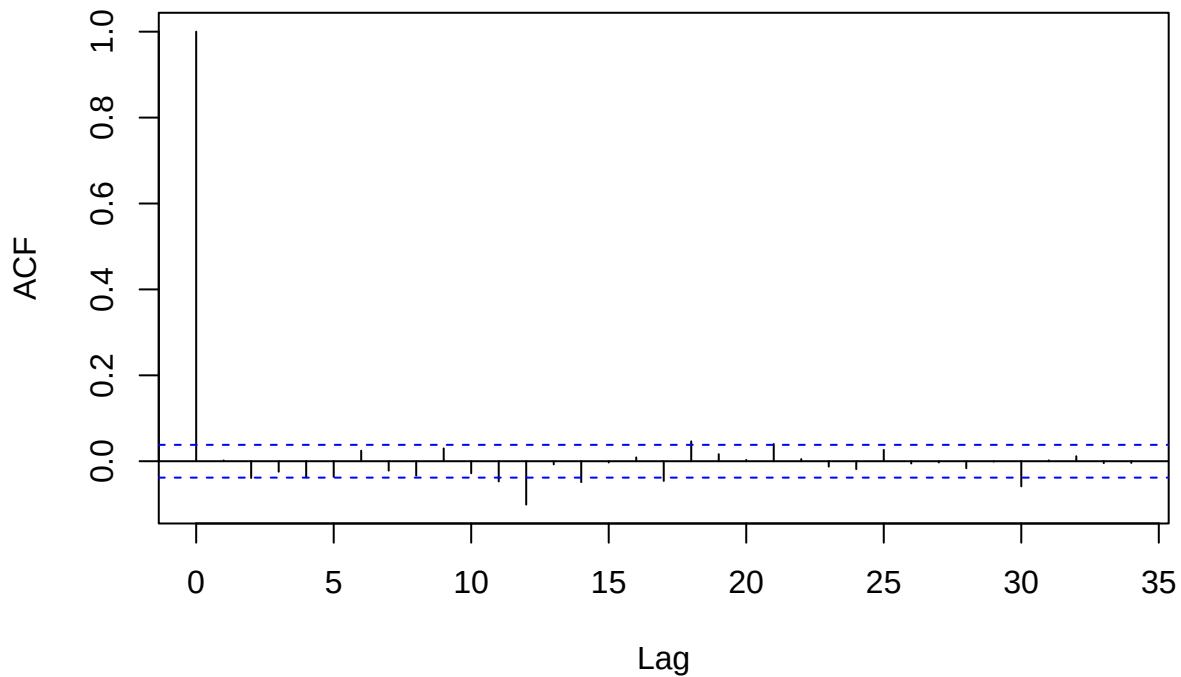
Normal residuals

```

mdl6_rs_session_num_res <- residuals(mdl6_rs_session_num_reml, type = "normalized")
acf(mdl6_rs_session_num_res, main = "ACF of BAU residuals")

```

ACF of BAU residuals



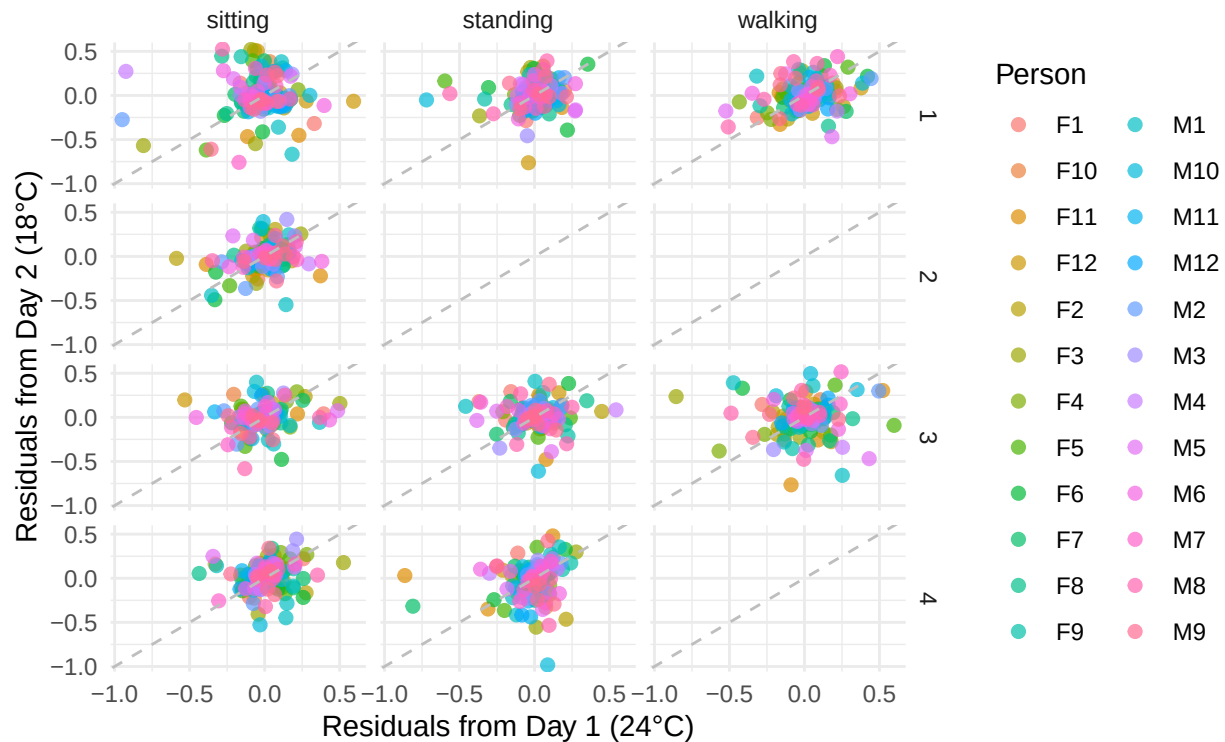
```
# Base model with all main effects, no interactions, no person random effect
df_clean$resids_fin <- residuals(mdl6_rs_session_num_reml)

df_pivot <- df_clean %>%
  pivot_wider(
    id_cols = c(person, session, activity, HF_type),
    names_from = day,
    values_from = resids_fin,
    names_prefix = "res_day"
  ) %>%
  # Remove cases where data is missing for one of the two days
  filter(!is.na(res_day1) & !is.na(res_day2))

ggplot(df_pivot, aes(x = res_day1, y = res_day2, color = factor(person))) +
  geom_point(alpha = 0.7, size = 2) +
  # Dashed line y=x represents perfect correlation
  geom_abline(slope = 1, intercept = 0, linetype = "dashed", color = "gray") +
  facet_grid(session ~ activity) +
  labs(
    title = "Residual Correlation: Day 1 vs Day 2 (Independent)",
    subtitle = "Checking for independence across different temperature environments",
    x = "Residuals from Day 1 (24°C)",
    y = "Residuals from Day 2 (18°C)",
    color = "Person"
  ) +
  theme_minimal() +
  theme(legend.position = "right")
```


Residual Correlation: Day 1 vs Day 2 (Independent)

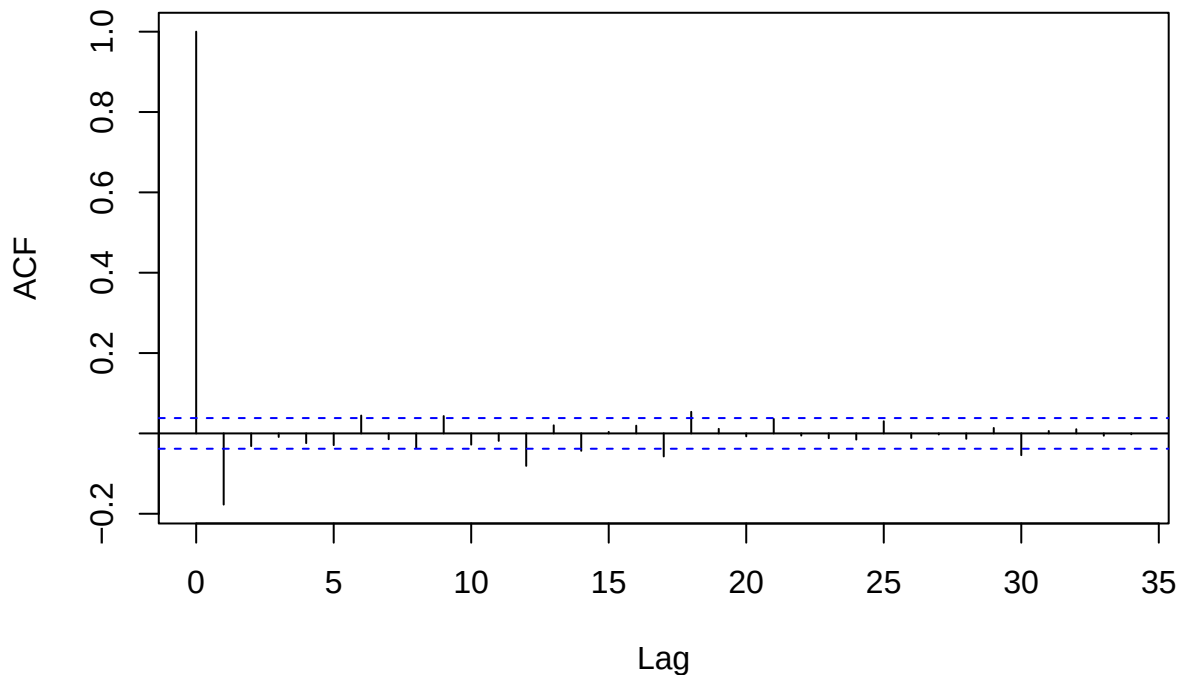
Checking for independence across different temperature environments



```
mdl6_rs_session_num_ar1temp_reml <- lme(
  fixed = log(HF_value) ~ (activity) + (HF_type) + (day) + (session) + (HF_type * activity) + (HF_t
  random = list(person = ~ session_num, HF_type = ~ 1, day = ~ 1),
  data = df_clean,
  method = "REML",
  correlation = corAR1(form = ~ 1 | person/HF_type/day),
  control = lmeControl(maxIter = 200,
    msMaxIter = 200,
    tolerance = 1e-6,
    niterEM = 100,
    opt = "nlminb") #basically, increase iterations
)
```

```
mdl6_rs_session_num_ar1temp_res <- residuals(mdl6_rs_session_num_ar1temp_reml, type = "normalized",)
acf(mdl6_rs_session_num_ar1temp_res, main = "ACF of AR1 residuals")
```

ACF of AR1 residuals



```
print(anova mdl6_rs_session_num_reml, mdl6_rs_session_num_ar1temp_reml))

##                                Model df      AIC      BIC    logLik    Test
## mdl6_rs_session_num_reml          1 36 -657.8396 -446.6514 364.9198
## mdl6_rs_session_num_ar1temp_reml    2 37 -724.6877 -507.6331 399.3438 1 vs 2
##                                L.Ratio p-value
## mdl6_rs_session_num_reml
## mdl6_rs_session_num_ar1temp_reml 68.84805 <.0001

# Base model with all main effects, no interactions, no person random effect
df_clean$resids_fin_ar1 <- residuals(mdl6_rs_session_num_ar1temp_reml)

df_pivot <- df_clean %>%
  pivot_wider(
    id_cols = c(person, session, activity, HF_type),
    names_from = day,
    values_from = resids_fin_ar1,
    names_prefix = "res_day"
  ) %>%
  # Remove cases where data is missing for one of the two days
  filter(!is.na(res_day1) & !is.na(res_day2))

ggplot(df_pivot, aes(x = res_day1, y = res_day2, color = factor(person))) +
  geom_point(alpha = 0.7, size = 2) +
  # Dashed line y=x represents perfect correlation
  geom_abline(slope = 1, intercept = 0, linetype = "dashed", color = "gray") +
  facet_grid(session ~ activity) +
  labs(
    title = "Residual Correlation: Day 1 vs Day 2 (AR1)",
```

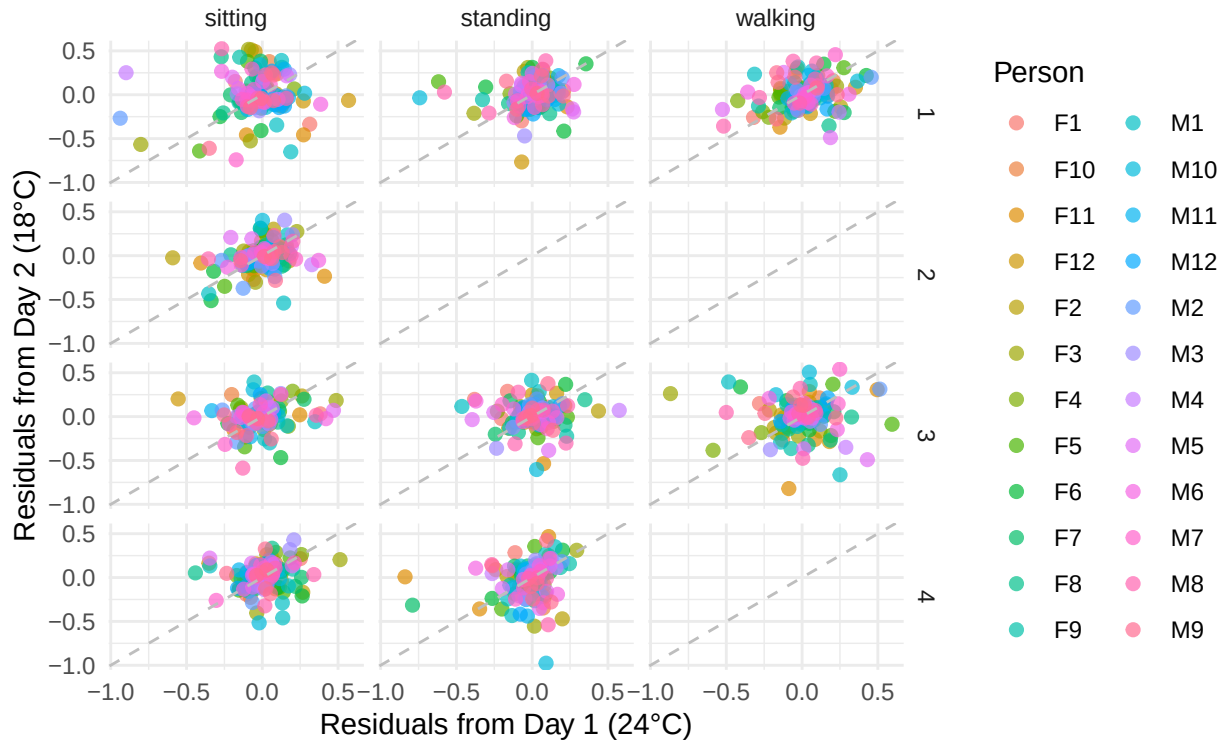
```

subtitle = "Checking for independence across different temperature environments",
x = "Residuals from Day 1 (24°C)",
y = "Residuals from Day 2 (18°C)",
color = "Person"
) +
theme_minimal() +
theme(legend.position = "right")

```

Residual Correlation: Day 1 vs Day 2 (AR1)

Checking for independence across different temperature environments



```

mdl6_rs_session_num_arh1temp_reml_day <- lme(
  fixed = log(HF_value) ~ (activity) + (HF_type) + (day) + (session) + (HF_type * activity) + (HF_t
  random = list(person = ~ session_num, HF_type = ~ 1, day = ~ 1),
  data = df_clean,
  method = "REML",
  correlation = corAR1(form = ~ 1 | person/HF_type/day),
  weights = varIdent(form = ~ 1 | day),
  control = lmeControl(maxIter = 200,
    msMaxIter = 200,
    tolerance = 1e-6,
    niterEM = 100,
    opt = "nlminb") #basically, increase iterations
)

print(anova(mdl6_rs_session_num_ar1temp_reml, mdl6_rs_session_num_arh1temp_reml_day))

```

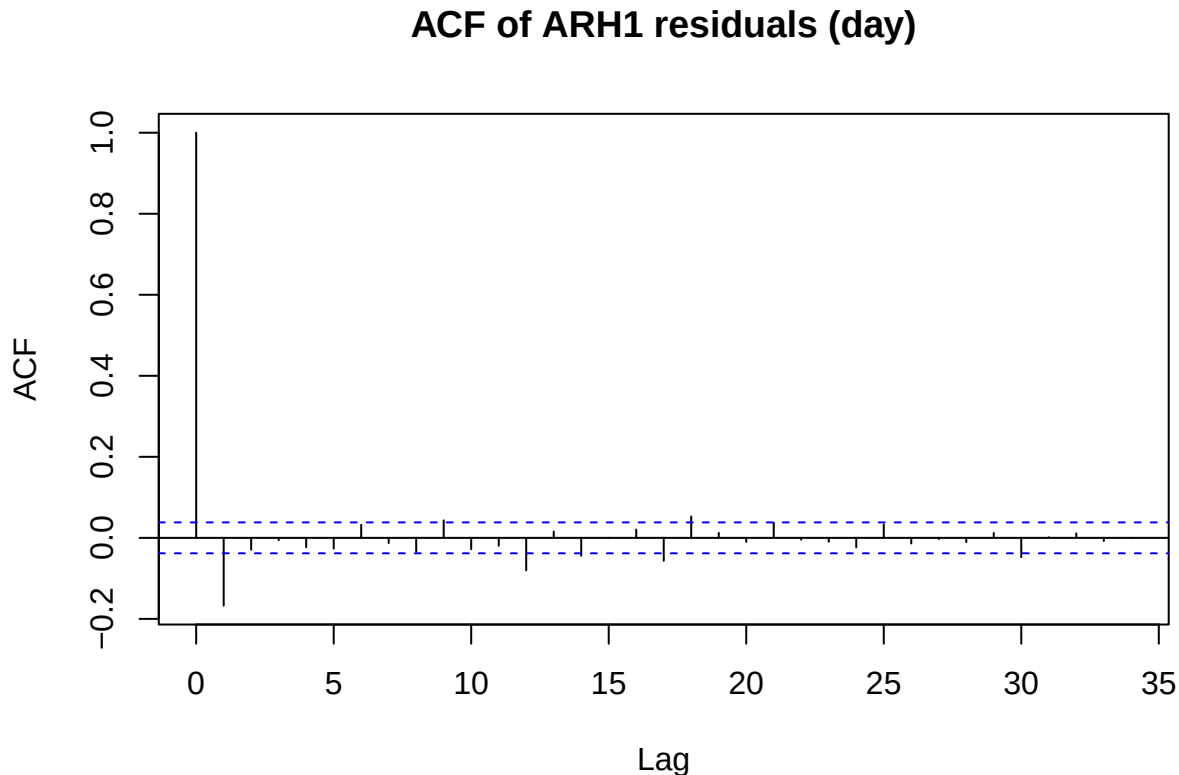
```

##               Model df          AIC          BIC    logLik
## mdl6_rs_session_num_ar1temp_reml      1 37 -724.6877 -507.6331 399.3438
## mdl6_rs_session_num_arh1temp_reml_day  2 38 -734.7629 -511.8420 405.3814

```

```
##                                     Test L.Ratio p-value
## mdl6_rs_session_num_ar1temp_reml
## mdl6_rs_session_num_ar1temp_reml_day 1 vs 2 12.0752    5e-04

mdl6_rs_session_num_ar1temp_reml_day_res <- residuals(mdl6_rs_session_num_ar1temp_reml_day, type = "n
acf(mdl6_rs_session_num_ar1temp_reml_day_res, main = "ACF of ARH1 residuals (day)")
```



```
# Base model with all main effects, no interactions, no person random effect
df_clean$resids_fin_arh1_day <- residuals(mdl6_rs_session_num_ar1temp_reml_day)

df_pivot <- df_clean %>%
  pivot_wider(
    id_cols = c(person, session, activity, HF_type),
    names_from = day,
    values_from = resids_fin_arh1_day,
    names_prefix = "res_day"
  ) %>%
  # Remove cases where data is missing for one of the two days
  filter(!is.na(res_day1) & !is.na(res_day2))

ggplot(df_pivot, aes(x = res_day1, y = res_day2, color = factor(person))) +
  geom_point(alpha = 0.7, size = 2) +
  # Dashed line y=x represents perfect correlation
  geom_abline(slope = 1, intercept = 0, linetype = "dashed", color = "gray") +
  facet_grid(session ~ activity) +
  labs(
    title = "Residual Correlation: Day 1 vs Day 2 (ARH1, Day)",
    subtitle = "Checking for independence across different temperature environments",
    x = "Residuals from Day 1 (24°C)",
```

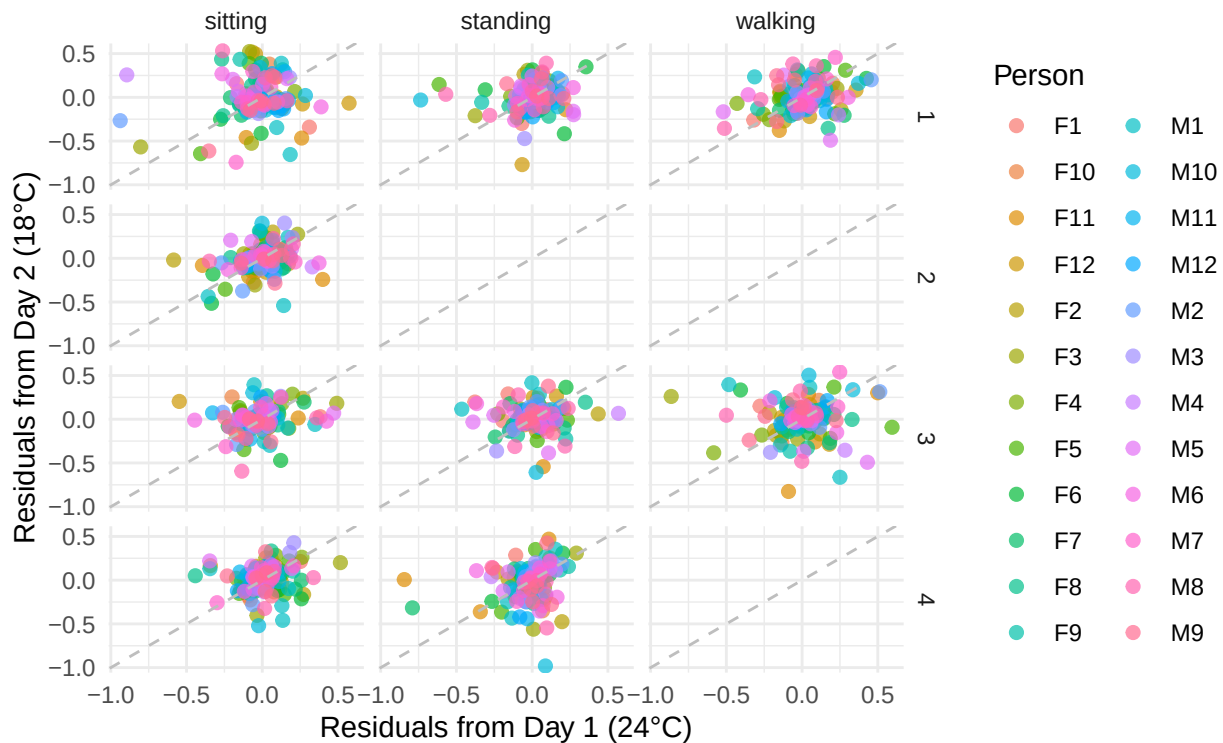
```

y = "Residuals from Day 2 (18°C)",
color = "Person"
) +
theme_minimal() +
theme(legend.position = "right")

```

Residual Correlation: Day 1 vs Day 2 (ARH1, Day)

Checking for independence across different temperature environments



```

mdl6_rs_session_num_arh1temp_reml_activity <- lme(
  fixed = log(HF_value) ~ (activity) + (HF_type) + (day) + (session) + (HF_type * activity) + (HF_t
  random = list(person = ~ session_num, HF_type = ~ 1, day = ~ 1),
  data = df_clean,
  method = "REML",
  correlation = corAR1(form = ~ 1 | person/HF_type/day),
  weights = varIdent(form = ~ 1 | activity),
  control = lmeControl(maxIter = 200,
    msMaxIter = 200,
    tolerance = 1e-6,
    niterEM = 100,
    opt = "nlminb") #basically, increase iterations
)

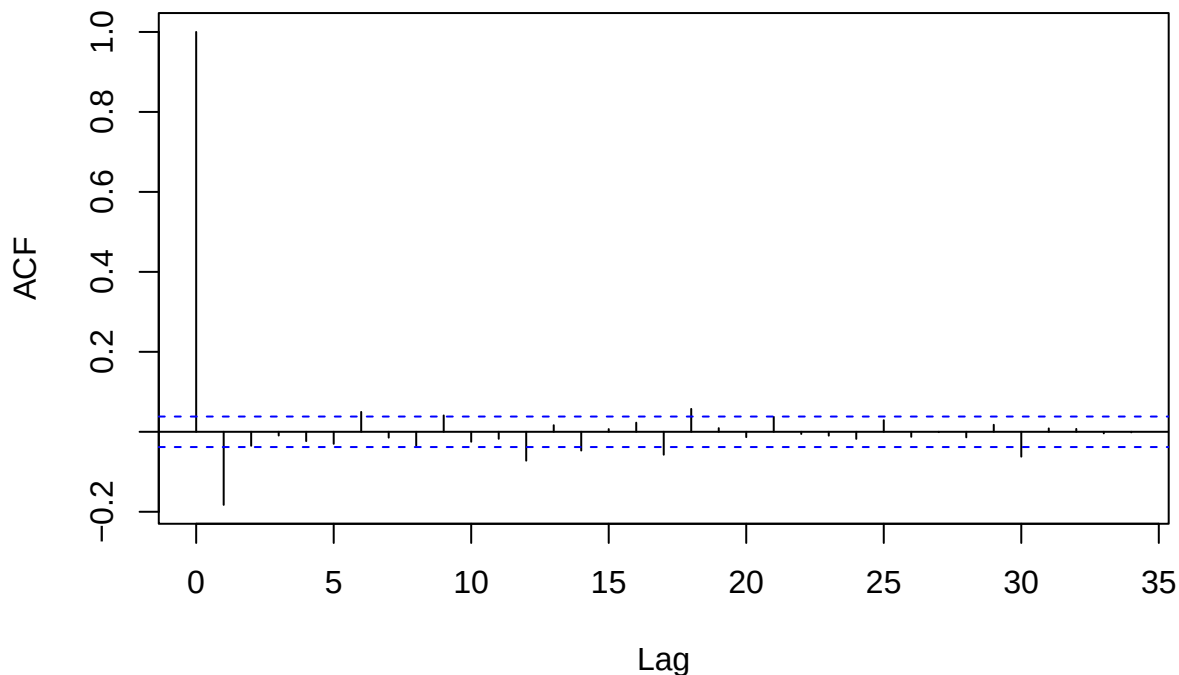
print(anova(mdl6_rs_session_num_ar1temp_reml, mdl6_rs_session_num_arh1temp_reml_activity))

```

##	Model	df	AIC	BIC
##	mdl6_rs_session_num_ar1temp_reml	1 37	-724.6877	-507.6331
##	mdl6_rs_session_num_arh1temp_reml_activity	2 39	-738.1507	-509.3635
##		logLik	Test	L.Ratio p-value
##	mdl6_rs_session_num_ar1temp_reml	399.3438		

```
## mdl6_rs_session_num_arh1temp_reml_activity 408.0754 1 vs 2 17.46307 2e-04
mdl6_rs_session_num_arh1temp_reml_activity_res <- residuals(mdl6_rs_session_num_arh1temp_reml_activity,
acf(mdl6_rs_session_num_arh1temp_reml_activity_res, main = "ACF of ARH1 residuals (actiity)"))
```

ACF of ARH1 residuals (actiity)



```
# Base model with all main effects, no interactions, no person random effect
df_clean$resids_fin_arh1_activity <- residuals(mdl6_rs_session_num_arh1temp_reml_activity)

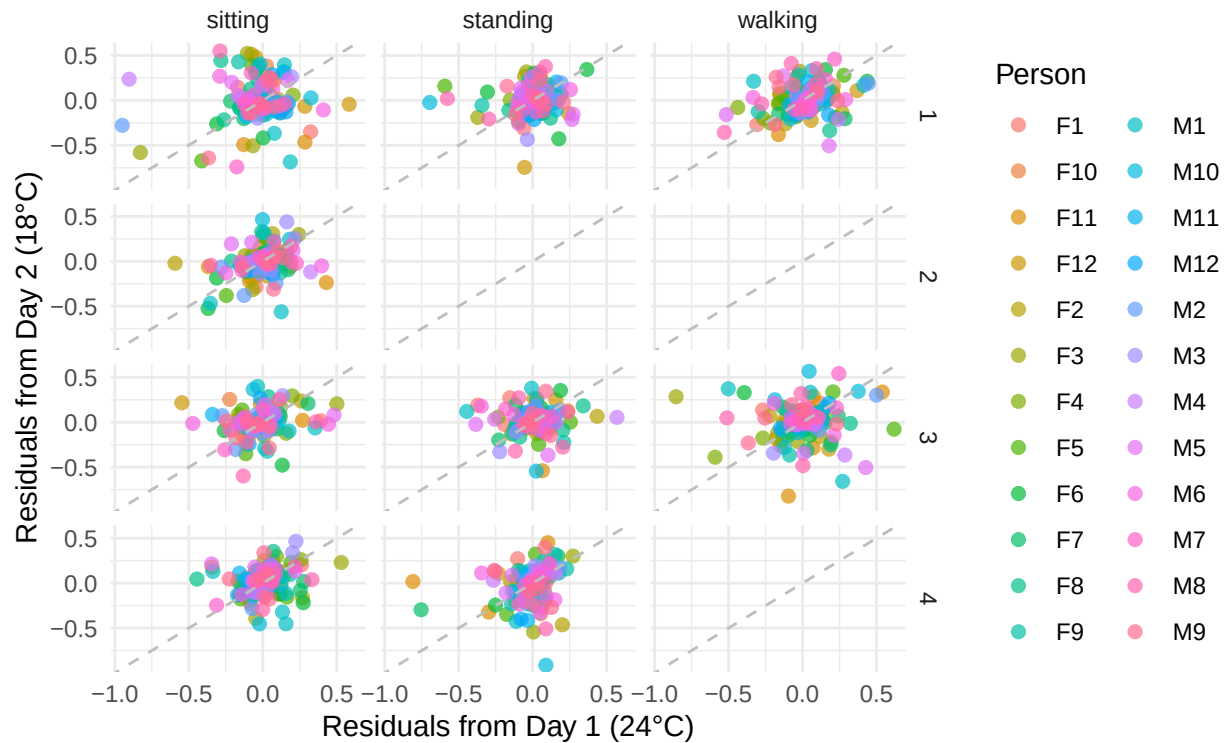
df_pivot <- df_clean %>%
  pivot_wider(
    id_cols = c(person, session, activity, HF_type),
    names_from = day,
    values_from = resids_fin_arh1_activity,
    names_prefix = "res_day"
  ) %>%
  # Remove cases where data is missing for one of the two days
  filter(!is.na(res_day1) & !is.na(res_day2))

ggplot(df_pivot, aes(x = res_day1, y = res_day2, color = factor(person))) +
  geom_point(alpha = 0.7, size = 2) +
  # Dashed line y=x represents perfect correlation
  geom_abline(slope = 1, intercept = 0, linetype = "dashed", color = "gray") +
  facet_grid(session ~ activity) +
  labs(
    title = "Residual Correlation: Day 1 vs Day 2 (ARH1, Activity)",
    subtitle = "Checking for independence across different temperature environments",
    x = "Residuals from Day 1 (24°C)",
    y = "Residuals from Day 2 (18°C)",
    color = "Person"
```

```
) +  
theme_minimal() +  
theme(legend.position = "right")
```

Residual Correlation: Day 1 vs Day 2 (ARH1, Activity)

Checking for independence across different temperature environments

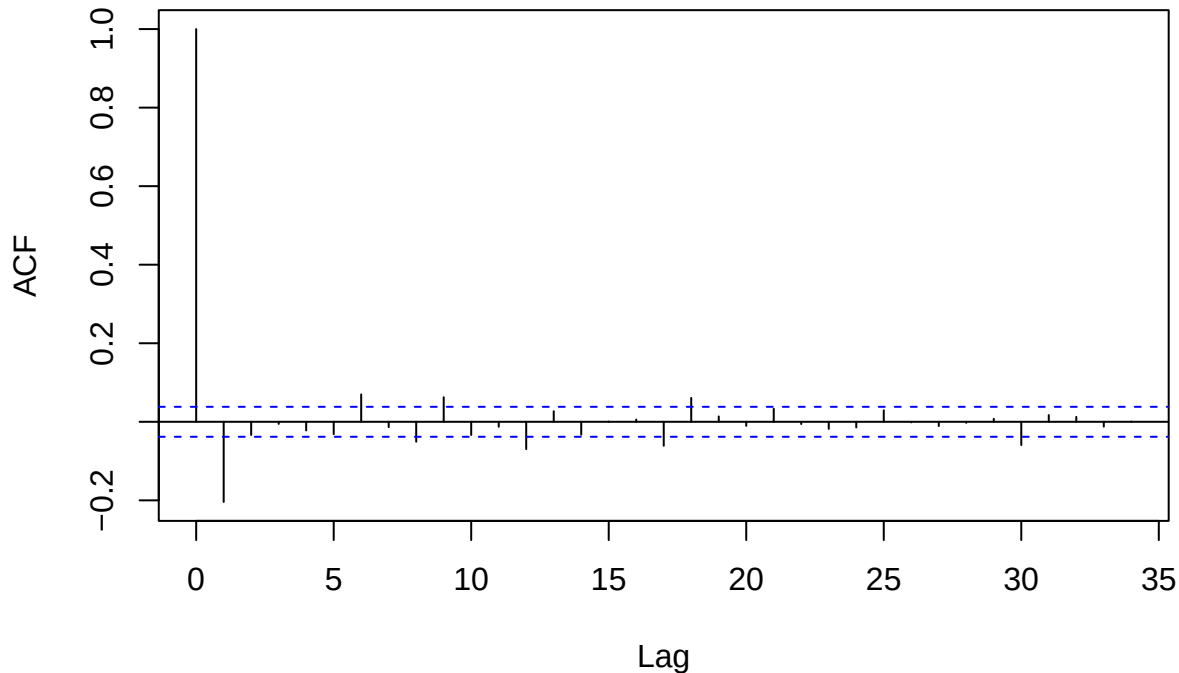


```
mdl6_rs_session_num_arh1temp_reml_sensor <- lme(  
  fixed = log(HF_value) ~ (activity) + (HF_type) + (day) + (session) + (HF_type * activity) + (HF_t  
  random = list(person = ~ session_num, HF_type = ~ 1, day = ~ 1),  
  data = df_clean,  
  method = "REML",  
  correlation = corAR1(form = ~ 1 | person/HF_type/day),  
  weights = varIdent(form = ~ 1 | HF_type),  
  control = lmeControl(maxIter = 200,  
    msMaxIter = 200,  
    tolerance = 1e-6,  
    niterEM = 100,  
    opt = "nlminb") #basically, increase iterations  
)  
  
print(anova(mdl6_rs_session_num_ar1temp_reml, mdl6_rs_session_num_arh1temp_reml_sensor))
```

```
##                                Model df      AIC      BIC  
## mdl6_rs_session_num_ar1temp_reml          1 37 -724.6877 -507.6331  
## mdl6_rs_session_num_arh1temp_reml_sensor    2 42 -1283.6735 -1037.2873  
##                                logLik    Test  L.Ratio p-value  
## mdl6_rs_session_num_ar1temp_reml          399.3438  
## mdl6_rs_session_num_arh1temp_reml_sensor 683.8368 1 vs 2 568.9859 <.0001
```

```
mdl6_rs_session_num_arh1temp_reml_sensor_res <- residuals(mdl6_rs_session_num_arh1temp_reml_sensor, type = "reml")
acf(mdl6_rs_session_num_arh1temp_reml_sensor_res, main = "ACF of ARH1 residuals (sensor)")
```

ACF of ARH1 residuals (sensor)



```
# Base model with all main effects, no interactions, no person random effect
df_clean$resids_fin_arh1_sensor <- residuals(mdl6_rs_session_num_arh1temp_reml_sensor)

df_pivot <- df_clean %>%
  pivot_wider(
    id_cols = c(person, session, activity, HF_type),
    names_from = day,
    values_from = resids_fin_arh1_sensor,
    names_prefix = "res_day"
  ) %>%
  # Remove cases where data is missing for one of the two days
  filter(!is.na(res_day1) & !is.na(res_day2))

ggplot(df_pivot, aes(x = res_day1, y = res_day2, color = factor(person))) +
  geom_point(alpha = 0.7, size = 2) +
  # Dashed line y=x represents perfect correlation
  geom_abline(slope = 1, intercept = 0, linetype = "dashed", color = "gray") +
  facet_grid(session ~ activity) +
  labs(
    title = "Residual Correlation: Day 1 vs Day 2 (ARH1, Sensor)",
    subtitle = "Checking for independence across different temperature environments",
    x = "Residuals from Day 1 (24°C)",
    y = "Residuals from Day 2 (18°C)",
    color = "Person"
  ) +
```



```
theme_minimal() +
theme(legend.position = "right")
```

Residual Correlation: Day 1 vs Day 2 (ARH1, Sensor)

Checking for independence across different temperature environments

